Credit Banks, Expectations, and the Cost of Layered Climate Policy: Evidence from U.S. Transportation Fuel

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Abstract

Congress must decide whether to extend 45Z, the federal production tax credit for clean transportation fuel, beyond 2029. The credit is paid on gallons that are already regulated: the federal Renewable Fuel Standard requires refiners to blend biofuel, and California, Oregon, Washington and New Mexico run clean fuel standards with bankable credits and capped prices. I ask what an extension would achieve on top of these policies. I build a dynamic model of U.S. fuel and vehicle markets in which the federal and state credit markets clear each year, and estimate investment, electric-vehicle and charger adoption, and credit holders' expectations from market data. As long as the federal mandate binds, it fixes how much biofuel is used, so an extension mainly lowers the price of RINs, the credits refiners buy to comply. If EPA keeps the mandate unchanged, about three quarters of each federal dollar reaches fuel buyers as cheaper fuel and emissions rise by 64 million tonnes over 2030–45; emissions fall only if EPA raises the mandate at the same time. Current law as a whole abates carbon dioxide at about $134 a tonne and achieves about 15% of the welfare gain from a carbon price set at the social cost of carbon. State credit prices have stayed far below what the statutes imply. Under the expectation rule I estimate from these prices, low prices mean that fewer credits are generated, so California's bank runs out and its credit price reaches the cap earlier and more abruptly than under perfect foresight.

Keywords: overlapping regulation, federalism, tradable credits, banking, expectations, subsidy incidence.

JEL codes: H23, H77, Q48, Q54, Q58.

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1 Introduction

U.S. climate policy for transportation fuel is made by several governments at once, and their instruments apply to the same gallon of fuel. Refiners must blend biofuel under the federal Renewable Fuel Standard (RFS), which EPA enforces with tradable credits known as RINs. Producers of low-carbon fuel can also claim a federal production tax credit, 45Z. On top of these, California, Oregon, Washington and New Mexico run their own clean fuel standards, whose credits can be banked for later use and whose prices are capped. All of these rules were rewritten in 2025 and 2026: Congress extended 45Z to 2029 and limited it to North American feedstock, EPA raised the RFS percentages for 2026 and 2027, and California tightened its standard. Congress must now decide whether to extend 45Z beyond 2029. The credits these policies created in 2026 were worth about $41 billion at observed prices.[1]

This paper asks what an extension of 45Z would achieve on top of these policies, and when its effects would arrive. Answering it requires knowing what the existing policies achieve together, so I estimate that too. My main finding is that, while the federal mandate binds, about 73 cents of each federal dollar would reach fuel buyers as cheaper fuel, and emissions would rise by 64 Mt over 2030–45.

The answer depends on which constraint binds. When the federal mandate binds, it fixes the quantity of biofuel, so a federal subsidy for biofuel mainly lowers the price of the mandate's credits, and the lower price is passed on to fuel buyers. For a single blend mandate this is a known result (de Gorter and Just, 2009; Lapan and Moschini, 2012). In the United States the binding constraint could instead be production capacity, the state standards or their price caps. The paper's main aim is to measure where a federal dollar goes in each of these cases. Which case applies depends partly on how EPA sets the mandate's percentages, although it is the Treasury, not EPA, that pays for the credit. The same problem comes up whenever two levels of government regulate one externality, for example when a state renewable portfolio standard meets a federal production tax credit. The federal blenders' credit shows the mechanism at work. Between 2014 and 2021 the $1 credit lapsed and was restored several times while the mandate bound, and each dollar of credit lowered the price of biodiesel and renewable diesel RINs by about 41 cents (standard error 13 cents). Congress often restored the credit after the fact, so its value was uncertain when fuel was sold. I use the episode as a benchmark for a temporary, uncertain credit, and Section 6 discusses what it implies for one expected to stay.

The timing depends on the state credit markets, where prices have stayed far below what the statutes imply. In a market with a bank, the credit price should rise toward the cap as the bank runs down (Salant, 2016), so each price implies a year in which holders expect the bank to run

[1] Credits generated in 2026 times their 2026 price: RINs at the observed January–August prices (about $1.50), state clean fuel credits ($2.6 billion), and the model's 2026 outlay on 45Z ($4.2 billion, excluding fuel from imported fat). At the model's 2026 RIN price of $2.29 the total is $59 billion.

out. In California between 2024 and 2025, while the state was tightening its standard, the horizon implied by fundamentals fell from about 10 years to about 7, but the horizon implied by the price rose from 29 to 31 years (Figure 1; Section 2.1). Yet when I run the model forward under current law, California's bank runs out by 2030–33 under every description of expectations I consider.

To study both questions I build a dynamic equilibrium model of U.S. fuel and vehicle markets for 2026–45 (Section 4). Each year the markets for fuel, feedstock, RINs and the four state credits clear together, because the same gallon earns credits in several of them, and credit banks, biofuel capacity, vehicle fleets and charging stations carry each year's outcome into the next. The model includes the four state programs with their caps and the rules that respond to their banks, the RFS with credit banking, and 45Z.

Three behaviors in the model are estimated from market data (Section 5). Investors' decisions to build or wait are estimated from 46 capacity decisions in 2014–25. Entry costs are about 3 to 11 times engineering cost, and investors value a credit that is in law as if it will continue. Electric-vehicle and charger adoption is estimated from state sales and station counts in 2021–25 (Appendix D.6). For credit holders I estimate a rule, fit to prices and banks in California and Oregon in 2016–25, that maps the credit balance into the year they expect the bank to run out. Holders respond to news about the bank rather than to the statute: a megatonne more credits than expected moves California's expected exhaustion year out by about three quarters of a year. Given the net credits that actually occurred, the rule tracks California's price more closely than a no-change forecast; when it must also forecast the balance, it does better only up to four quarters ahead. Since the rule describes how holders update rather than why, I also report every result under perfect foresight and with a weaker response to the balance.

The paper has four main results. First, while the federal mandate binds, extending 45Z beyond 2029 is largely a transfer. About 73 cents of each federal dollar reach fuel buyers as cheaper fuel, and the extra fuel use raises emissions by 64 Mt over 2030–45. Capacity barely changes, because investors already value the policy stack as if it would continue.

Second, moving the binding constraint is much cheaper than paying for biofuel within it. For the same additional 100 Mt of abatement, a higher RFS requirement costs $163 a tonne, pairing it with the 45Z extension raises the cost to $235, and a carbon price costs $32. These are costs of a small addition to current law, so they should not be compared with the average cost of current law as a whole.

Third, current law abates 1.93 Gt of carbon dioxide over 2026–45 at about $134 a tonne and achieves 15% of the net welfare gain of a carbon price set at the social cost of carbon. Over 2030–45, more than half of the abatement the state standards record inside the four states comes from renewable diesel drawn from the rest of the country, and as a share of income they cost the lowest income quintile five times as much as the highest, although the gap is much smaller as a share of spending.

Fourth, expectations matter mostly for when these effects arrive. At low prices firms generate fewer credits, so a market that expects scarcity to be far off exhausts its bank sooner. Under the estimated rule California's bank runs out in 2031, two years earlier than under perfect foresight, and the price then jumps to the cap instead of rising toward it. The cost of the stack and the incidence of a federal dollar change little.

The 73 cents assumes a credible, permanent credit; if an extension were only partly capitalized, as the blenders' credit was, less would reach fuel buyers and emissions would rise by less. The results also hold the RFS percentages fixed. If EPA raises them to offset the credit, partly or fully, the extension still lowers welfare relative to the same mandate without it once climate damages are counted.

For federal policy, these results mean that, while the RFS binds, extending 45Z mainly lowers fuel prices, and that whether federal credits reduce emissions at all depends on how EPA sets the RFS percentages. For the states, the results imply that when the market doubts a standard, prices stay low for longer and then jump to the cap instead of rising toward it, which happens earlier than it would under perfect foresight.

1.1 Related literature and contributions

The paper is closest to work on overlapping climate policy. Goulder and Stavins (2011) and Perino et al. (2019) show that overlapping climate policies can cancel each other, as in the waterbed effect of emissions trading (Rosendahl, 2019), and Fischer and Preonas (2010) show the same for combined renewable energy policies. For biofuels, a subsidy on top of a binding blend mandate lowers fuel prices and leaves biofuel use unchanged (de Gorter and Just, 2009; Lapan and Moschini, 2012). Holland et al. (2009) show that an intensity standard subsidizes the fuels it regulates, and Larson (2026) studies two linked cap-and-trade programs with price limits. I derive the incidence of a federal dollar when a national mandate, a national credit and banked state standards with price caps act on the same gallon, and I measure each channel. Relative to this literature, the binding constraint here is determined in equilibrium, and it depends on choices made by EPA, an agency other than the one that pays for the credit.

The timing results relate to work on policy expectations and banked credits. Since Kydland and Prescott (1977), economists have recognized that policy works through what people expect the government to do, and that climate policy faces a credibility problem because its benefits arrive long after its costs (Helm et al., 2003). Uncertainty about policy has been measured from news text (Baker et al., 2016) and inferred from investment (Chen, 2024). A banked compliance credit gives a more direct measure, because its price indicates how soon the market expects the policy to bind, although reading expectations from it requires assumptions about the holding cost, the terminal price and the risk of rule changes. Work on emissions markets has shown that banking

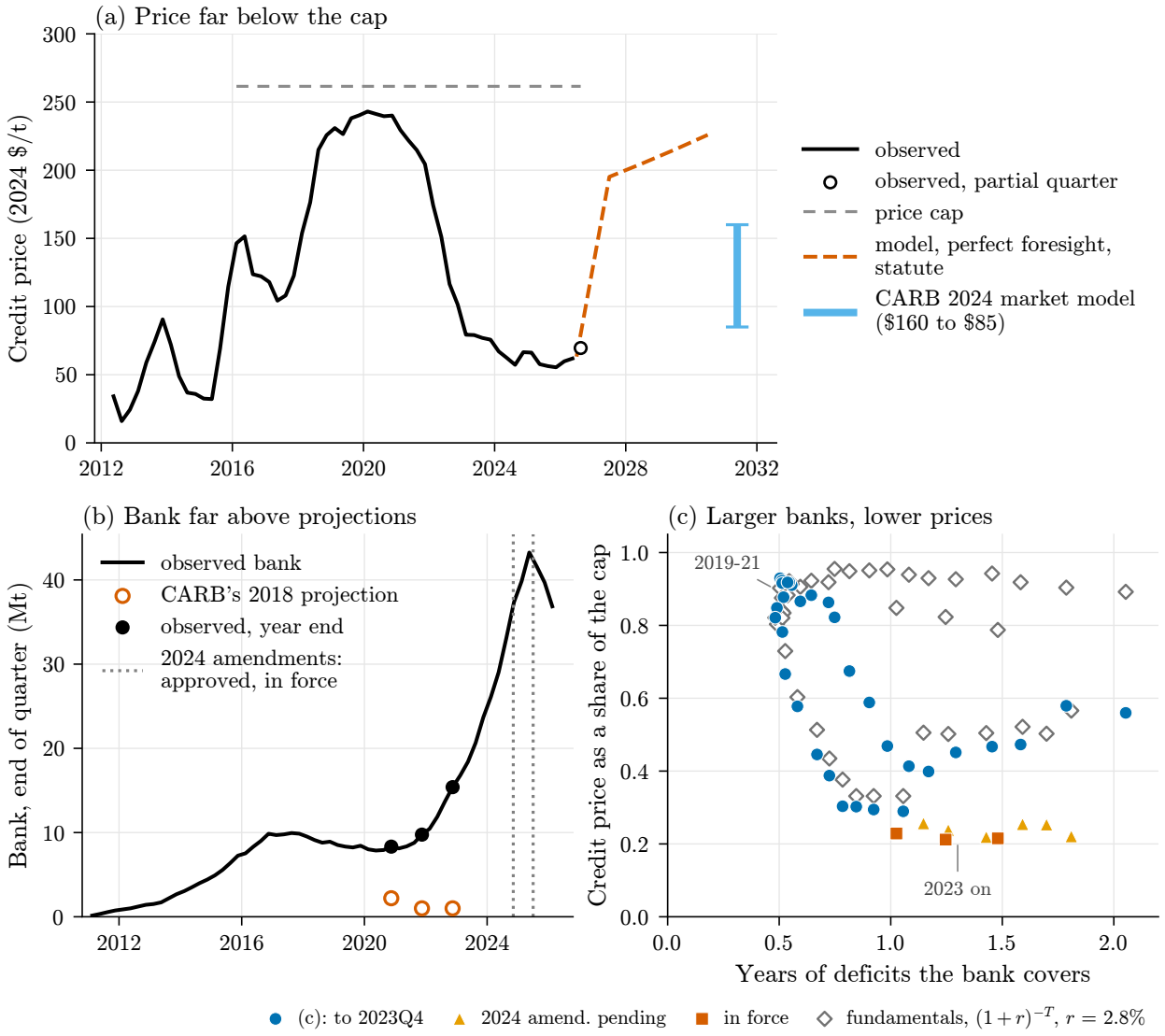


Figure 1: California's credit market: the price sits far below the cap, the bank far above the regulator's projections, and larger banks come with lower prices

Notes: (a) Quarterly average credit price, 2024 dollars, with the price cap and the model's perfect-foresight path under the statute; the bar marks the range of CARB's 2024 market-model projection (California Air Resources Board, 2024). (b) End-of-quarter bank, with CARB's 2018 projections (California Air Resources Board, 2017); dotted lines mark the approval (8 November 2024) and entry into force (1 July 2025) of the 2024 amendments. (c) Credit price as a share of the cap against the years of deficits the bank covers, California, quarterly, 2016Q1–2026Q1, by period of the 2024 amendments. Diamonds: the price that would reach the cap at the fundamentals exhaustion date, $P/\bar{P} = (1+r)^{-T}$, with T from Appendix Table 14 and $r = 2.8\%$; in 2023Q2–Q4 T is censored at 40 years, so those diamonds are upper bounds. Sources: CARB LCFS data and transfer reports; CPI-U.

and expectations shape prices (Salant, 2016; Borenstein et al., 2019; Quemin and Trotignon, 2021; Chen, 2018; Perino, 2024). I show that California's credit prices were consistent with a binding statute when the bank was small and with exhaustion decades away once the bank was large, and that standard descriptions of expectations do not fit this history. I estimate an expectation rule that is consistent with it, and under that rule the gap changes when federal support reaches fuel buyers and when the price reaches the cap, while the cost of the policies changes little.

The quantitative model builds on studies of U.S. fuel policy. Recent U.S. climate policy relies on subsidies more than on prices (Bistline et al., 2023). Studies of the RFS and the state standards have examined RIN prices and pass-through (Lade et al., 2018b; Irwin et al., 2020; Knittel et al., 2017; Mazzone et al., 2022), the design of cost containment (Lade and Lin Lawell, 2021), the distributional effects of transportation policies (Holland et al., 2015), the interaction of the RFS with clean fuel standards (Lade et al., 2018a), the outlook for California's credit balance (Bushnell et al., 2020), and policies for aviation fuel in 2030 (Wu et al., 2026). This paper models the four state programs jointly with the federal instruments over twenty years, estimates investment, electric-vehicle and charger adoption and the mapping from the bank to credit prices, validates investment and the price rule on history, and compares the stack with a carbon price at equal abatement, as Goulder et al. (2016) do for a clean energy standard. The comparison depends on how fuel taxes are treated, a question studied by Parry and Small (2005) and Bento et al. (2009); I report it under each convention.

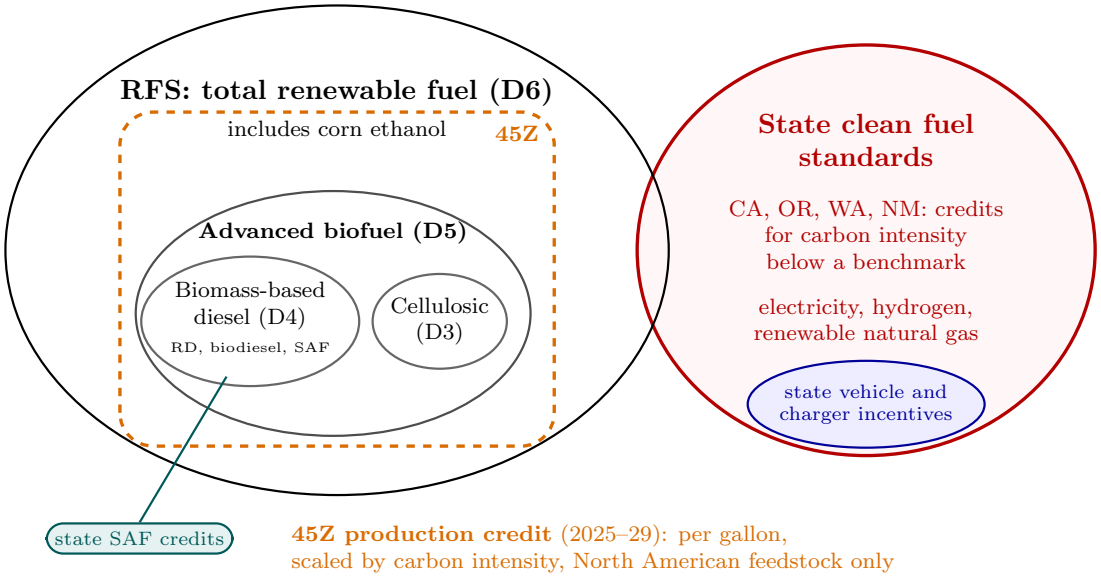
The rest of the paper is organized as follows. Section 2 describes the policies and what credit prices say about the statute. Section 3 presents the framework, Section 4 the model, and Section 5 the data, estimation and validation. Section 6 reports the cost and incidence of the stack and of additional federal support, Section 7 the role of expectations, and Section 8 concludes.

2 Policy Background

Federal and state instruments regulate the same gallons of U.S. transportation fuel, and Figure 2 shows how they fit together. The federal instruments reward gallons of qualifying fuel everywhere, while the state standards reward low carbon intensity and apply only in four states. Because all of them draw on the same feedstocks and the same fuel pools, they interact through prices. Appendix A.1 gives the details.

Two federal instruments apply everywhere. The Renewable Fuel Standard (RFS) requires refiners and importers to retire credits (RINs) in proportion to the gasoline and diesel they sell. The requirements are nested: biomass-based diesel (D4) counts toward the advanced requirement, which counts toward the total. As a result, the D4 price is at least the D6 price, and it exceeds the D6 price only when the biomass-based diesel requirement binds on its own. EPA's 2026–27 rule raised the total requirement from 13.1% of gasoline and diesel in 2025 to 15.8% in 2027 (U.S.

(a) The instruments and what they reward



(b) Rules that respond to the state bank

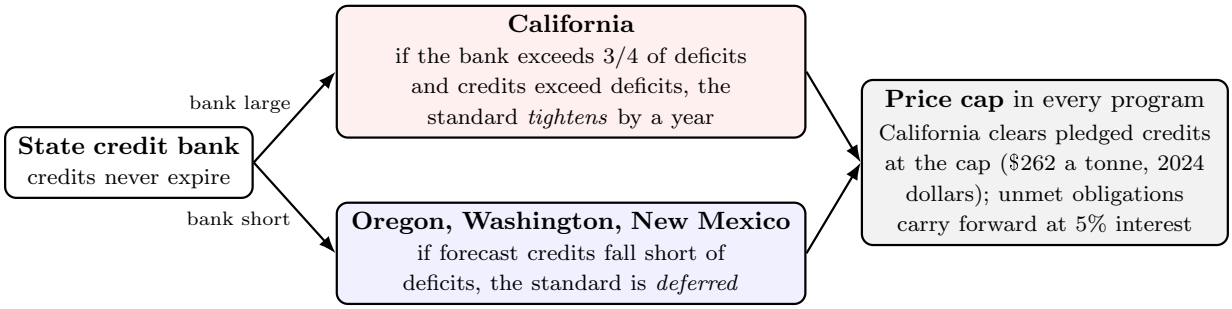


Figure 2: The policy stack for U.S. transportation fuel, October 2026

Notes: (a) The Renewable Fuel Standard's nested requirements, the 45Z production credit, the four state clean fuel standards and the state credits for SAF and for vehicles and chargers. (b) The rules that make the state standards respond to their own credit banks. Sources: 40 CFR 80; 26 U.S.C. 45Z as amended by U.S. Congress (2025); 17 CCR 95480 ff.; OAR 340-253; RCW 70A.535; 20.2.92 NMAC. Appendix A.1 and Appendix Table 7 give the details.

Environmental Protection Agency, 2026c); I hold the 2027 percentages for later years. The 45Z credit pays up to $1.00 a gallon, scaled by lifecycle emissions. The 2025 budget law extended it through 2029 and limited it to feedstock grown in North America (U.S. Congress, 2025).

Four states add clean fuel standards. California (since 2011), Oregon (2016), Washington (2023) and New Mexico (2026) set a declining benchmark carbon intensity. Fuel above the benchmark generates deficits, fuel below it generates credits, and credits trade and never expire. Electricity used in vehicles earns credits. California's 2024 amendments cut the benchmark 30% below 2010 by 2030 and 90% by 2045 (California Air Resources Board, 2025). Each program caps the credit

price; California runs a credit clearance market at the cap ($262 a tonne in 2024 dollars), in which holders sell pledged credits, and carries obligations still unmet forward at 5% interest. Washington's statute holds its standard at 20% from 2030 until its regulator certifies new biofuel capacity; the main specification keeps the 20% limit and treats the path to 45% as a scenario. Renewable diesel was 72% of California's diesel in 2025Q1; further deliveries are limited by the cost of bringing more of it into the state, since no blend limit binds.

The state programs also respond to their own banks, in opposite directions (Figure 2b). Oregon, Washington and New Mexico must defer their standards when forecast credits fall short of deficits. California instead tightens its standard by one year when the bank exceeds three quarters of the year's deficits and credits exceed deficits; the first test is in May 2027. No deferral has ever been ordered, so these rules work, if at all, through expectations.

2.1 What credit prices say about the statute

While a bank is held, a credit's price should rise at the holding cost and reach the cap when the bank runs out (Cronshaw and Brown-Kruse, 1996; Schennach, 2000); in a market with a price ceiling, the price is likely to settle at the ceiling (Borenstein et al., 2019). Each price therefore implies a year in which holders expect the bank to run out, $t + \ln(\bar{P}/P_t)/\ln(1+r)$. California's history has two regimes. In 2019–21 the bank covered less than a year of deficits and the price was within about a fifth of the cap. The implied horizon, 1.7–3.8 years at a 5% holding cost, was close to, and if anything shorter than, a projection of fundamentals (3–7 years). By 2024 the bank had quadrupled, to 37 Mt, and the price had fallen to $64. In 2024–25 the price implied that the bank would last another 29–31 years, while the model's projection of fundamentals implies 7–10, or 15–25 if the recent growth in net credits is extrapolated. The gap widened as the 2024 amendments tightened the standard: from 2024 to 2025 the horizon fundamentals imply fell from about 10 years to about 7 (from 25 to 15 with the growth extrapolated), while the price-implied horizon rose from 29 to 31 years (Appendix Table 14). The price did not respond when the 2024 amendments took effect in July 2025, and it averaged $63 in January–August 2026 although deficits now exceed credits by about 10.7 Mt a year against an end-2025 bank of 39.7 Mt. Panel (c) of Figure 1 shows both regimes. Since 2023 the price has been a fifth to a third of the cap while the bank covered one to two years of deficits; a market that believed the statute would have priced credits close to the cap. In both regimes the price moved further than fundamentals, as it would if holders extrapolated the current balance.[2]

[2]The regulator expected scarcity sooner than the market did. In 2018 CARB projected a bank of about 2.2 Mt in 2020 and 1.0 Mt in 2021–22, against 8.3 Mt observed in 2020 and 9.8 and 15.4 Mt in 2021–22 (California Air Resources Board, 2017). A projection-free bound makes the same point: at 2026 flows the bank lasts fewer than four years, so a price near $60 for decades requires about 10 Mt a year of extra net credits, more than replacing all of California's fossil diesel with renewable diesel would generate (8.2 Mt). At a 2.8% holding cost taken from market interest rates the 2024–25 horizon is 51–54 years. Oregon shows the same pattern.

The price does respond to news about the regulator. California's price fell 16% in the week the Office of Administrative Law rejected the 2024 amendments and rose 14% in the week it approved them, and Washington's rose 26% (0.23 log points) in the week its Senate passed HB 1409, while CARB's quarterly data releases move prices little. The price-expectation rule of Section 4 captures the response to the balance but not the response to regulatory news.

3 A Framework: National Support on a Banked State Standard

This section presents a stylized model of a federal fuel subsidy in a country where one region has a bankable credit market. It adapts the analysis of overlapping climate policies in Perino et al. (2019) to an intensity standard with a price cap and a federal mandate, and adds banking. The framework defines the objects that the quantitative model measures and shows which features of the markets determine their size. The central object is what happens to a marginal dollar of national support, that is, whether it becomes a rent, a transfer to fuel buyers or new abatement, and in which region.

3.1 Set-up

There are two regions, $j \in \{S, N\}$. Region S has a carbon intensity standard; region N has none. Fuel is measured in energy units and sells at price p_j in region j; demand is $Q_j = D_j(p_j)$, with $D_j' < 0$, and $D = D_S + D_N$ is national demand. Fossil fuel is supplied perfectly elastically at cost m and has carbon intensity c_0. Biofuel is a perfect substitute in use, has intensity $c_1 < c_0$, and is produced by competitive firms with national cost $C(q)$, where $q = q_S + q_N$, $C'' > 0$, and capacity $q \leq H$. Delivering biofuel to S costs $\tau(q_S)$ per unit at the margin, with $\tau' > 0$; region N takes biofuel at no extra cost up to a blend or logistics limit $\bar{q}_N$. I call N's room to absorb more biofuel the *outlet*.

Three policies act on this market. The federal mandate requires $q \geq M$. It is met with tradable credits (RINs) at price ρ: each unit of biofuel generates one RIN, and each unit of fossil fuel carries an obligation $\pi = M/Q^F$, where $Q^F = (Q_S - q_S) + (Q_N - q_N)$ is national fossil fuel use. The federal subsidy pays s per unit of biofuel. The state standard $\sigma \in (c_1, c_0)$ gives each unit of fossil fuel sold in S a deficit of $c_0 - \sigma$ and each unit of biofuel a credit of $\sigma - c_1$; electric vehicles generate Γ credits. State credits trade at price λ, which cannot exceed the cap $\bar{P}$; when credits fall short at the cap, the state issues the shortfall X^+ at $\bar{P}$, as in the quantitative model.

Fuel prices therefore are

$$p_N = m + \pi\rho, \qquad p_S = m + \pi\rho + (c_0 - \sigma)\lambda, \tag{1}$$

and a unit of biofuel earns $p_N + \rho + s$ if sold in N and $p_S + (\sigma - c_1)\lambda + \rho + s - \tau(q_S) = p_N + (c_0 -$

$c_1)\lambda + \rho + s - \tau(q_S)$ if sold in S. The state's net credit generation, credits less deficits, is

$$b = (\sigma - c_1)\, q_S - (c_0 - \sigma)(Q_S - q_S) + \Gamma = (c_0 - c_1)\, q_S - (c_0 - \sigma)\, Q_S + \Gamma. \tag{2}$$

3.2 Equilibrium

Definition 1. An equilibrium is a set of quantities (q_S, q_N, Q_S, Q_N) with $Q_j = D_j(p_j)$, prices (ρ, λ) satisfying (1), a capacity rent ψ, a shadow value of the outlet ν and issuance X^+ such that

$$C'(q) + \psi + \nu = p_N + \rho + s, \tag{3}$$

$$C'(q) + \psi + \tau(q_S) = p_N + (c_0 - c_1)\lambda + \rho + s, \tag{4}$$

$$0 \le \psi \perp H - q \ge 0, \qquad 0 \le \nu \perp \bar{q}_N - q_N \ge 0, \qquad 0 \le \rho \perp q - M \ge 0, \tag{5}$$

$$b + X^+ \ge 0, \quad \lambda \ge 0, \quad \lambda\,(b + X^+) = 0, \qquad 0 \le X^+ \perp \bar{P} - \lambda \ge 0. \tag{6}$$

Equations (3) and (4) are the zero-profit conditions for biofuel sold in N and in S (I assume $q_N > 0$). Conditions (5) are the three federal-level constraints: capacity, the outlet and the mandate. Each has a price that is positive only when the constraint binds: ψ is a rent to producers, ν is the discount at which the marginal unit is sold in N, and ρ is the RIN price. Condition (6) clears the state credit market, with the cap as a safety valve. Subtracting (3) from (4),

$$\tau(q_S) = (c_0 - c_1)\,\lambda + \nu. \tag{7}$$

When the outlet is slack ($\nu = 0$), biofuel is sold in both regions up to the point where the state credit premium alone covers the cost of delivery to S. The federal subsidy and the RIN are paid in both regions and so drop out of the condition.

Lemma 1 (The state credit market)**.** *Suppose the outlet is slack. Then $q_S = \tau^{-1}\big((c_0 - c_1)\lambda\big)$, and net credit generation is a function $b(\lambda; \rho)$ that does not depend on s, with*

$$\frac{\partial b}{\partial \lambda} = \frac{(c_0 - c_1)^2}{\tau'} - (c_0 - \sigma)^2 D_S' > 0. \tag{8}$$

The credit price is $\lambda^* = b^{-1}(0)$ if $b(\bar{P}) \ge 0$; otherwise $\lambda^* = \bar{P}$ and the state issues $X^+ = -b(\bar{P})$. When the standard binds below the cap, emissions in S are

$$Z_S = c_0(Q_S - q_S) + c_1 q_S = \sigma Q_S + \Gamma. \tag{9}$$

The two terms in (8) are the two ways a higher credit price creates credits: it pulls biofuel into S (first term) and raises the price of fuel in S, which lowers fuel use and deficits (second term). Equation (9) follows from setting $b = 0$ in (2): a binding intensity standard fixes emissions per

unit of fuel, so emissions in S change only through fuel use and credits from outside the fuel pool (Holland et al., 2009). Appendix Figure 11 draws the allocation of national biofuel between the regions and the state credit market.

3.3 Where additional national support goes

Consider a marginal increase ds in the subsidy. Where it goes depends on which of the constraints in (5)–(6) binds. Let $\varpi \equiv -(c_0 - \sigma)^2 D_S'/(c_0 - c_1) > 0$, the additional biofuel the state needs to stay in balance when its credit price falls by one dollar.

Proposition 1 (Where additional support goes)**.** *To first order:*

(i) Capacity binds *($q = H$, mandate slack). $d\psi/ds = 1$: the dollar is a rent to producers, and no quantity or other price changes.*

(ii) The mandate binds *($q = M$, capacity slack), and the outlet is slack or the state is at its cap. $d\rho/ds = -1/(1+\pi)$, and national biofuel does not change. At a given state credit price, the price of every unit of fuel falls by $\pi/(1+\pi)$ and fuel buyers gain $M\,ds$, the whole cost of the subsidy. Below the state's cap, part of the gain to fuel buyers in S is offset by a higher state credit price; this effect is of the order of the demand response and small when demand is inelastic.*

(iii) Neither binds and the outlet is slack. *$dq_N/ds = 1/C''$ and $dq_S = d\lambda = 0$: the extra biofuel goes to N, and national emissions fall by $(c_0 - c_1)/C''$. The state's credit price, fuel use and emissions do not change, whether or not the state is at its cap.*

(iv) The outlet binds and $\lambda < \bar{P}$.

$$\frac{d\lambda}{ds} = -\frac{1}{(c_0 - c_1) + \varpi\,(C'' + \tau')} < 0, \qquad \frac{dq_S}{ds} = -\varpi\frac{d\lambda}{ds} > 0, \qquad \frac{dZ_S}{ds} = \sigma(c_0 - \sigma)D_S'\frac{d\lambda}{ds} > 0. \tag{10}$$

The dollar lowers the state's credit price, a transfer to its fuel buyers, and the state's emissions rise.

(v) The outlet binds and $\lambda = \bar{P}$.

$$\frac{dq_S}{ds} = \frac{1}{C'' + \tau'}, \qquad \frac{dZ_S}{ds} = \frac{dX^+}{ds} = -\frac{c_0 - c_1}{C'' + \tau'}. \tag{11}$$

Biofuel displaces fossil fuel in S one for one, and the state issues fewer credits.

Figure 3 summarizes the five cases, and Appendix F gives the proofs. Cases (i) and (ii) of Proposition 1 concern the federal constraints. When capacity or the mandate binds, the subsidy does not increase biofuel output, and in case (ii) the whole dollar reaches fuel buyers through a

lower RIN price.[3] The level of the biodiesel RIN price has moved with the value of the blenders' credit in this way, including in the years the credit lapsed and was restored (Irwin et al., 2020). Because the dollar lowers the price of every unit of fuel, fuel use rises. With biofuel fixed at M, fossil fuel use increases by $-D'(p)\,\pi/(1+\pi)$ per dollar and national emissions rise, so a subsidy on top of a binding mandate subsidizes fossil fuel consumption (de Gorter and Just, 2009; Lapan and Moschini, 2012). Cases (iii)–(v) concern the state, and which of them applies depends on the outlet. When biofuel can still go to N, the arbitrage condition (7) already sends the state the biofuel that its standard pays for, and a subsidy paid in both regions leaves that allocation unchanged. When the outlet is full, the extra biofuel must go to S, and the sign of the interaction depends on the state's regime. Below the cap the standard fixes emissions per unit of fuel, so the extra biofuel lowers the credit price and allows more fossil fuel to be sold in the state; the subsidy becomes a transfer to the state's fuel buyers, and emissions in S rise. This is the waterbed effect of overlapping instruments under a binding target (Goulder and Stavins, 2011; Perino et al., 2019). At the cap the target is not met, so in the terminology of Perino et al. (2019) the waterbed is punctured and the extra biofuel is new abatement. Case (iii) is the fuel-market counterpart of their internal leakage, since the extra biofuel goes to the region where no standard binds. In this framework, relocation (the reallocation of a fixed national volume of biofuel toward S) is caused by the state standard and not by the subsidy. With the mandate binding, tightening σ raises q_S and lowers q_N one for one.

3.4 When the transfer arrives: banks and expectations

Now let time run, $t = 1, 2, \ldots$, and let the state bank credits. The standard σ_t tightens, so net generation $b_t(\lambda)$ falls over time and eventually becomes negative. The bank evolves as

$$B_t = B_{t-1} + b_t(\lambda_t) + X_t^+, \qquad B_t \geq 0. \tag{12}$$

Credit holders pay a holding cost r, so as long as the bank is positive they hold credits only if the price is expected to rise at rate r. The price is then determined by the year T in which the bank is expected to run out and the price to reach the cap:

$$\lambda_t = \bar{P}\,(1+r)^{-(T-t)}, \qquad t \leq T. \tag{13}$$

This is Hotelling's rule with the cap as a backstop (Hotelling, 1931; Heal, 1976), and it leaves open how holders form their expectation of T. Under perfect foresight, T is the year in which the bank

[3] Case (ii) holds the obligation per unit of fossil fuel, π, fixed, which is exact when fuel demand is inelastic. EPA sets the RFS as percentages of fuel sold, and so does the quantitative model. With a fixed percentage, the lower fuel price raises fossil fuel use, and the required biofuel rises with it by π times that increase; with a fixed volume M, the obligation per unit falls instead. Both corrections are proportional to the demand response to the RIN pass-through and are small when demand is inelastic.

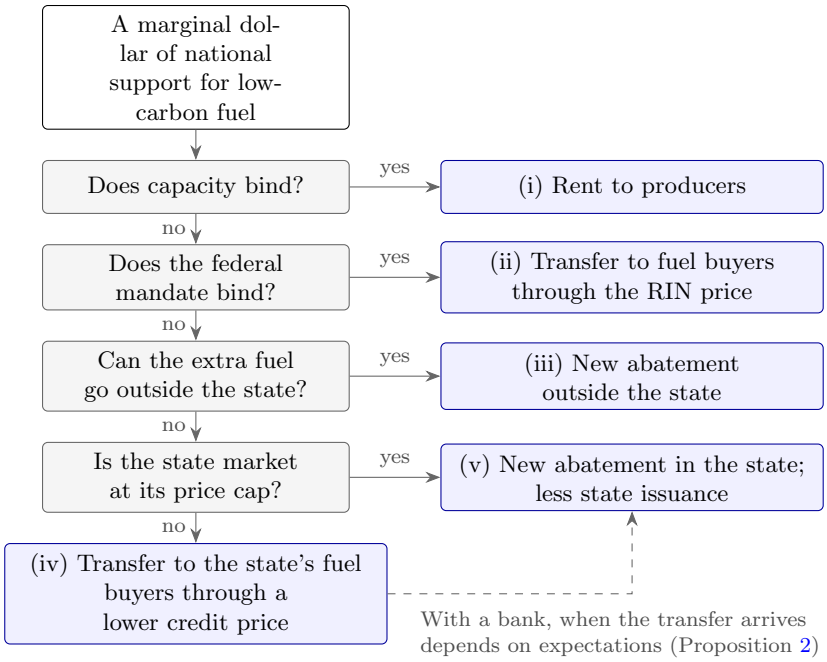


Figure 3: The five cases of Proposition 1. Each question refers to the constraint that binds at the margin. In the quantitative model several apply at once, so Section 6 reports shares.

actually runs out given the path (13):

$$B_{t_0-1} + \sum_{t=t_0}^{T} b_t\big(\bar{P}(1+r)^{-(T-t)}\big) = 0. \tag{14}$$

Under the adaptive rule that the paper estimates, holders revise T in proportion to the surprise in the year's net generation,

$$T_t = T_{t-1} + \gamma\big(b_t - E_{t-1}[b_t]\big), \qquad E_{t-1}[b_t] = \bar{b}_{t-1} + \Delta_t^{sched} + \Delta_t^{known}, \tag{15}$$

where $\bar{b}_{t-1}$ is last year's net generation (Rule A) or the average of the last three years (Rule B), Δ_t^{sched} the effect of the scheduled tightening of the standard, and Δ_t^{known} other changes known in advance. The price is (13) with $T = T_t$, solved within the year together with b_t. The quantitative model adds a term that raises the price as the bank runs down, with weight $e^{-Y_t/K}$, where Y_t is the number of years the bank would last at the expected shortfall.

Appendix F.2 works out how a permanent subsidy that raises the state's net generation changes the price path under each description (Proposition 2 and Figure 12). Under perfect foresight the price falls from the first year. In the numerical example, under the adaptive rule it falls by less if the subsidy is a surprise, and not for many years if the subsidy was anticipated, because the part that holders do not treat as news first accumulates in the bank and lowers the price only as the bank runs down. The timing of the transfer therefore depends on what holders count as news.

Holding the policy fixed, the same accounting compares any two descriptions of expectations.

Corollary 1 (Doubt brings scarcity forward)**.** *Take two descriptions of expectations, e and e', that start from the same bank and face the same net generation functions $b_t(\lambda)$, and let T^e and $T^{e'}$ be the first years in which the bank cannot cover the year's shortfall at the cap under each. If the credit price under e is no higher than under e' in every year before $\min(T^e, T^{e'})$, then the bank under e is no larger in every such year, and $T^e \leq T^{e'}$.*

Because net generation rises with the price (Lemma 1), a market that expects scarcity to be distant generates fewer credits each year and uses up its bank sooner. Under perfect foresight the price reflects the coming scarcity from the start. A market that extrapolates from a large bank keeps the price low until little of the bank is left, and the price jumps to the cap when the bank runs out. When holders doubt the statute, the bank therefore runs out earlier and the price rises more abruptly.

The framework has a single state and a single subsidy paid at the same rate on every gallon, and only one constraint binds at a time. The quantitative model departs from it in three ways that matter for the results. First, 45Z varies with carbon intensity and, from 2026, with feedstock origin, so a change in the subsidy also changes the feedstock mix and the state's credit generation per gallon; Lemma 1 then holds only approximately, and case (iii) can move the state's credit price a little. Second, several cases apply at once, in different markets and years: the RFS can bind nationally while one state is at its cap and another is not. Third, the states' codified responses (acceleration in California, deferral in Oregon, Washington and New Mexico) make the standard itself respond to the bank, which in the framework would make σ a function of B. The quantitative model therefore has to determine four things: which constraints bind in each year and market; the slopes C'', τ' and D'_S that set the size of each case (τ', the cost of delivering more renewable diesel to the clean fuel states, is the least anchored); how credit holders' expectations map the bank into prices; and how investors value capacity, which determines whether capacity binds.

4 Quantitative Model

The quantitative model applies the framework to the U.S. fuel and vehicle markets from 2026 to 2045. It has five regions: California, Oregon, Washington and New Mexico, each with its own clean fuel standard, bank, cap and credit price, and the rest of the country, which has none. Credits do not trade across programs, but fuel moves between states. Each year a fuel market clears fuels, feedstocks, RINs and LCFS credits for given capacity, fleets and credit prices; banks, holders' expectations, the state rules, capacity and the vehicle stock link each year to the next. The fuel-market block shares its feedstock and crediting structure with Wu et al. (2026); the dynamics, the expectations, investment, vehicles and the four state programs are new. Appendix B states the

model in full and summarizes how each block is disciplined by data.

4.1 Credit markets, rules and expectations

In each program j, credits from biofuels, from electricity ($\Gamma^{EV}_{j,t}$, from the vehicle block) and from fixed sources, less deficits from fossil gasoline and diesel, give the year's net credit generation $N_{j,t}$. The market clears as a complementarity problem with a price floor at zero and a ceiling at the cap $\bar{P}_{j,t}$,

$$N_{j,t} + B^{draw}_{j,t} = s^{+}_{j} - s^{-}_{j}, \qquad 0 \le P^{L}_{j} \perp s^{+}_{j} \ge 0, \qquad 0 \le \bar{P}_{j,t} - P^{L}_{j} \perp s^{-}_{j} \ge 0, \tag{16}$$

where B^{draw} is the draw on the bank that holders' expectations allow. Standards follow each program's adopted schedule; Washington's is capped at 20% from 2030 until Ecology makes the demonstration the statute requires. When a program is short at its cap, I close it by issuing the shortfall at the cap price and count the proceeds as state revenue, whereas the statutes carry the unmet obligation forward as a deficit with interest (17 CCR §95485(c)); Section 6.1 states the conditions under which the two treatments give the same prices and quantities.

Credit holders do not know when the bank will run out. They price credits by Hotelling's rule, rising to the cap in their expected exhaustion year at a holding cost $r = 5\%$ (set rather than estimated),

$$P^{L}_{j,t} = \min\{\bar{P}_{j,t},\ \bar{P}_{j,t}/(1+r)^{\mathrm{gap}_{j,t}}\}, \tag{17}$$

where $\mathrm{gap}_{j,t}$ is the expected number of years to exhaustion. Holders revise it in response to the surprise in the year's net credit generation,

$$G_{j,t}(s) = T_{j,t-1} - t + \gamma_j\big(s - E_{t-1}[s_{j,t}]\big), \tag{18}$$

so a year with more credits than expected moves exhaustion out by γ_j years per Mt and lowers the price. Expected net generation is the average of the last three years plus the changes holders know about in advance: the scheduled tightening of the standard, the crop-oil cap and the end of the 2026 transitional terms. As the bank runs down, holders also weigh how many years it would last, $Y_{j,t}$:

$$\mathrm{gap}_{j,t} = (1 - w_{j,t})\, G_{j,t} + w_{j,t}\, \min\{G_{j,t}, Y_{j,t}\}, \qquad w_{j,t} = e^{-Y_{j,t}/K}. \tag{19}$$

K is the horizon, in years of bank left, at which holders start to price exhaustion; with a large bank the price follows the surprises, and as the bank shrinks it rises toward the cap even without them. I treat the rule as a forecasting rule for the credit price, in the sense of Krusell and Smith (1998) and Gillingham et al. (2022), and not as a model of how holders form beliefs. Its role is to supply the mapping from credit balances to prices that the counterfactuals need.[4] γ, K and the

[4] It has the form of the steady state of a Kalman filter for holders who read each year's net credits as a noisy signal of a persistent balance (Appendix D.2).

initial expected exhaustion years are estimated on California's and Oregon's prices and banks for 2016–25 (Section 5.2).

The other agents form expectations differently, because their markets differ. The RIN market is national, its rules are public and set by one agency, and its bank is limited by regulation, so RIN holders store with foresight, the standard description of such a market (Lade et al., 2018b); a deficit can be carried one year but not two in a row. Fleets buying trucks see LCFS prices on the holders' path; households see current prices (Anderson et al., 2013). The state credit markets, by contrast, are small, their rules change often, their banks have never run out, and their prices do not follow the statutory path. The model is therefore solved as a temporary equilibrium, in which each year's markets clear given the expectations each group holds. All groups share one assumption: every agent expects the current policy stack, including the responses written into the statutes, to continue, and no agent prices in a scheduled expiry or a proposal that has not been enacted.

The RFS's nested requirements (biomass-based diesel, advanced, total) are EPA's percentages applied to the fossil base; the 2026–27 percentages are EPA's final standards and are held at their 2027 level afterwards, so obligations fall as fossil demand falls. Every obligated gallon carries the cost of the full obligation, whether its RINs come from this year's production or from the bank. California's acceleration mechanism advances later benchmarks by a year when the bank exceeds three quarters of the year's deficits and credits exceed deficits (17 CCR §95484(b)); Oregon, Washington and New Mexico defer scheduled tightening when the forecast bank falls short, which I model as the smallest temporary standard that keeps the bank non-negative at the cap. Neither rule has been triggered so far, so whether either is triggered in the forward runs is a model outcome that the data have not tested.

4.2 The fuel market in year t

Gasoline, diesel and jet fuel in each region combine a fossil component with biofuels: ethanol in gasoline; biodiesel and renewable diesel in diesel; HEFA and alcohol-to-jet SAF in jet. Fossil fuel is supplied elastically at a national price plus state differentials and compliance costs. Biofuel producers are price takers, and a pathway produces only if its cost (operating cost, feedstock, utilization cost, delivery) covers its value: the fuel it replaces plus its RINs, its LCFS credits at the program's carbon intensity, 45Z and state credits. Lipids form two groups (vegetable oils; fats and used cooking oil) with upward-sloping supply, and the feedstock mix responds to their relative margins with one estimated response.

The federal–state mechanism of Proposition 1 enters the model through one feature: renewable diesel delivered to a clean fuel state bears a cost that rises with the volume delivered there, which stands in for terminal capacity, logistics and contracts. Without this cost, renewable diesel would move all-or-nothing to whichever program pays most. The slopes are estimated on quarterly de-

liveries since 2023 and determine whether additional national support raises deliveries to the clean fuel states or lowers their credit prices (cases (iii) and (iv)); they are the least anchored parameters in the fuel market, so I report results at half and at twice the estimated slopes. In California the slope steepens once renewable diesel exceeds 72% of the diesel pool, the highest share in 29 quarters of data.

4.3 Capacity investment

Hydroprocessing capacity is endogenous from 2029. In each year, potential investors (91 refineries that could convert, 19 operating plants that could expand, 20 greenfield entrants, and existing plants that could convert to SAF) decide whether to commit now or wait. Each faces a competitive, single-agent optimal stopping problem with logit shocks, as in Rust (1987), with entry costs estimated from 46 capacity decisions. Investors take prices as given, which is reasonable for a national commodity market whose prices are set by the RFS and the credit markets. Strategic interaction among entrants and differences among firms of the same investor type are not modelled. The value of committing per gallon of capacity is

$$V^{c}_{\ell,t} = \sum_{s=L}^{L+19} \delta^{s} \Big[u^{e}_{t+s} \big(\mathbb{E}_t \, m_{t+s} + F_{\ell}/u_0 \big) - F_{\ell} \Big] - \kappa_{\ell,t}, \tag{20}$$

with a three-year construction lag, a 20-year life, the operating margin m (fuel value plus RIN, LCFS and production credits, less feedstock, operating and delivery costs), expected utilization u^{e} from the model's own solution, fixed operating cost F_{ℓ} and entry cost κ_{ℓ}. Because expected utilization falls as capacity is added, each addition lowers the value of the next. Investors hold RIN and feedstock prices at current levels, follow the holders' LCFS path, and expect statutory credits due to expire to continue with probability p. Waiting has value because the margin may rise, and a potential investor commits with probability

$$P_{\ell,t}(z) = \Big[1 + \exp\big(-(V^{c}_{\ell,t}(z) - V^{w}_{\ell,t}(z))/\zeta \big) \Big]^{-1}. \tag{21}$$

The parameters of the rule are estimated on 46 investment decisions (Section 5.2).

4.4 Vehicles and charging

The vehicle block extends Cole et al. (2023) to light-, medium- and heavy-duty vehicles in the five regions. The electric share of new sales is logit in the ratio of the costs of ownership and, for light-duty vehicles, in the public charging network; public stations enter when their margin, including the LCFS credits they earn, covers the cost of building. Station entry is free entry at the regional level, as in the models of Zhou and Li (2018) and Springel (2021), with the entry

elasticities estimated by simulated method of moments (Appendix D.6). The network effects run both ways, from stations to sales and from the electric stock to stations, as in the estimates of Li et al. (2017). Competition among nearby stations is not modelled. Entry on the vehicle supply side, such as a new manufacturer in the electric truck market, is not endogenous either: vehicle prices follow an exogenous path, and the Tesla Semi price scenario bounds it. The block returns the electric shares of the fleet, which shift fuel demand, and electricity credits, which enter the LCFS balance. Electricity credits are about a fifth of California's credit generation, so the bank after 2030 depends on the pace of electrification.

4.5 Equilibrium and welfare

Each year the fuel, feedstock, RIN and LCFS markets clear jointly as a mixed complementarity problem (Appendix B.3). Because LCFS holders do not have rational expectations, the object computed is a temporary equilibrium path (Definition 2): each year's prices clear that year's markets given expectations formed from the past, the RIN bank solves its storage problem on the path, capacity and fleets follow their rules at the path's prices, and the state rules act on the path's banks. The blocks are coupled by damped iteration over 2026–45 (Appendix B.6).

Welfare is the sum of the surpluses of fuel users (by state), vehicle buyers, biofuel producers, feedstock suppliers, the federal and state governments and owners of capital, less climate damages at EPA's 2023 social cost of carbon (2% discount rate). Transfers cancel in the total; costs are resource costs, with capital at its capital cost rather than the investment hurdle. California's and Washington's cap-and-trade programs enter as a fixed cost per tonne on road gasoline and diesel sold in those states, at the 2025 auction prices ($27 and $59 a tonne in 2024 dollars, about $0.22 and $0.47 a gallon of gasoline). The cost does not rise after 2025, jet fuel is not covered, and allowance revenue is not counted in welfare. The programs are therefore treated as taxes, and the abatement they cover is counted in full. Fuel taxes are a pre-existing wedge, since a gallon not burned loses its tax and avoids the congestion, accident and local air-pollution costs of fuel use (Parry and Small, 2005; Parry et al., 2006; Holland et al., 2016). I report the resource cost, the forgone fuel-tax revenue and the change in these other external costs as separate lines; the cost per tonne in the text includes the first two, and Appendix H gives the values. The benchmarks are a carbon price at the social cost of carbon on all fuels and vehicles in the model, and a carbon price that delivers the same cumulative abatement; both are first best only within the model's partial-equilibrium scope. For federal instruments I report outcomes per dollar of federal cost.

5 Data, Estimation and Validation

This section covers everything except the feedstock and vehicle blocks, which are estimated in Appendix D.6. Data from 2026 are used in three ways, and no observation is used in more than

one: the 2026 credit prices and banks are the starting point, four 2026 quantities are calibration targets, and every other 2026 observation is held out for the fit to history. Structural parameters are estimated on data through 2025.

5.1 Data

For the state programs I use the programs' quarterly credit and deficit data by fuel and feedstock and their transfer prices: CARB's LCFS data (2011–2026Q1) and monthly credit transfer reports (California Air Resources Board, 2026c,d), Oregon DEQ's quarterly summaries (2016–2026Q1) (Oregon Department of Environmental Quality, 2026) and Washington Ecology's (2023–2026Q1) (Washington Department of Ecology, 2026). Annual prices are volume-weighted transfer prices; for 2026 I use January–August, with data cut off on 11 October 2026. New Mexico's program began in April 2026 and has no price history. For the RFS I use EPA's EMTS data on RIN generation, retirements and holdings, and EPA's RIN transaction prices.

Fuel volumes, prices and biofuel production come from EIA (monthly production and capacity of renewable diesel and biodiesel; product supplied; wholesale prices). Feedstock use and prices come from EIA, USDA ERS and farmdoc's monthly series; feedstock shares by state come from the programs' quarterly data. The investment data are 46 dated decisions to build or convert hydroprocessing capacity in 2014–2025 (16 refinery conversions, 5 expansions, 19 greenfield plants and 6 SAF conversions), each dated at its public announcement.

Data on light-duty electric sales and stocks by state and on public stations from AFDC cover 2020–2025. Truck sales by class use 2025 zero-emission counts from ICCT, the California Energy Commission and Washington's Department of Licensing, with class 3 counts completed for Oregon, New Mexico and the rest of the country (International Council on Clean Transportation, 2026). Incentive funding comes from program records: HVIP voucher data (as of 31 July 2026), the EnergIIZE award list, utility make-ready program filings and state budgets (CALSTART / California HVIP, 2026; CALSTART and California Energy Commission, 2026).

Oil prices follow EIA's Short-Term Energy Outlook to 2027 and the Annual Energy Outlook 2025 reference case from 2030, interpolated in between and shifted to the model's price definitions (U.S. Energy Information Administration, 2026b,a); the AEO high- and low-price cases are sensitivities. Fuel demand in 2026–27, electricity prices and inflation follow the same sources.

5.2 Estimation

I estimate two sets of expectations: what credit holders expect about the end of their bank, and what investors expect about the policy stack. The credit supply and the margins come from the historical version of the fuel market model, and Appendix D gives the details. Credit holders expect the bank to run out in some year, and the credit price rises toward the cap as that year approaches.

Under the price-expectation rule they project net credits forward from the recent past and revise the expected year when the balance surprises them. I estimate the rule by maximum likelihood on the quarterly prices and banks of California and Oregon, 2016–25.[5] California's response to a balance surprise is $\hat{\gamma}_{CA} = 0.753$ years per Mt, and the smaller programs share a response of 0.109 per percentage point of the obligation (0.134 for California on that scale). Holders start to price exhaustion when about $\hat{K} = 2.33$ years of bank are left (95% interval [1.58, 3.14]); the main specification uses $K = 2$. No program in the sample came close to exhausting its bank, so behavior near exhaustion is extrapolated, and the estimate of K depends on three annual shift terms in the history (Appendix D). The forward runs use $K = 2$, within the interval of the estimate with the shift terms; without them the estimate is 3.68 ([2.93, 4.76]). A year with a megatonne more credits than expected thus moves California's expected exhaustion year out by about three quarters of a year and lowers the price.

Given the net credits that actually occurred, the rule forecasts California's price two and three years ahead with errors of $40 and $39 a tonne, against $87 and $117 for a no-change forecast, and it gets the direction of the price right in 83–100% of quarters one to three years ahead. When it must also forecast the balance, it beats a no-change forecast up to four quarters ahead but not at eight and twelve, where rules with regulatory risk or a rolling horizon also do better (Figure 4 and Appendix Table 12), as is common for asset prices (Meese and Rogoff, 1983; Alquist et al., 2013). The rule describes how holders revise the expected exhaustion year; it does not explain why the gap of 2023–25 opened, and it ignores news about the regulator. I therefore report every result also under perfect foresight and with the response to the balance set at its model-free value, $\gamma = 0.32$.

The second expectation is investors'. I estimate a competitive, single-agent optimal-stopping logit, as in Rust (1987), on the 46 capacity decisions, with the expected margin of each investor-year computed by the historical model from information public at the time. Entry costs are 20.25 dollars per gallon of annual capacity for conversions, 10.5 for expansions and new plants and 3.0 for SAF retrofits (Appendix Table 15), about 11 times engineering cost for conversions and 3 times for new plants. The estimate should be read as a hurdle rather than a construction cost, since it also absorbs permitting, feedstock contracts, the cost of financing first-of-a-kind plants and the risk that the policies change. I therefore do not count the excess over capital cost as a social cost, and I report capacity fixed at its main path as a bound. Estimated on 2014–19, the rule predicts the investment of 2020–25 better than a constant hazard or a static logit.

Investors value a credit that is in law as if it continues and a credit that has lapsed almost as if it were gone. The lapses of the blenders' credit, which cover 15 of the 46 decisions, separate the

[5] The holding cost is set at 5%; the data identify the product $\gamma \ln(1 + r)$, which is what the counterfactuals use. OPIS quotes for adjacent December deliveries imply a real carry of 3.0–3.4% a year. With $r = 3.4\%$ in the rule (product unchanged), in perfect foresight and in RIN storage, the share of the 45Z extension reaching fuel buyers changes by 0.002 and the cost per tonne of current law by less than 0.1% (Appendix G.7). Appendix D gives the forward-quote evidence and the alternative rules.

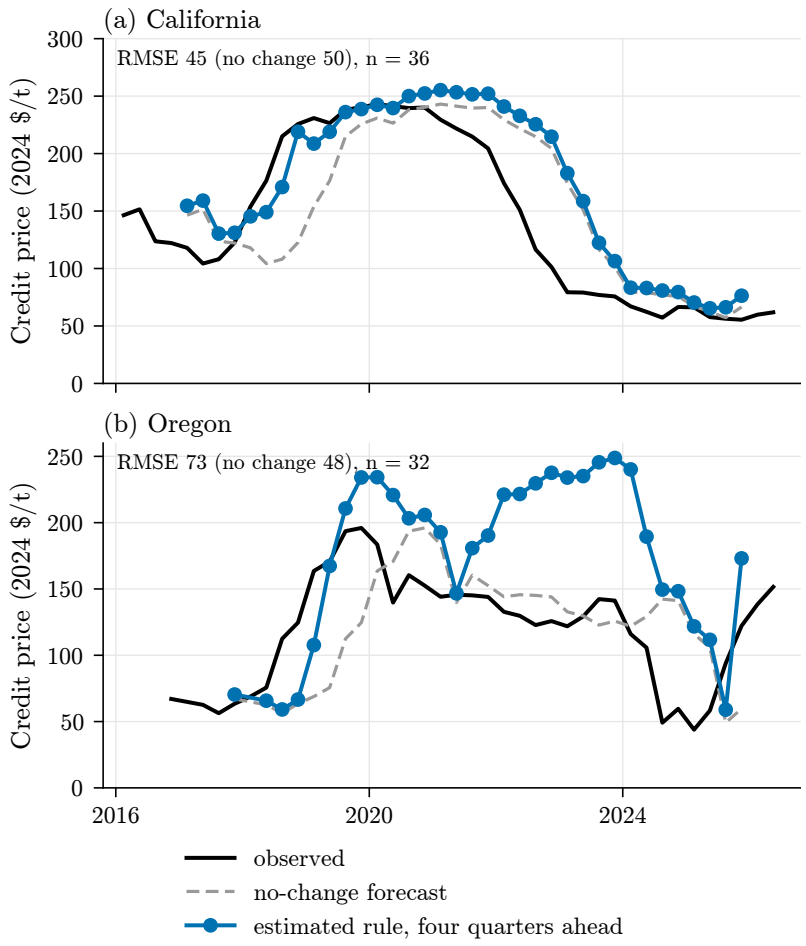


Figure 4: Four-quarter-ahead forecasts of the credit price from the observed state

Notes: California and Oregon. Forecasts by the estimated price-expectation rule from each quarter's observed price and bank, against the observed price and a no-change forecast (grey). Root mean squared errors of the rule and of the no-change forecast in each panel. Appendix Table 12 reports other horizons and descriptions of expectations. 2024 dollars.

two: the estimated probability of continuation is 1 while the credit is in law (95% interval [0.67, 1.00]) and close to zero during lapses. The credit's history fits this reading, since it lapsed four times and was restored retroactively each time. As in Liu et al. (2018), the credit is a switching policy, but where that paper sets the switching probabilities, I estimate p from the 46 decisions. Almost all of the identification comes from 13 decisions in the 2017–19 lapse, so I also report the counterfactuals with a one-half probability of continuation ($p = 0.5$). A 45Z extension changes a credit that is in law, and investors who already value it as continuing see little news in it.

5.3 Other parameters

The remaining parameters are calibrated, taken from the literature or set by statute; Appendix Table 11 lists the ones that matter most for the results and where each value comes from, and Ap-

pendix C gives the details. Four levels are set to 2025, the last complete year: the hydroprocessing utilization curve, the feedstock-mix intercepts, the corn oil supply and the RIN bank (3.21 billion RINs, from EPA's published bank and 2025 flows). Because a single year identifies a level but not a slope (Heckelei and Wolff, 2003, pp. 29–30), slopes come from longer histories; the delivery slopes into the clean fuel states, for example, are estimated on quarterly data since 2023. In 2026 the credit prices are pinned at their observed levels and four 2026 targets identify four terms that fade out by 2028: the renewable diesel delivery terms of California, Oregon and Washington, and the supply of imported fat. RIN prices are not inputs; the model determines them.

5.4 Fit to history

Run on the history of 2017–25 from start years 2017, 2019, 2021 and 2023, with the oil prices, capacity and standards that occurred, the model tracks California's credit price with a smaller error than a random walk (29.6 against 43.9), although with nine annual errors the difference is not significant; for Oregon its error is larger than the random walk's. RIN prices and fuel quantities also track history better than a random walk and the Annual Energy Outlook. Six of the eight 2026 observations that were not used as targets fall within their tolerances. The two misses are the D6 RIN price before EPA's exemption decisions of 31 August and the fat share of California's renewable diesel, which fell with the new origin rule and tariffs, a channel the model does not include. Appendix E reports every comparison.

6 Cost and Incidence of the Policy Stack

In the quantitative model several cases of Proposition 1 can apply at once, so I report where each dollar goes as shares and note for each year which constraints bind. All results in this section use the estimated price-expectation rule.

6.1 Current law: the path, the cost and who pays

Figure 5 shows the main run from 2026 to 2045: LCFS prices and banks by program, biofuel volumes by fuel, RIN prices, the shortfall at the state caps, the electric share of new vehicle sales and U.S. emissions with and without fuel policy. The electric share of new light-duty sales rises from 20% to 47% in the clean fuel states and from 7% to 12.5% elsewhere. California's bank runs out in 2031 and its price reaches the cap in 2032. California's acceleration mechanism never triggers, and the deferral rules take effect in New Mexico from 2033 and in Oregon from 2034. Washington keeps its statutory 20% limit. The RFS binds in every year. The D6 price is \$2.29–2.52 in 2026–28 and falls gradually to about \$1.14 in 2045. Appendix G.1 describes the path in more detail: trucks, the 2029 step in the RIN price, hydroprocessing capacity, SAF and Washington's path to 45%.

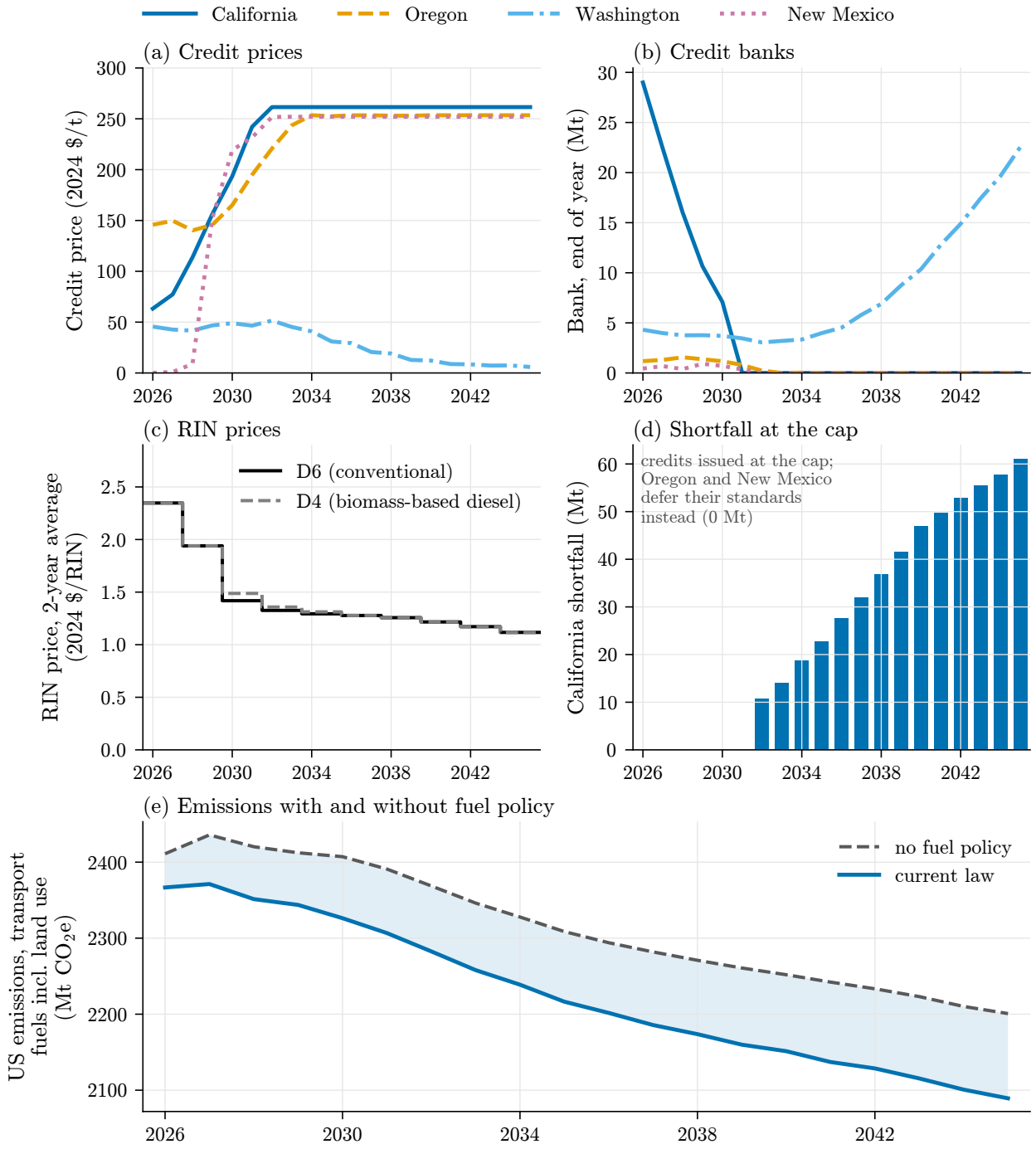


Figure 5: Current law, 2026–2045 (main run)

Notes: (a) Credit prices and (b) end-of-year banks by state; (c) D6 and D4 RIN prices, two-year averages; (d) California's shortfall at its cap, met by credits issued at the cap (Oregon and New Mexico defer their standards instead); (e) U.S. lifecycle emissions of transportation fuels including indirect land-use change, under current law and without fuel policy. 2024 dollars. Biofuel volumes and electric vehicle shares are in Appendix Figure 13.

From 2032 California is at its cap every year, and the shortfall at the cap grows from 11 Mt to 61 Mt in 2045, $2.8 to $15.9 billion a year valued at the cap. With the compliance options in the model the gap cannot be closed: the standard falls to 90% below 2010 by 2045, fossil gasoline falls only from 9.0 to 5.2 billion gallons, and from the late 2030s renewable diesel and corn ethanol themselves generate deficits. From 2032 California's standard therefore acts in the model like a carbon price of $262 a tonne on carbon intensity above the benchmark. The model closes each year by issuing the shortfall as credits at the cap, and this closure is a maintained assumption of the main scenario. The statute instead carries an unmet obligation forward with interest; the two give the same prices and quantities if the cap binds every year to 2045 and fuel sellers expect carried obligations to be settled at the cap or never enforced, and then the choice only decides who holds the money (Appendix G.1).

I compare current law with the run without fuel policy, which removes the RFS, the four LCFS programs, 45Z, 45Q and the state biofuel incentives from 2026 while keeping existing capacity, banks and vehicle incentives. Over 2026–2045 current law abates 1.93 Gt at a resource cost of $259 billion (present value at 2%), $134 a tonne,[6] and a net fiscal cost of $40.8 billion, $21 a tonne, net of $5.2 billion of duties on imported fat, before the credits issued at the state caps. Those credits, mostly California's, are worth $102 billion at the caps. Counted as state revenue, they make the governments net receivers of $61.5 billion; under the statute fuel sellers hold them as an unpaid obligation instead.

Current law passes a cost–benefit test at EPA's values. Its break-even social cost of carbon is about $150 a tonne in 2030 (2024 dollars) at a discount rate of either 2% or 3% ($152 and $150): 0.55 times EPA's central path and 0.88 times its path for a 2.5% near-term rate, the path consistent with discounting at 3%. The discount rate has little effect on the break-even value but a large one on EPA's estimate of the damage, which falls from $279 to $170 a tonne in 2030 (U.S. Environmental Protection Agency, 2023). At 3% the resource cost falls to $233 billion, $121 a tonne, and current law gains $31 billion. Appendix G.2 reports the other results at 3% and the treatment of duties on imported fat.

The layers differ in cost. Removing the RFS from 2030 while keeping the other layers raises emissions by 705 Mt over 2030–45 and lowers the resource cost by $78 billion, so the RFS abates at about $111 a tonne, below the $132 of the whole stack over the same years, while the state standards cost $254 per national tonne. Even so, current law achieves only 15% of the welfare gain, net of climate damages, of a carbon price at the social cost of carbon ($215 against $1,444 billion over 2026–45), within the transportation fuel and vehicle markets the model covers; the carbon price also abates at a lower cost per tonne ($123 against $134).

Relocation is caused by the state standards themselves, and I measure it by comparing current

[6] This is the cost per tonne used throughout: resource cost including forgone fuel-tax revenue, in present value at 2%, over undiscounted tonnes including indirect land-use change. Appendix G.2 reports it under other definitions.

law with the run without the state standards. Over 2030–45 the four standards cut emissions in their own states by 1,059 Mt, but raise them in the rest of the country by 565 Mt, as renewable diesel that would have displaced diesel elsewhere is drawn into the clean fuel states; net of this relocation the standards abate 494 Mt at about $254 a tonne. Per gross in-state tonne the cost is $119. The difference reflects relocation, a form of domestic leakage, under which each in-state credit yields only 0.47 net national tonnes. The cap, $262, is the price of the marginal credit and should not be read as the average cost of the standard. The relocation share depends on the delivery slopes τ', the parameter the data identify least precisely, which determine how much of a state's compliance comes from drawing in renewable diesel from elsewhere and how much from other credits. At half and at twice the estimated slopes, the relocated share is 55% and 52%, against 53%, and the cost per national tonne $242 and $253, against $254. At both slopes, 0.74 of the 45Z extension's cost reaches fuel buyers. The states' rules that defer or tighten their standards could offset part of the extension, but they do not, because the extension barely moves the state banks (Appendix G.4).

Table 1 shows who gains and who pays under each layer. Each column compares current law with current law without that layer. The federal mandate takes $215 billion from fuel users over 2030–45, almost all of it outside the clean fuel states, at a resource cost of $78 billion. Feedstock suppliers gain $90 billion, $84 billion of it U.S. farmers; biofuel producers gain $58 billion and credit holders $12 billion. The state standards take $308 billion from fuel users, with a resource cost of $126 billion. Biofuel producers gain $66 billion, credit holders $27 billion and buyers of electric vehicles $17 billion, while feedstock suppliers lose $10 billion, because the standards move renewable diesel into the states without raising its output. Almost all of the 45Z credit of 2026–29 reaches fuel users ($14 billion of its $17 billion), as expected for a credit on a binding mandate.

Every layer that raises fuel prices is regressive. I use the Consumer Expenditure Survey shares of fuel spending by income quintile. Through fuel prices, vehicle incentives and taxes, the federal mandate costs the lowest quintile 0.37% of pre-tax income a year and the highest 0.08%. The state standards, averaged over the whole country, cost the lowest quintile 0.70% and the highest 0.14%. For households in the four states the burden is several times larger: in California the state standards cost the lowest quintile 4.96% of income a year and the highest 0.77%, and in the other three states 1.38% and 0.25%. Measured income in the lowest quintile is low relative to its spending; as a share of spending the California burden is 1.94% and 1.41%, still regressive but much less so. The 45Z extension has the opposite effect. Cheaper fuel lowers the burden on the lowest quintile by 0.08% of income, and the taxpayer cost falls mostly on the highest. These figures leave out producer and credit-holder rents and the credit revenue that returns to households in the four states through utility programs, worth $33.0 billion over 2026–45. Appendix G.2 gives the allocation of taxes by quintile, the treatment of the credits issued at the caps and the use of credit revenue. Holland et al. (2015) also show that such transportation policies have large regional distributional effects.

Table 1: Incidence of each policy layer, against current law without it ($ billion, present value at 2%)

	Federal mandate	45Z credit, 2026–29	State standards	45Z extension
Fuel users	−214.7	13.8	−307.8	45.0
of which in the clean fuel states	−18.9	0.8	−316.7	3.6
Buyers of vehicles	−4.6	0.0	17.1	0.1
Biofuel producers	57.7	0.7	66.4	2.7
Feedstock suppliers	90.2	3.4	−9.7	9.9
of which U.S. farmers	84.4	2.8	−10.4	8.0
Credit holders and other generators	12.4	−1.0	27.3	−1.6
Charging station owners	0.4	0.0	0.3	0.0
Taxpayers (federal and state budgets)	1.3	−16.9	−21.7	−53.2
Credits issued at the caps	−21.2	−0.1	102.3	0.7
Resource cost ($-W$)	78.4	0.2	125.7	−3.5
Abatement (Mt)	705	−14	494	−64
Burden through fuel prices, vehicles and taxes, U.S. households, lowest / highest quintile (% of income a year)	0.37 / 0.08	−0.02 / 0.01	0.70 / 0.14	−0.08 / 0.03
California households			4.96 / 0.77	
Oregon, Washington and New Mexico households			1.38 / 0.25	

Notes: Fuel users include buyers of non-road diesel outside the model. Positive: the layer makes the group better off. Federal mandate, state standards and the 45Z extension over 2030–45 (the extension against current law); the 45Z credit of 2026–29 over 2026–45. Taxpayers exclude the credits issued at the caps, which appear on their own row: state revenue if the state issues them, an unpaid obligation of fuel sellers under the statute. Burden: positive is a cost; the extension lowers the burden on low-income households. Negative abatement means the layer raises emissions: the 45Z credit of 2026–29 raises them by 14 Mt at a resource cost of $0.2 billion, so with climate damages counted it lowers welfare by $3.8 billion. Fuel-price burden: gasoline by the households' own state, diesel and jet fuel through the prices of goods, vehicles and taxes by U.S. quintile shares, from the Consumer Expenditure Survey; Oregon, Washington and New Mexico use the Census West profile. The lowest quintile's measured pre-tax income is low relative to its spending, which raises its shares. The reference runs have no ethanol blend floor.

6.2 A federal dollar on a binding mandate

The additional support I study is an extension of 45Z beyond 2029, over 2030–45, at its current-law rate, measured per dollar of its federal cost. I compare three runs: current law, the extension, and the extension with LCFS prices held on their current-law path. The difference between the last run and current law is the effect at unchanged state credit prices; the difference between the extension and the last run is the part that works through the state credit prices.

Over 2030–45 the extension's federal credit outlays are $53.5 billion in present value at 2% ($68.0 billion undiscounted; $53.2 billion for the net change in federal and state budgets, which adds forgone fuel taxes). Table 3 divides that cost among the groups that receive it; the components sum to the federal cost. The total RFS requirement binds in every year, so the extension falls in

case (ii) of Proposition 1: the RIN value biofuel producers receive falls by $64.0 billion. The credit replaces RIN value rather than adding to it. The mandate binds in every forward year because the RIN bank is at its floor from 2030 and, without RIN value, biofuel would cost more than the fossil fuel it replaces on the main oil price path. On the record it bound in 45 of the 50 quarters from 2013Q4 to 2026Q1 that can be classified. Two identified reasons account for part of the difference between the RIN value lost and the federal cost: fuel made from imported fat earns no 45Z after 2025, so the Treasury does not pay the credit that fuel would carry and foreign fat suppliers bear it through a lower import price (7.7% of the federal cost), and cheaper fuel raises the obligated volume (1.6%).

The lower RIN price lowers the price of every obligated gallon, and fuel buyers in the model gain $38.9 billion net, 73% of the federal cost; another 12% reaches buyers of non-road diesel outside the model. Feedstock suppliers receive 19% and biofuel producers 5%.

The 73 cents is a benchmark: it is what a binding mandate implies for a credible, permanent credit in a competitive market, with RIN prices passed fully into fuel prices, as the evidence on wholesale prices supports (Knittel et al., 2017). The record on the credit itself implies less. Over 2014–21, when the $1 blenders' credit lapsed and was restored, a dollar of credit moved the D4 price by −0.41 (s.e. 0.13, 86 months in which the mandate bound; Table 2), well short of the −1 that full capitalization implies. Part of the gap may come from traders who expected each lapse to be reversed, but two other measures of that expectation imply that only 40–50% of the credit was capitalized, while a third, from weekly transaction prices, is consistent with full capitalization (Appendix G.3). The two numbers describe different policies. The blenders' credit was repeatedly allowed to lapse and then restored retroactively, so in many months its value was uncertain when fuel was sold, whereas the extension I study would be a standing credit on production. The data therefore cannot settle whether a permanent extension would be capitalized in full. If it were capitalized only in part, the RIN price would fall by less, the part not priced in would stay with biofuel producers, and fuel prices would fall by less, so less than 73 cents would reach fuel buyers and emissions would rise by less than 64 Mt.

Hydroprocessing capacity barely changes, falling by 0.07 billion gallons a year on average over 2030–45 (0.9%); capacity that can make SAF falls by 0.16 billion gallons and other capacity rises. This follows almost directly from the investment estimate. Investors who already value the credit as continuing ($p = 1$, estimated at [0.67, 1.00]) receive no news from the extension, and the lower RIN price slightly lowers their margins. At $p = 0.5$ the extension is news, and capacity rises by 0.13 billion gallons a year (1.9%); the shares barely move.

The state credit markets play a small part. The transfer through the four state credit prices is −1.6% of the federal cost and the change in the shortfall at the caps +1.4%. Holding the state credit prices on their current-law path changes the share reaching fuel buyers from 0.727 to 0.739, and removing the state standards altogether raises it to 0.744. Through the RIN price alone fuel

Table 2: Capitalization of the blenders' credit into the D4 price

Sample	Coefficient (s.e.)	Months	Implied q [95% interval]
All months, 2014–21	−0.39 (0.12)	89	0.61 [0.37, 0.84]
Months in which the mandate binds	−0.41 (0.13)	86	0.59 [0.34, 0.84]
2014–16	−0.10 (0.11)	33	0.90 [0.69, 1.12]
2017–21	−0.59 (0.13)	53	0.41 [0.15, 0.67]
Months with D4 above D6 by $0.05 or more	−0.53 (0.11)	82	0.47

Notes: q: the probability of retroactive restoration that would make the estimate consistent with full capitalization, $-(1-q)$; it is a reading of the estimate, not a test. With the D4–D6 spread as the dependent variable the coefficient is not significant in the samples above; it is −0.28 (s.e. 0.10) in 2014–19. In the event windows, the retroactive restoration of December 2019 moved the spread by −0.04 to −0.07 dollars per RIN over ten days, against a standard deviation of 0.076 in non-event windows; neither the 2022 extension nor the 2025 extension of 45Z moved it detectably.

buyers gain 1.081 of the federal cost, more than one because 45Z is not paid on fuel from imported fat and because the obligated volume rises; they give back 0.338 through the prices and quantities of the biofuel they buy.

Table 3: Incidence of a 45Z extension, 2030–2045 (shares of the federal cost; present value at 2%)

	Main	With EPA offset
RIN transfer to fuel buyers in the model	1.081	−0.378
Fuel buyers, other (biofuel prices and quantities)	−0.338	0.268
Transfer through the state credit prices	−0.016	0.034
Share to fuel buyers in the model	*0.727*	−0.076
RIN transfer to non-road diesel buyers (outside the model)	0.115	−0.041
Feedstock suppliers	0.185	0.463
Biofuel producers (net of capital costs)	0.050	0.246
State revenue from credits issued at the caps	0.014	−0.013
Other (duties and other government lines, holders, exempt refiners, vehicle buyers)	−0.025	0.004
Net resource cost	−0.066	0.418
Federal cost ($ billion)	53.5	57.6
Emissions change (Mt)	+64.3	−102.2
of which outside the clean fuel states	+59.7	−90.4
Resource saving per extra tonne / cost per tonne abated ($)	55	235
Binding case of Proposition 1	ii (v)	ii (v)

Notes: Extension of 45Z from 2030 to 2045 at its current-law rate, against current law (the main specification). Appendix Table 19 adds perfect foresight, $\gamma = 0.32$ and $p = 0.5$. "With EPA offset": the extension together with an RFS requirement raised by 6.4%, the increase that returns the D6 price to its current-law path, against current law. Shares sum to one. Case (v), California at its cap, applies in the state market from 2032. Under the alternative RIN deficit pattern the share to fuel buyers changes by 0.1% and the RIN transfer by 3%. Memo: 45Z booked on fuel from imported fat, borne by foreign fat suppliers through a lower import price, 0.077 of the federal cost (main column).

Because the extension makes fuel cheaper, fuel use and emissions rise. Over 2030–45 emissions increase by 64 Mt, 60 Mt of it outside the clean fuel states, and almost all of it (62 Mt) from fossil gasoline, whose price falls with the D6 price; the extension saves $3.5 billion of resource cost, about $55 per additional tonne emitted. Cheaper imported fat accounts for $4.2 billion of the saving, a terms-of-trade gain, and the rest of the economy's resource cost rises by $0.7 billion; the $1.7 billion of duty revenue the Treasury loses as fat imports fall is a transfer. Valued at the EPA social cost of carbon, the additional emissions cost $15.8 billion, so the extension lowers welfare by $12.2 billion once climate damages are counted. The decomposition does not depend on whether the biomass-based diesel requirement binds on its own or on which years the market borrows RINs (Appendix G.4).

The result assumes that EPA holds those percentages fixed, and EPA has changed them in both directions (Appendix A.2).

Two runs separate what the credit does from what EPA does. If EPA raised the requirement by enough to return the D6 price to its current-law path, 6.4%, the extension and the higher requirement together would abate 102 Mt at $235 a tonne, while a higher requirement on its own, raised by 3.8%, abates the same 100 Mt at $163 a tonne. The abatement of the pair therefore comes from the requirement; the credit changes who pays for it and raises its cost, and EPA could raise the requirement without the extension at lower cost. Figure 6 shows the whole range of offsets. Emissions stop rising at an increase of about 2.6%, roughly the size of the revisions EPA has made to its own proposals, and at that size the pair still lowers welfare once climate damages are counted. Appendix G.4 gives the shares and welfare at each point.

Two other federal tools fall in the same case and are left for future work: a restored 45Z bonus for SAF, and a federal year-round E15 law, which loosens the ethanol blend limit under the same binding total requirement.

6.3 Extension: federal dollars that reach the state markets

The 45Z extension works almost entirely through the federal mandate. As an extension, I also consider federal support that reaches the state credit markets directly. Support for electric vehicles is the main example: every vehicle charged in a clean fuel state earns credits, so a federal dollar spent on chargers or vehicles adds to the bank and, while the bank is large, lowers the state credit price rather than adding abatement in the state (cases (iii)–(v) of Proposition 1). In this part of the stack the outcome depends on banks and expectations. Table 4 reports a federal fast-charging credit (50% of installed cost, 2028–32) and a federal purchase credit ($7,500 per light-duty vehicle, 2028–32), each at its proposed scale. The charging credit abates 171 Mt over 2028–45. Its federal credit outlays are $8.2 billion in present value at 2% ($14.0 billion for the net change in federal and California budgets, which adds forgone fuel taxes), and it costs $31 a tonne in resources. The

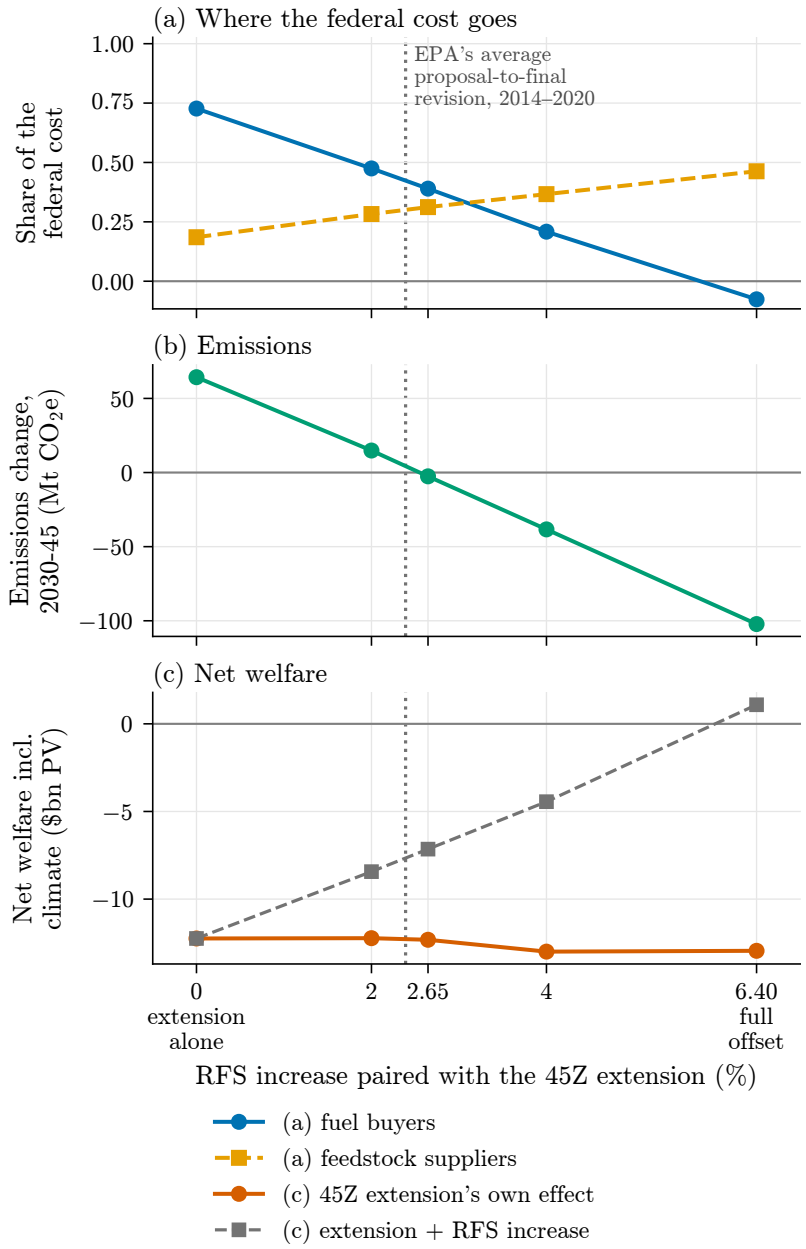


Figure 6: What the 45Z extension does as EPA raises the RFS requirement, 2030–2045

Notes: The 45Z extension paired with an RFS requirement raised by 0, 2, 2.65 (about where the extra emissions disappear), 4 and 6.40% (full offset, the increase that returns the D6 price to its current-law path), each against current law. (a) Shares of the federal cost reaching fuel buyers and feedstock suppliers. (b) Change in emissions, including indirect land-use change. (c) Net welfare including climate damages, $ billion, present value at 2%. The federal cost rises from $53.5 billion to $57.6 billion across the range. Panel (c) compares each pair with current law; the extension's own effect, against the same requirement without it, is −$12.2 billion at no offset and −$13.0 billion at the full offset. Dotted line: EPA's average revision from proposal to final rule, 2014–20, on the total percentage standard. It shows the size of adjustments EPA has made, not a prediction of how EPA would respond to the extension.

credit expires in 2032, so stations are built ahead of the expiry: 59,800 fast-charging stations in 2032, against 26,400 under current law. Stations cannot be withdrawn; outside the clean fuel states they recover only 33–76% of their cost in 2033–45, and the resource cost includes those losses. Part of the rush comes from the one-period entry rule, so 171 Mt is an upper bound. [7] The purchase credit abates 433 Mt over 2028–45 at $101 a tonne of resource cost, for federal credit outlays of $99.4 billion ($105.5 billion including forgone fuel taxes); 66% of the federal dollar reaches vehicle buyers and 7% fuel buyers. Of the 15.6 million vehicles subsidized in 2028–32, 6.7 million (43%) are additional, about $17,500 per additional vehicle; the implied price elasticity of electric-vehicle demand, −2.9, is at the high end of the literature, so this cost per tonne is a lower bound. Without the state standards the credit abates 437 Mt at $80 a tonne. The difference between each pair measures how much of the federal effort the state markets absorb.

Table 4: The policy menu: abatement, cost and incidence by instrument, 2030–2045

	Abatement	Cost per tonne ($)			Fiscal	Share to
	(Mt)	Resource	+ tax wedge	+ other ext.	cost ($bn)	fuel buyers
Current law against no fuel policy	1,675	118	132	108	20.4	—
2026–45, the headline in Section 6.1	1,925	121	134	111	40.8	—
45Z credit as legislated, 2026–29	−14	costs 51	costs 11	costs 88	16.9	0.71
45Z extension	−64	saves 24	saves 55	costs 12	53.2	0.73
45Z extension with RFS +2.65%	3	n.m.	n.m.	n.m.	55.2	0.39
45Z extension with EPA offset	102	237	235	238	58.2	−0.08
Reaching the state markets						
Fast-charging credit, 2028–45	171	−3	31	33	14	0.18
Purchase credit, 2028–45	433	61	101	104	105.5	0.07
Moving the constraint (100 Mt target)						
Higher RFS requirement (+3.8%)	99.5	153	163	139	0.3	—
Carbon price	101	4	32	−32	−55	—

Notes: 2030–45 unless stated. Present value at 2% over undiscounted tonnes. Resource: resource cost, with forgone fuel-tax revenue counted as a transfer. + tax wedge: forgone fuel-tax revenue counted as a cost, the convention used in the text. + other ext.: adds the change in congestion, accident and local air-pollution costs (Section 4.5). For the 45Z extension, which raises emissions, the entry is the resource saving (or cost) per additional tonne emitted. The last cost column uses central values (Section 4.5); with the ranges there, current law costs $89–120, the carbon price −$83 to $1, and the purchase credit up to $131 if electric vehicles cause the local damages of the 2014 grid. Fiscal cost: federal and state budgets, excluding the credits issued at the state caps; including them as state revenue, it is −81.9 for current law, 16.3 for the charging credit, 114.2 for the purchase credit and 1.5 for the higher requirement. The share to fuel buyers is the share of the federal cost ($53.5 billion for the extension and $57.6 billion for the extension with the EPA offset). The 45Z credit of 2026–29 is over 2026–45. n.m.: not meaningful; with RFS +2.65% the extension abates only 2.6 Mt at a resource cost of $7.8 billion. Negative cost: a net gain. The run without fuel policy has no ethanol blend floor.

[7] California's LCFS already gives infrastructure credits to fast chargers, but under the rules studied by Bushnell et al. (2020) these credits are capped at 5% of the prior quarter's deficits (2.5% for fast charging and 2.5% for hydrogen); the small effect they find is on LCFS compliance, not on station counts.

6.4 Moving the constraint

An alternative to subsidizing fuel within a binding constraint is to change the constraint itself. Table 4 also compares instruments at a common target of 100 Mt of additional abatement over 2030–45: a higher RFS requirement and a carbon price on transportation fuels. The run without 45Z in 2026–29 measures the effect of the existing credit, which costs $16.9 billion in federal credit outlays (the same including forgone fuel taxes) and raises emissions by 14 Mt, almost all in 2026–29, with 71% of its cost reaching fuel buyers; removing it saves $0.2 billion of resources over 2026–45 and raises welfare by $3.8 billion once climate damages are counted.

The 100 Mt target is about 6% of current-law abatement over 2030–45, so these are costs at a small margin and not comparable with the $134 average cost of the stack. The carbon price that reaches the target is small: $1.83 a tonne in 2030, rising at 2% a year to $2.46 in 2045. Its resource cost is nevertheless $32 a tonne, because almost all of the abatement comes from burning less fuel, and that fuel already carries federal and state taxes. The lost tax revenue on this pre-existing wedge costs $28 a tonne, while the carbon price's own deadweight triangle costs under $1; without the fuel-tax wedge, the cost is $3.5 a tonne. Whether the wedge is a cost depends on the other external costs of driving, which a gallon not burned also avoids. Table 4 therefore reports the cost per tonne under three conventions: resource cost alone, with forgone fuel-tax revenue, and with the change in the other external costs as well. The ranking of instruments does not depend on the convention: a carbon price costs $4 or $32 a tonne and a higher RFS requirement $153 or $163. With the other external costs counted, the carbon price yields a net gain of $32 a tonne, because the fuel it saves avoids congestion and accidents worth more than its tax, and the higher requirement costs $139. The convention matters most for instruments that change how much fuel is used. Counting the tax wedge favors the 45Z extension, which makes fuel cheaper: more fuel use raises tax revenue, which the wedge counts as a gain, but it also raises congestion, accidents and pollution, which the wedge ignores.

The largest change to the constraint replaces the stack with a single national standard. In this experiment Congress announces a national low carbon fuel standard in 2028. It takes effect in 2030, replaces the four state programs and repeals the RFS. The standard declines at California's rate and has California's cap, $262 a tonne. It reaches the cap in 2031 and stays there every year, so it works like a carbon price of $262 a tonne on carbon intensity above the benchmark. Against no fuel policy, over 2030–45, it abates 5,177 Mt at $92 a tonne, against 1,675 Mt at $132 for current law. With jet fuel obligated as well, it abates 5,691 Mt at $89. Net welfare including climate damages is +$784 billion, against +$189 billion for current law. The gain is larger because a national market prices carbon in every state, whereas the current stack subsidizes fuel within a binding federal mandate and prices carbon only at the state caps. Appendix G.6 gives the definition, the path at the cap and the version with jet fuel.

Appendix G.7 reports the headline results with the delivery slopes at half and at twice their estimated values, a real holding cost of 3.4% (with γ rescaled so that $\gamma \ln(1+r)$ is unchanged), a stronger and a weaker response of holders to the balance, Advanced Clean Cars II restored and direct air capture with 45Q extended; present values at a 3% discount rate are in Appendix G.2.

7 Expectations and the Timing of Effects

7.1 Totals and timing

This section shows that expectations mainly affect when the effects of the policies arrive, and much less how large they are. The year the state banks run out depends on how holders form expectations, and it is earliest when they doubt the statute (Table 5). California's bank runs out in 2033 under perfect foresight, in 2031 under the estimated rule and in 2030 under the price-level bound described below, and its price first reaches the cap in 2033, 2032 and 2031, an ordering that follows the level of early prices. Holders who foresee the scarcity price it in from the start, so California's price in 2028 is \$205 under perfect foresight against \$114 under the rule. The higher price draws more renewable diesel into the state and lowers fuel use, and the bank lasts longer. Holders who extrapolate a large bank keep the price low, generate fewer credits and draw the bank down faster, as Corollary 1 implies; when it runs out, the price jumps to the cap. Across these descriptions of expectations the totals change little.

Table 5 reports the headline results under five descriptions of expectations: the estimated price-expectation rule, the rule with the model-free response to the balance ($\gamma = 0.32$), perfect foresight, investors who give the policy stack a one-half chance of continuing ($p = 0.5$), and the price-level bound described below. The cost per tonne of current law is \$134–137 in every column, the share of the extension reaching fuel buyers 0.727–0.742 and the increase in emissions 64–66 Mt. The timing varies more. California's bank runs out between 2030 and 2033, Oregon begins to defer its standard between 2034 and 2036, and New Mexico in 2033, or 2037 under perfect foresight. In no column does the 45Z extension change the year either state begins to defer, but expectations do affect when its transfer reaches fuel buyers. Under perfect foresight, 0.71–0.77 of the 45Z extension reaches fuel buyers in every year from 2030. Under the rule, the share is 0.70 in 2030 and 0.58 in 2031. In 2030 the biomass-based diesel requirement binds on its own and the extension lowers the D4 price more than the D6 price, while in 2031 the extension brings California to its cap a year early. The cumulative share reaches 0.727 only by 2045 (Figure 7b), so the total that reaches fuel buyers is about the same under both descriptions and only its timing differs.

Because the 45Z extension works mainly through the federal mandate, and its effect on the state credit prices is small (Table 3), expectations in those markets change its effects very little. A federal dollar that does reach them might behave differently. Table 6 repeats the federal fast-

Table 5: Main results under five descriptions of expectations

	Estimated rule (main)	$\gamma = 0.32$	Perfect foresight	$p = 0.5$	Price-level bound
Current law against no fuel policy, 2026–45					
Abatement (Mt)	1,925	1,929	1,949	1,925	1,918
Resource cost ($ billion, PV)	258.8	259.5	267.4	257.9	256.9
Cost per tonne / levelized ($)	134 / 168	135 / 168	137 / 172	134 / 168	134 / 168
45Z extension against current law, 2030–45					
Share of the federal cost to fuel buyers	0.727	0.742	0.735	0.728	0.739
RIN transfer / federal cost	1.081	1.107	1.092	1.118	1.068
Emissions change (Mt)	+64.3	+65.7	+65.2	+64.0	+66.0
Timing under current law					
First year at the cap: California	2032	2032	2033	2032	2031
First year of deferral: Oregon	2034	2035	2036	2034	2034
New Mexico	2033	2033	2037	2033	2033
California's bank runs out	2031	2032	2033	2031	2030

Notes: Each column against the same run without fuel policy, in which expectations play no role. Washington, held at its 20% limit, reaches its cap in no column. Oregon and New Mexico are dated by their first deferral: once they defer, their prices stay just below the cap and the year they first touch it is not a stable measure. With $\gamma = 0.94$ the extension's share is 0.740, emissions rise by 65.4 Mt and California first reaches its cap in 2032. With the states' codified responses switched off (estimated rule only), the extension raises emissions by 63.8 Mt and the share to fuel buyers is 0.720. Present values at 2%, 2024 dollars. The reference runs have no ethanol blend floor. Price-level bound: holders expect a permanent relaxation of California's standard of 10.0 Mt a year from 2027 that never comes (Section 7.2).

charging credit, the federal purchase credit and the national standard under the estimated rule, perfect foresight and the price-level bound, each against current law under the same expectations. The estimates are close under all three: the charging credit abates 171–172 Mt at $31–33 a tonne, the purchase credit 432–433 Mt at $98–102, and the national standard 3,496–3,504 Mt at $72–73. Only the year California reaches its cap shifts, by one or two years, as it does under current law.

Figure 7 shows California's credit price under the rule and under perfect foresight, with and without the 45Z extension, together with the observed price. Once California is at its cap the two paths coincide. Under perfect foresight Washington's price falls to zero after the early 2030s. RIN prices barely differ (at most $0.06 in any two-year average). Over 2026–45, perfect foresight abates 23.4 Mt more, mostly inside the clean fuel states and in 2027–29, when its higher early price raises abatement in California, at a resource cost $8.6 billion higher (present value at 2%; Table 5), and the shortfall at the caps is $6.8 billion smaller. Over 2030–45 alone the difference is 12.1 Mt, almost all outside the clean fuel states, at $4.6 billion. Under the rule the extension barely moves California's price ($194 against $193 in 2030) and brings the first year at the cap forward from 2032 to 2031, when the shortfall is negligible (0.09 Mt). Since holders, like investors, treat 45Z as continuing, an extension changes their expectations only through the credit balance it produces. Allowing an announced extension to move the expected exhaustion year directly (at 0.59 of its effect

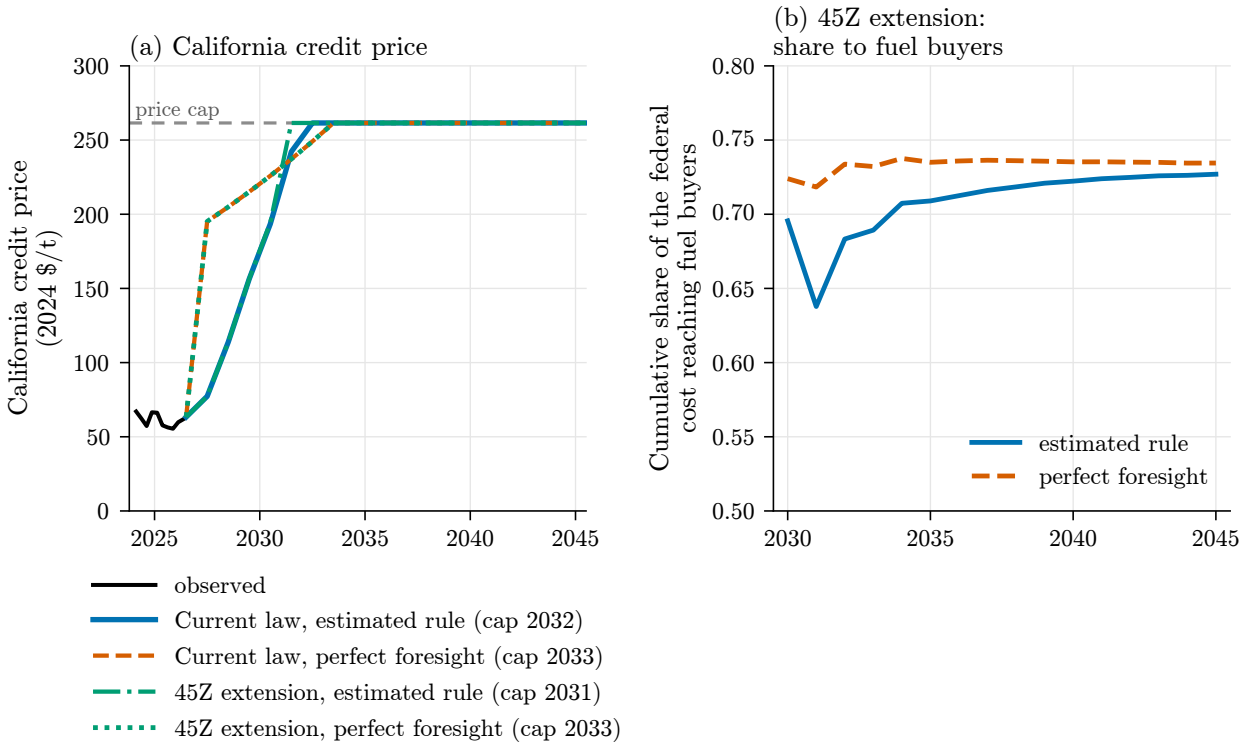


Figure 7: California's credit price and the timing of the 45Z transfer under the estimated rule and under perfect foresight

Notes: (a) California's credit price under current law and the 45Z extension, each under the estimated price-expectation rule and under perfect foresight; observed price 2024–26; the legend gives the first year at the cap. 2024 dollars. (b) Cumulative share of the extension's federal cost that has reached fuel buyers, by year.

on the perfect-foresight exhaustion year, the upper end of the share estimated for announcements in Appendix D) changes the extension's effect on California's 2030 credit price from \$0.86 to \$0.91 a tonne, and California still reaches its cap in 2032. Under perfect foresight the extension leaves California's price path unchanged (\$226 in 2030 and the cap in 2033 either way), because the price is set backward from a year at the cap that the extension does not move; its only effect is on the bank (23.9 against 23.4 Mt at the end of 2030).

7.2 Credibility

Every column of Table 5 prices more scarcity than the market has since 2023, which raises the question of whether credit holders believe the statute. Perfect foresight prices the bank under the statutes as written, and the rule reproduces the observed level only through the history it was fitted to. If holders expect a relaxation that the regulatory record does not support, actual prices will stay below both columns. The transfer to fuel buyers then arrives later, and more of each federal dollar remains a transfer through the state credit price and less of it produces abatement. To gauge

Table 6: Federal support that reaches the state markets under three descriptions of expectations

		Abatement (Mt)	Cost per tonne ($)	Share to fuel buyers	California at the cap
Fast-charging credit, 2028–45	Estimated rule	171	31	0.18	2032
	Perfect foresight	172	33	0.14	2033
	Price-level bound	172	32	0.14	2031
Purchase credit, 2028–45	Estimated rule	433	101	0.07	2033
	Perfect foresight	432	98	0.09	2035
	Price-level bound	433	102	0.07	2032
National standard, 2030–45	Estimated rule	3,502	73	—	2031
	Perfect foresight	3,504	73	—	2031
	Price-level bound	3,496	72	—	2031
with jet fuel obligated	Estimated rule	4,016	71	—	2031
	Perfect foresight	4,018	71	—	2031
	Price-level bound	4,010	70	—	2031

Notes: Each row against current law under the same description of expectations; under current law California reaches its cap in 2032 (rule), 2033 (perfect foresight) and 2031 (price-level bound). Abatement is undiscounted and includes indirect land-use change; cost per tonne is the present value at 2% of resource cost including forgone fuel-tax revenue, over undiscounted tonnes. The national standard has almost no federal outlay, so no share is reported.

how far prices could fall, I compute a bound. Suppose that from 2027 California's holders expect a permanent relaxation of the standard, worth D Mt of extra credits a year, and that the relaxation never comes. I choose D so that California's price in 2027–29 is close to its 2023–25 level, $66.75 a tonne, compared with $116 in the main run. The bank, the market, the cap and the codified responses are as in the main run, and only expectations differ. The required misperception is 10.0 Mt a year, a relaxation of the 2027 standard by 5.2 g/MJ and about a quarter of California's 2027 deficit, and it gives an average price of $69 in 2027–29. This value is consistent with the forward market. Fitting the rule's expected path to OPIS forward quotes for 2023–26 favours any D of at least 9.4 Mt a year, which includes 10.0, although even then the rule's expected prices in 2026 stay $68–83 above the quotes, because the rule starts from too early an expected exhaustion year.[8]

Figure 8 shows the result. Because holders expect credits that never arrive, they do not build the bank. The price stays near $65 until 2029, the bank runs out in 2030, and the price jumps to $214 in 2030, against $193 in the main run. California reaches its cap in 2031, a year earlier. As in Corollary 1, a lower price means fewer credits, so the bank runs out sooner. The totals change little (last column of Table 5), so doubt about the statute affects when scarcity arrives and how

[8] At $D = 3.85$ and 7.7 the price averages $96 and $80, and the results below lie between the main run and the bound.

abruptly, while the cost of the stack and the incidence of a federal dollar stay about the same.

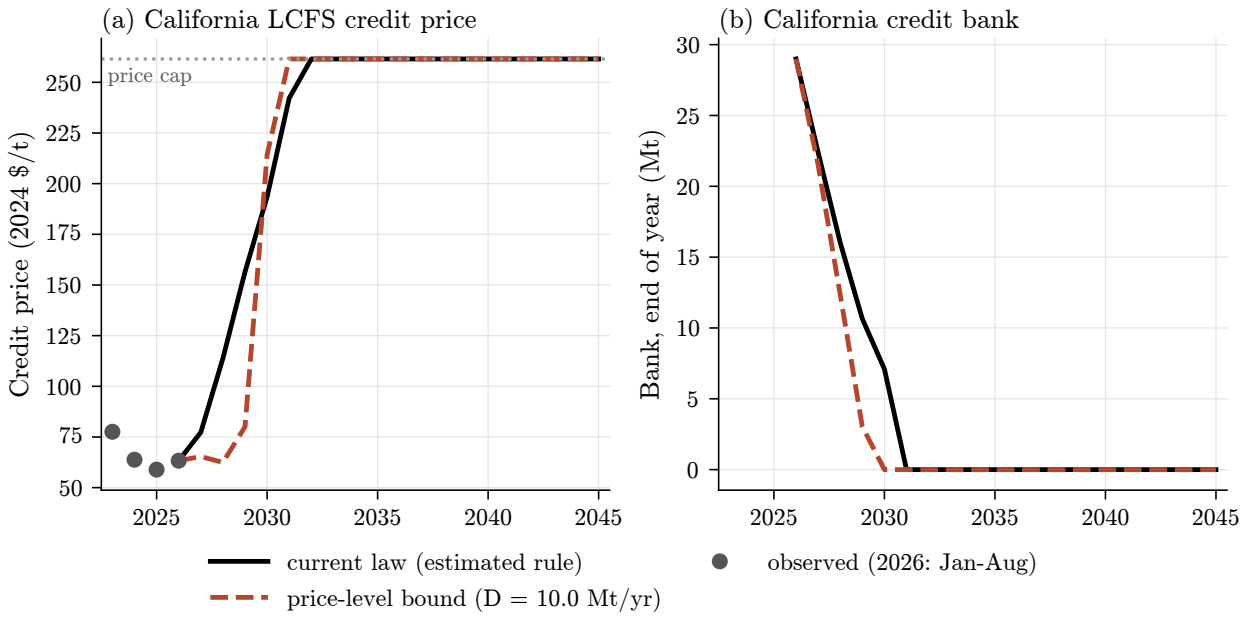


Figure 8: California's credit price and bank when holders expect a relaxation that never comes

Notes: Current law under the estimated price-expectation rule, and the price-level bound, in which holders expect from 2027 a permanent relaxation of 10.0 Mt a year that never comes. Observed prices 2023–26 (2026: January–August). 2024 dollars. The dotted line is the cap.

Investors' beliefs matter less. Because investors value a credit in law as if it continues, a guarantee of the credit changes investment little. At $p = 0.5$ capacity able to produce SAF is lower (0.88 against 1.03 billion gallons in 2030), but SAF output is about the same (0.2 billion gallons in 2030, 0.01 from 2033). Wu et al. (2026) give the comparison for SAF in 2030.

8 Conclusion

This paper treats U.S. fuel policy as one system in which federal and state instruments act on the same gallon, and asks what additional federal support achieves within it. Because the RFS binds, a permanent extension of 45Z is largely a transfer: in the quantitative model about 73 cents of each dollar go to fuel buyers, and emissions rise by 64 Mt. A higher RFS requirement that abates 100 Mt over 2030–45 costs $163 a tonne, and a carbon price that achieves the same abatement $32. Current law as a whole abates 1.93 Gt of carbon dioxide over 2026–45 at about $134 a tonne. It passes a cost–benefit test at EPA's values, comfortably at 2% and narrowly at 3%, but it achieves only 15% of the welfare gain of a carbon price, and over 2030–45 more than half of the abatement the state standards record inside their borders comes from renewable diesel drawn from other states. State credit prices are far below what the statutes imply. In the model this mainly affects timing: at low prices fewer credits are generated, California's bank runs out sooner and its price reaches the cap

abruptly, while the cost of the stack and the incidence of federal support change little.

As long as the RFS binds, extending 45Z lowers fuel prices and slightly raises emissions, and the credit reduces emissions only if EPA raises the RFS percentages in response. Whether federal fuel credits reduce emissions therefore depends largely on EPA. More predictable percentages and exemption decisions would also make the other layers of policy easier to plan around. 45Z also serves goals this paper does not evaluate, such as farm income and domestic feedstock. As a transfer to feedstock suppliers, however, it is costly: they receive about one dollar for every $5.40 of federal outlays (the inverse of their share in Table 3). Abatement alone could be obtained much more cheaply with a carbon price, and a national standard at California's cap that replaced the state programs and the RFS would abate more than three times as much as current law, at about 70% of the cost per tonne.

In California the binding constraint from 2032 is the price cap. Once the cap binds, lowering the trajectory has no effect on the credit price, and new credit sources such as direct air capture add abatement at a cost no higher than the cap. The statute carries unmet obligations forward and requires them to be retired within five years, which cannot happen if the cap binds every year. Selling credits at the cap, as California's cap-and-trade program already does with its price ceiling units (California Air Resources Board, 2026a), would make the cap a hard ceiling (Fell et al., 2012) without changing prices or abatement, and the revenue would go to the state instead of remaining an obligation of fuel sellers that may never be enforced. In Oregon and New Mexico the deferral rules take effect in 2034 and 2033 respectively, and from then on their standards tighten only as fast as new credit sources become available. Federal tax credits also sit on top of state standards in electricity, through renewable portfolio standards, and in state cap-and-trade programs and zero-emission vehicle credit markets (Goulder and Stavins, 2011; Fischer and Preonas, 2010). The same question, which constraint binds, determines the incidence of the federal credit in each of these settings.

Several limitations should be kept in mind. The model is partial equilibrium, so imported fats drawn into the state standards may be replaced abroad by other oils. With the land-use emissions of soybean oil, that leakage is 16 Mt over 2026–45 if half of the fat is replaced and 32 Mt if all of it is, against 1,925 Mt of abatement.[9] The price-expectation rule also contains no regulatory risk, although prices move on news about the regulator. California is assumed to issue credits at its cap from 2032, which matches the statute's carry-forward only if fuel sellers do not expect the carried obligations to be enforced at more than the cap. Finally, California's and Washington's cap-and-trade programs are treated as fixed carbon prices. If their caps bind, the abatement of the state standards is lower, while the 45Z results are almost unchanged (Borenstein et al., 2019).

[9] The emission factor, 0.256 kg CO_2e a pound, is the difference between the model's carbon intensities of soybean-oil and animal-fat renewable diesel. It is my construction, not a published leakage factor.

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A Institutional Details

The federal purchase credits for new and commercial clean vehicles (30D, 45W, 25E) ended for vehicles acquired after 30 September 2025, and the refueling property credit (30C) ends for property placed in service after 30 June 2026. Before OBBBA, the Inflation Reduction Act had extended 30C through 2032; the model dates each change by when it became law (Section 4.4).

Credits for residential charging accrue to the distribution utility, which must return at least half of them to drivers and may direct the rest to equity programs and charger grants. Credits from public and depot charging accrue to the equipment owner. The state standards therefore fund electric vehicle and charger subsidies out of fuel-buyer payments, a channel the vehicle block makes explicit.

The largest state program for zero-emission trucks is California's HVIP, a point-of-sale voucher whose amount depends on the vehicle class (California Air Resources Board, 2026b). Vouchers are requested when a truck is ordered and redeemed when it is delivered, so trucks delivered in 2025 were funded mostly by vouchers requested in 2023–24; the funding committed after HVIP reopened in September 2025 pays for trucks delivered in 2026–27. Combined purchase incentives may not exceed 90% of the vehicle's price. Depot charging for trucks is supported by California's EnergIIZE grants (CALSTART and California Energy Commission, 2025), by utility make-ready programs and, through mid-2026, by 30C. Washington exempts zero-emission trucks from sales tax and runs WAZIP; Oregon and New Mexico have smaller rebate and tax-credit programs (New Mexico Legislature, 2024). California's Advanced Clean Trucks sales requirement is not being enforced, and the model removes it. In the model, each incentive is included only to the extent of announced funding (Section 4.4).

Several states outside the clean-fuel bloc offer per-gallon SAF tax credits (Illinois, Minnesota, Nebraska, Iowa and Hawaii), mostly with annual budget caps. Washington has a SAF credit in statute with no certified in-state production.

Figure 9 shows California's market by quarter: the credit price, the bank, and the quarter's deficits and credits, with the dates of the 2018 and 2024 amendments.

A.1 The policy stack in detail

Table 7 summarizes the instruments that the model represents.

Federal instruments

The RFS requires refiners and importers to retire renewable identification numbers (RINs) in proportion to the gasoline and diesel they sell. The obligations are nested: biomass-based diesel (D4) counts toward the advanced requirement, which counts toward the total requirement, so the D4 price is at least the D6 (conventional ethanol) price and exceeds it only when the biomass-based

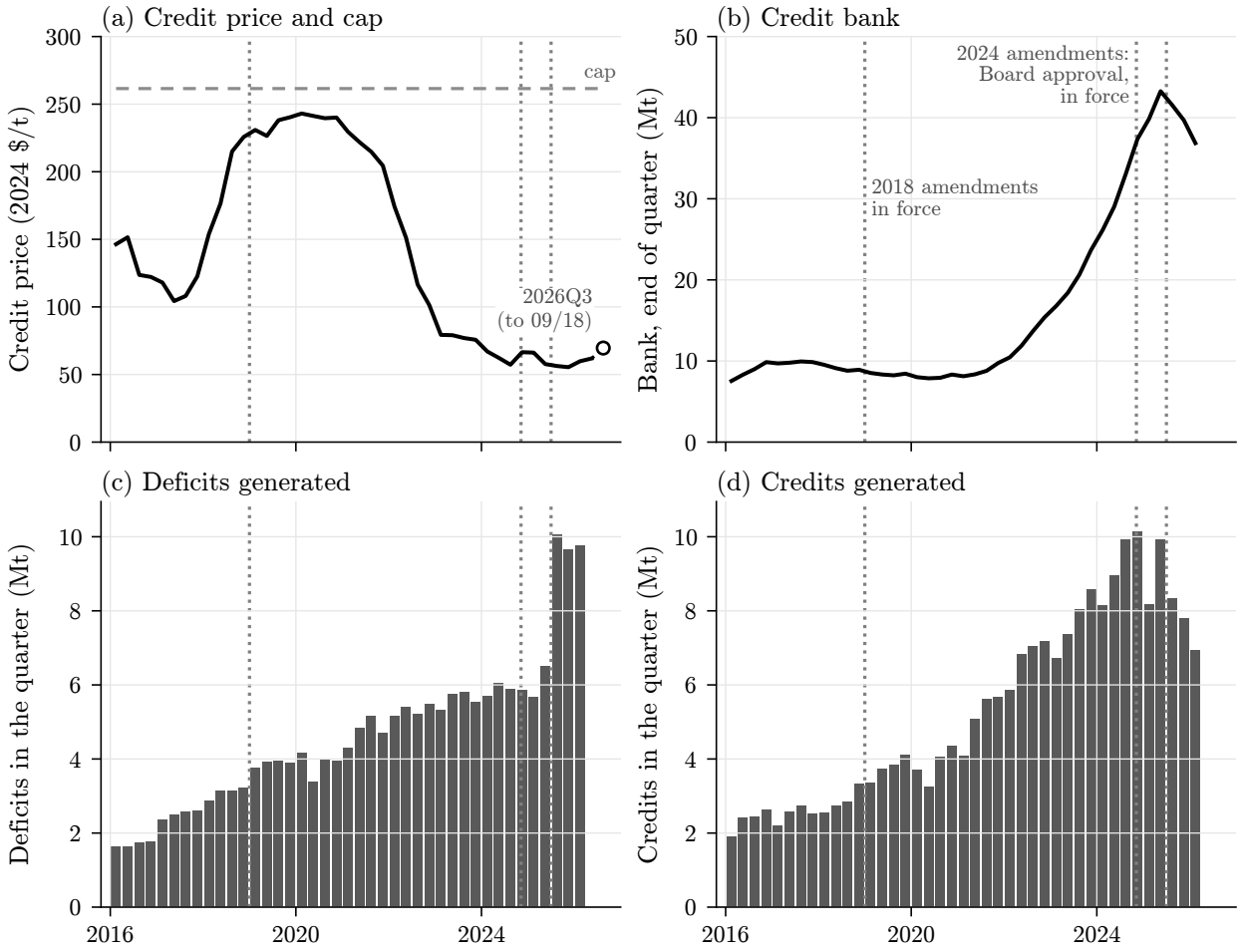


Figure 9: California's LCFS market by quarter, 2016–2026

Notes: (a) Credit price, 2024 dollars, with the cap; hollow point: 2026Q3 to 18 September. (b) Bank at the end of the quarter. (c) Deficits and (d) credits in the quarter. Dotted lines: the 2018 amendments in force, Board approval of the 2024 amendments, and the 2024 amendments in force. Source: CARB LCFS quarterly data summaries (to 2026Q1) and weekly transfer reports.

diesel requirement binds on its own. Each gallon earns RINs according to its equivalence value. EPA's final rule for 2026–2027 (U.S. Environmental Protection Agency, 2026c) raised the biomass-based diesel percentage from 3.15% of gasoline and diesel in 2025 to 5.24% in 2026 and 5.37% in 2027, and the total from 13.13% to 15.50% and 15.78%, and lowered the equivalence value of renewable diesel and sustainable aviation fuel (SAF) from 2027. The RFS has no statutory volumes after 2022; I hold the 2027 percentages for later years. Because the obligations are percentages of fossil gasoline and diesel, the required volume falls as electric vehicles reduce fuel demand. RINs can be used in the year they are generated or the next, carried-over RINs may cover at most 20% of an obligation, and a deficit may be carried forward one year but not two in a row (40 CFR 80.1427).

Since 2025 the federal blenders' credits are replaced by a production credit of up to $1.00

Table 7: The policy stack for U.S. transportation fuels, October 2026

Instrument	What it rewards	Who pays	Bank and price limits	Status
RFS (EPA)	Gallons of renewable fuel, by nested category (RINs)	Obligated refiners and importers, passed to fuel buyers	RINs usable for two years; carry-over $\leq$20%; one-year deficit	Percentages set for 2026–27
45Z (Treasury)	Gallons, scaled by lifecycle intensity; North American feedstock only	Federal taxpayers	—	Expires after 2029
LCFS: CA, OR, WA, NM (state agencies)	Carbon intensity below a declining benchmark, including electricity	Fuel buyers in the state	Unlimited banking; capped prices	Schedules to 2035–2045; CA tightens automatically, OR/WA/NM may defer
Vehicle and charger credits (30D, 45W, 30C)	Electric vehicles and chargers	Federal taxpayers	—	Ended September 2025 (vehicles) and June 2026 (chargers)

per gallon (2022 dollars) scaled by $(50 - \text{ER})/50$, where ER is the fuel's lifecycle emissions in kgCO_2e/MMBtu. The 2025 budget law (U.S. Congress, 2025, OBBBA) extended the credit through 2029, removed the higher SAF rate for fuel produced from 2026, excluded indirect land-use change from the emissions rating, and limited eligibility to fuel made from feedstocks grown in the United States, Mexico or Canada. The origin rule matters for the clean fuel states, whose renewable diesel had come increasingly from imported used cooking oil and tallow. Fuel from facilities claiming the carbon capture credit (45Q) cannot claim 45Z in the same year.

State clean fuel standards

California (LCFS, since 2011), Oregon (Clean Fuels Program, 2016), Washington (Clean Fuel Standard, 2023) and New Mexico (Clean Transportation Fuel Program, April 2026) set an annual benchmark carbon intensity for gasoline and diesel and their substitutes. Fuel above the benchmark generates deficits, fuel below it generates credits, and credits trade. Electricity used in vehicles earns credits, adjusted by an energy economy ratio. Fossil jet fuel carries no obligation, but alternative jet fuel may opt in and earn credits. Credits can be banked without limit and do not expire.

California's 2024 amendments (effective 1 July 2025) cut the benchmark to 22.75% below the 2010 base in 2025, 30% in 2030 and 90% in 2045 (California Air Resources Board, 2025). Oregon targets 20% by 2030 and 37% by 2035, New Mexico 20% by 2030 and 30% by 2040 (against 2018), and Washington, after HB 1409 (signed 17 May 2025), 45% below 2017 by 2038, with annual reductions of 3–4 percentage points in between. The 2028–37 rule is due by 1 January 2028; the model uses the 2025–27 standards in force and the Department of Ecology's draft path for

2028–37, which lies within the statutory range. The statute also bars Ecology from tightening the standard beyond 20% from 2030 until it demonstrates that at least one new or expanded biofuel production facility has received a siting, operating or environmental permit after 1 January 2025 (RCW 70A.535.025). A qualifying county permit appears to have been issued in December 2025, but as of October 2026 Ecology had not made the demonstration, so the main specification caps Washington's standard at 20% from 2030 and treats the statutory path to 45% as a scenario. A further provision letting Ecology relax a year's standard by up to two percentage points to avoid a forecast deferral is not modeled.

Each program limits the credit price, but in different ways. California prohibits any credit transfer above an inflation-indexed maximum price ($261.52 in 2024 dollars; 17 CCR §95487(a)(2)(D)) and runs a credit clearance market at that price; obligations still unmet carry over as an accumulated deficit with 5% compound interest, to be repaid within five years (§95485(c)). Oregon's and Washington's maxima bind sellers in their clearance markets (Oregon Department of Environmental Quality; Washington Department of Ecology), and unmet deficits carry forward with 5% added. New Mexico opens a clearance market at the agency's discretion, with an initial maximum of $270 (New Mexico Environment Department).

Rules that respond to the market

The programs also differ in how the standard itself responds to market conditions, in opposite directions. Oregon, Washington and New Mexico pair their standards with statutory forecast deferral. Each year a state agency forecasts whether banked plus newly generated credits will cover projected deficits; if the forecast falls below 100 percent, the regulator *shall* defer compliance, by temporarily relaxing the standard, holding it at the prior year's level, or suspending deficit accrual (State of Oregon; Oregon Department of Environmental Quality; State of Washington; New Mexico Environment Department, ORS 468A.273; OAR 340-253-2100; RCW 70A.535.110; 20.2.92.601 NMAC). Oregon additionally requires a root-cause review, and permits an emergency deferral, if credit prices rise 100 percent or more over three months, and Washington's 2025 amendments let the regulator loosen annual targets by up to two percentage points to avoid a forecast deferral. California has no deferral provision. Since July 2025 it instead *tightens* its benchmarks automatically: the automatic acceleration mechanism advances the whole benchmark schedule by one year when the bank exceeds three quarters of the year's deficits and credits exceed deficits (17 CCR §95484(b)); the first assessment is in May 2027.

In practice, Oregon's forecast deferral has never been triggered: every annual fuel supply forecast from 2018 to 2027 concluded that banked plus projected credits sufficed (Oregon Office of Economic Analysis), and I found no record of a deferral order in Oregon or Washington. Deferral therefore operates, if at all, through expectations: it bounds how far scarcity can be priced before the

standard itself adjusts. I treat this asymmetry as one possible reason why Oregon's credit prices stayed well below its ceiling in its tighter years, not as an established mechanism; I found no market commentary that links Oregon or Washington prices to deferral. In the model, deferral and acceleration are both applied to the model's own forecasts (Section 4.1).

A.2 EPA's record on the RFS volumes

In 2015 it set the 2014–16 totals below the statute with its general waiver (U.S. Environmental Protection Agency, 2015), and in 2017 the D.C. Circuit vacated the part of the 2016 cut based on inadequate domestic supply (U.S. Court of Appeals for the D.C. Circuit, 2017). For 2018 it exempted 31 small refineries, about 1.4 billion RINs or close to 7.5% of that year's requirement, and RIN prices fell (U.S. Energy Information Administration, 2019). In 2026 it set record volumes for 2026–27 and reallocated 70% of the volumes exempted for 2023–25 (U.S. Environmental Protection Agency, 2026a). After its exemption decisions of 31 August 2026 the D6 price rose from a January–August average of $1.49 to $2.13.

Table 8 lists each year's total renewable fuel requirement, how EPA set it, and the small refinery exemptions it later granted. From 2010 to 2013 EPA kept the statutory totals and cut only the cellulosic standard. In the rule of December 2015 it cut the 2014–16 totals with its general waiver as well, and the D.C. Circuit vacated part of the 2016 cut in 2017. From 2017 to 2020 it again used only the cellulosic waiver, while exemptions rose to 7–9% of the requirement in 2017–19 before falling to 4% in 2020. From 2023 it sets the volumes itself, and in 2026 it set record volumes and reallocated most of the exempted volume.

B The Quantitative Model

Table 9 lists the policies and rules the model represents and those it leaves out.

B.1 Credit markets, rules and expectations

In each program $j \in \mathcal{L}$, credits from biofuels, from electricity ($\Gamma^{EV}_{j,t}$, from the vehicle block) and from fixed sources (biomethane, off-road electricity, refinery projects and infrastructure credits), less deficits from fossil gasoline and diesel, give the year's net credit generation $N_{j,t}$. Infrastructure credits are a share $\iota_{j,t}$ of fossil deficits, set at the observed share in 2026 and rising to the regulatory limit. The balance is

$$N_{j,t} + B^{draw}_{j,t} = s^{+}_{j} - s^{-}_{j}, \qquad 0 \le P^{L}_{j} \perp s^{+}_{j} \ge 0, \qquad 0 \le \bar{P}_{j,t} - P^{L}_{j} \perp s^{-}_{j} \ge 0, \tag{22}$$

where B^{draw} is the draw on the bank allowed by holders' expectations (below) and $\bar{P}_{j,t}$ the program's price cap (2024 dollars: CA $261.5, OR $253.5, WA $245.5; NM $270 nominal, indexed from

Table 8: RFS total renewable fuel requirements and exemptions, 2010–2027 (billion RINs)

Year	Statute	Final	Rule published	Reduction authority	D6 at final ($)	Exemptions (bn RINs)
2010	12.95	12.95	26 Mar 2010	cellulosic waiver	—	—
2011	13.95	13.95	9 Dec 2010	cellulosic waiver	0.03	24 (—)
2012	15.20	15.20	9 Jan 2012	cellulosic waiver	0.02	23 (—)
2013	16.55	16.55	15 Aug 2013	cellulosic waiver	0.94	8 (0.19)
2014	18.15	16.28	14 Dec 2015	cellulosic and general	0.50	8 (0.21)
2015	20.50	16.93	14 Dec 2015	cellulosic and general	0.50	7 (0.29)
2016	22.25	18.11	14 Dec 2015	cellulosic and general[a]	0.50	19 (0.79)
2017	24.00	19.28	12 Dec 2016	cellulosic waiver	0.84	35 (1.82)
2018	26.00	19.29	12 Dec 2017	cellulosic waiver	0.89	31 (1.43)
2019	28.00	19.92	11 Dec 2018	cellulosic waiver	0.19	29 (1.46)
2020	30.00	20.09[b]	6 Feb 2020	cellulosic waiver	0.15	24 (0.82)
2021	33.00	18.84	1 Jul 2022	set after the year	1.32	25 (1.06)
2022	36.00	20.63[c]	1 Jul 2022	statutory table	1.32	22 (0.83)
2023	—	20.94[c]	12 Jul 2023	"set" authority	1.53	26 (0.93)
2024	—	21.54	12 Jul 2023	"set" authority	1.53	28 (1.15)
2025	—	22.33	12 Jul 2023	"set" authority	1.53	29 (1.76)
2026	—	25.82[d]	1 Apr 2026	"set" authority	1.28	—
2027	—	25.98[d]	1 Apr 2026	"set" authority	1.28	—

Notes: Statute: Clean Air Act §211(o)(2)(B); after 2022 EPA sets the volumes. Rule published: Federal Register publication date of the final rule. Reduction authority: the waivers EPA used to reduce the total or advanced requirement below the statute. D6 at final: EMTS volume-weighted price, mean of the five trading days before the announcement (nominal). Exemptions: petitions granted (exempted RINs) by compliance year; 2013–18 counts from U.S. Government Accountability Office (2022), 2017–18 volumes from EPA as reported by EIA (U.S. Energy Information Administration, 2019); other entries from EPA's exemption dashboard as of September 2026 (current status, after later reconsiderations). [a] The general-waiver cut of about 0.5 billion was vacated in U.S. Court of Appeals for the D.C. Circuit (2017). [b] Revised to 17.13 in July 2022. [c] Plus a supplemental 0.25 billion (2022 and 2023) on the remand of U.S. Court of Appeals for the D.C. Circuit (2017). [d] Plus 70% of the 2023–25 exempted volumes, reallocated (26.81 and 27.02) (U.S. Environmental Protection Agency, 2026a). Sources: Federal Register final rules 2010–2026; U.S. Environmental Protection Agency (2015).

2027). Standards follow each program's adopted schedule; Washington's 2028–37 schedule is the Department of Ecology's draft, within the statutory range, capped at 20% from 2030 until Ecology makes the demonstration the statute requires (Section 2). After a program's last statutory year, the standard keeps falling at the average annual rate of its last five statutory years.

When a program is short at the cap, the state issues the shortfall as credits at the cap price, the bank ends at zero, and the revenue goes to the state. The statutes instead carry an unmet obligation forward as a deficit with interest (17 CCR §95485(c) in California); when the cap binds year after year that deficit is never repaid, so issuance at the cap is the safety valve a regulator would have to adopt. The statutory ledger is reported separately.

LCFS credit holders in program j do not know when the bank will run out and the price reach the cap. Their expectation of that year, $T_{j,t}$ (the expected exhaustion year), prices credits by Hotelling to the cap at the holding cost $r = 5\%$,

$$P^{L}_{j,t} = \min\{\bar{P}_{j,t},\ \bar{P}_{j,t}/(1+r)^{\text{gap}_{j,t}}\}, \tag{23}$$

Table 9: Policies and rules in the model

Layer	Represented	Not represented
RFS	Nested biomass-based diesel, advanced and total requirements on a fossil base; RIN banking with the two-year life, 20% carry-over and one-year deficit; small-refinery exemptions as three scenarios; EPA's 2026–27 percentages and reallocation	EPA's discretion as a choice; RIN market power
45Z	Credit scaled by tax carbon intensity; North American feedstock only from 2026; expiry after 2029 (in law; agents treat it as continuing); inflation adjustment	Prevailing-wage conditions; transferability
State standards	CA, OR, WA, NM schedules; unlimited banking; price caps and clearance (CA deficit carry-forward replaced by issuance at the cap); California's acceleration mechanism and crop-oil cap; Washington's 20% limit from 2030 (the 45% path as a scenario); Oregon, Washington and New Mexico forecast deferral; electricity and infrastructure credits	Washington's 2-point deferral relief; New Mexico's program review
Other incentives	State SAF and biodiesel credits as budget-capped steps; 45Q; truck purchase incentives (HVIP stacking)	Local and utility programs beyond those listed
Markets	Gasoline, diesel and jet pools in five regions; ethanol, biodiesel, renewable diesel, HEFA-SAF, alcohol-to-jet; two lipid groups and corn; hydroprocessing investment; electric vehicles and chargers by class	World feedstock prices, indirect land use, refinery closures

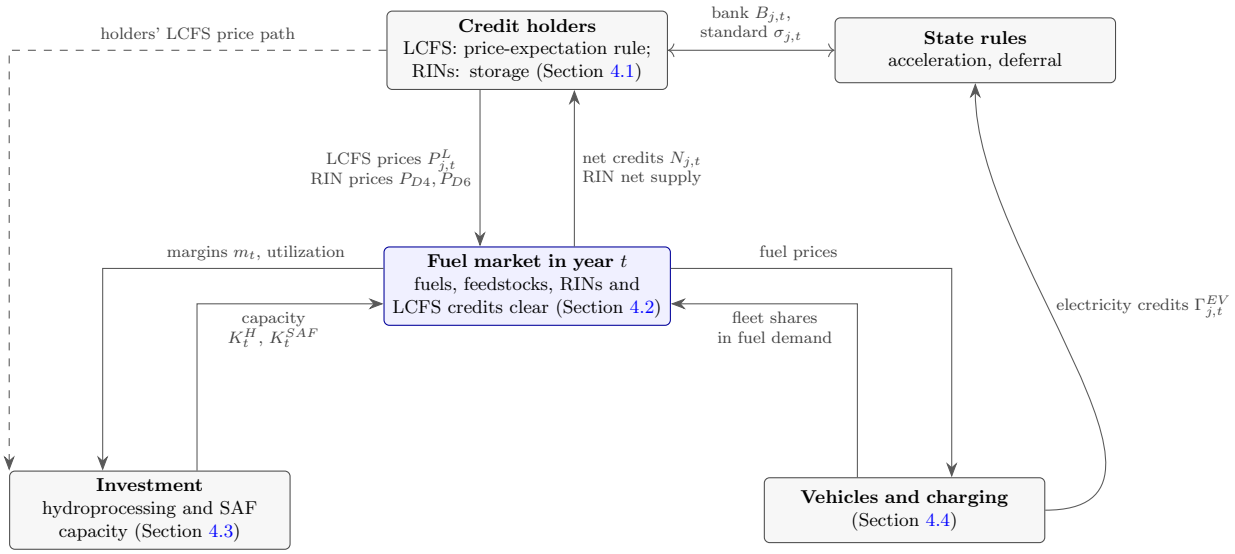


Figure 10: Structure of the quantitative model. Within a year, the fuel market clears given credit prices, capacity and fleets. Across years, credit holders' expected exhaustion years and banks, the state rules, capacity and the vehicle stock carry the state of the economy forward. The blocks are solved jointly over 2026–2045 (Section 4.5).

where $\text{gap}_{j,t}$ is the expected number of years to exhaustion. Holders update it by the surprise in the year's net credit generation $s_{j,t}$ relative to what they expected,

$$G_{j,t}(s) = T_{j,t-1} - t + \gamma_j\big(s - E_{t-1}[s_{j,t}]\big). \tag{24}$$

A year with more credits than expected moves the exhaustion date out by γ_j years per Mt and lowers the price.[10]

Expected net generation has two parts: a baseline and the changes holders know about in advance,

$$E_{t-1}[s_{j,t}] = \bar{s}_{j,t-1} + \Delta^{sched}_{j,t} + \Delta^{known}_{j,t}, \tag{25}$$

where Δ^{sched} is the mechanical effect of this year's change in standards at last year's quantities (including any acceleration or deferral already decided) and Δ^{known} the effect of other changes known in advance: the crop-oil cap, scheduled import volumes and the fading out of the 2026 transitional terms (Section 5.3). Scheduled expiries of federal credits are not among them: like investors, holders treat 45Z as continuing. Counting its expiry after 2029 as a known change would move California's 2030 price by about \$2/t. Expected changes use the information available at the time; in forward years there are no rule-change jumps, so changes in the rules enter only through Δ^{sched} and Δ^{known}. I consider two baselines $\bar{s}_{j,t-1}$:

- *Rule A (last year)*: $\bar{s}_{j,t-1} = s_{j,t-1}$;
- *Rule B (three-year average)*: $\bar{s}_{j,t-1} = \frac{1}{3}(s_{j,t-1} + s_{j,t-2} + s_{j,t-3})$, adaptive expectations in the sense of Nerlove (1958), with the available years at the start of the sample.

Holders also look at how many years the bank would last at the expected rate of shortfall, $Y_{j,t} = \max(B_{j,t-1}, 0)/(-E_{t-1}[s_{j,t}])$ (and $Y_{j,t} = \infty$, so $w_{j,t} = 0$, when $E_{t-1}[s_{j,t}] \geq 0$), and put more weight on it as it shrinks:

$$\text{gap}_{j,t} = (1 - w_{j,t})\, G_{j,t} + w_{j,t}\, \min\{G_{j,t}, Y_{j,t}\}, \qquad w_{j,t} = e^{-Y_{j,t}/K}. \tag{26}$$

K is the horizon, in years of bank left, over which holders start to price exhaustion. With a large bank the price follows the rule (18); as the bank runs down, the price rises toward the cap even without new surprises. The blend can only raise the price. Within a year, the price and the net generation it induces are a fixed point, and the bank evolves as $B_{j,t} = B_{j,t-1} + s_{j,t} + s^{-}_{j,t} - s^{+}_{j,t}$, so a year that clears at the cap ends with an empty bank. If the expected price would overdraw the bank, the year clears instead: the whole bank is available, equation (16) sets the price, and the expected exhaustion year is reset to match that price.

[10] One interpretation is the steady state of a Kalman filter for holders who read each year's net credits as a noisy signal of a persistent component of the balance (Appendix D.2).

γ_j, K and the initial expected exhaustion years are estimated on California's and Oregon's prices and banks for 2016–25. Oregon, Washington and New Mexico share one response γ per unit of their obligation, $\gamma_m = 100\,g/O_m$ with O_m the program's 2025 obligation in Mt, and the change in T is limited to 20 years per year in Oregon and Washington and 5 in New Mexico. The main specification uses Rule B with $K = 2$ in all four programs; Rule A is the alternative.[11]

The RIN bank is held in the conventional (D6) market. Holders are competitive and store RINs when the expected return covers the interest rate, which is the solution of a planner's problem that minimizes the discounted cost of meeting the obligations,

$$\min_{\{B_t, z_t\}} \sum_{t=2026}^{2045} d_t\big[C_t(B_t - B_{t-1}) + \Psi_t(B_t)\big] + d_{2046}\, C_{2046}(-B_{2045}) \quad \text{s.t.} \quad B_t \geq -k\, R_t\, z_t,\ z_t + z_{t-1} \leq 1, \tag{27}$$

where $C_t(n)$ is the area under the year's inverse net-supply curve of RINs (built from the year's solves of the fuel market), $d_t = 1.05^{-(t-2025)}$, Ψ_t a soft penalty for carrying more than 20% of the next year's obligation, and z_t marks a year in deficit. A deficit can be carried forward one year, not two in a row (40 CFR 80.1427(b)), and is limited to 10% of the obligation, applied to the industry as a whole. The 20% carry-over limit applies to each obligated party; the model's aggregate bank stays below 10% of the next year's obligation in 2026–30, so the limit does not bind in aggregate. At an interior bank, prices rise at 5% a year. The terminal year 2046 is a static copy of 2045 in which the bank is drawn down. A version that does not allow deficits is a sensitivity.

RIN markets are national and nested. Obligations are EPA's percentage standards applied to the obligated base,

$$\text{Base}_t = \textstyle\sum_j Q_{G0,j,t} + (1 - f_{D,t})\big(\sum_j Q_{D0,j,t} + X_t\big), \tag{28}$$

where Q_{F0} is fossil gasoline and diesel, X_t is obligated non-road, locomotive and marine diesel outside the model, and $f_{D,t}$ the share of diesel exempted for small refineries in EPA's 2026–27 rule (U.S. Environmental Protection Agency, 2026c). The total requirement (net of cellulosic, which has no supply in the model), the advanced requirement and the biomass-based diesel (BBD) requirement each bind as a complementarity with its price: P_{D6}, the advanced premium μ_t^{ADV}, and the biomass-based diesel premium $P_{D4} - P_{D6} - \mu_t^{ADV}$. The 2026–27 percentages are EPA's final standards, including the reallocated exempted volumes; from 2028 the reallocation is removed and the percentages are held at their 2027 level (net of reallocation) and applied to each year's base, so obligations fall as fossil demand falls. RIN equivalence values are 1.5 for BD and 1.6 for RD and SAF (RD 1.7 in 2026). Every obligated gallon carries the cost of the full obligation,

$$w_t^{RFS} = \pi_t^{BBD} P_{D4} + (\pi_t^{RF} - \pi_t^{BBD}) P_{D6} + (\pi_t^{ADV} - \pi_t^{BBD}) \mu_t^{ADV}, \tag{29}$$

[11] Under Rule A, the bank-years blend (19) is used in California only, because Oregon's bank never fell in the sample and the blend is not identified there; a version with the blend in all four programs is a sensitivity. Under Rule B the blend applies to all four programs.

whether the RINs come from this year's production or from the bank. Exempt refiners keep this cost as rent.

California's 2024 amendments advance all later benchmarks by one year, effective the following January, when the bank exceeds three quarters of the year's deficits and credits exceed deficits in the same year (17 CCR §95484(b)). The model applies the rule to its own bank and flows from 2027 on. Whether the mechanism triggers depends on small differences in the 2027 balance, so I do not treat a trigger, or its absence, as a finding.

The other three programs must defer scheduled tightening when the agency forecasts that banked and new credits will fall short of the obligation, by adopting a temporary standard that better reflects the forecast supply of credits. I model the temporary standard as a weighted average of this year's scheduled standard and the one in force last year,

$$\sigma_{j,t}(d) = (1-d)\,\sigma_{j,t}^{sched} + d\,\sigma_{j,t-1}^{\text{in force}}, \qquad d \in [0,1], \tag{30}$$

where d is the smallest deferral that makes the bank plus this year's net credits non-negative when the credit price is at the cap. Deferral is considered only when the bank is expected to run short at the scheduled standard, with expected net credits that include the changes already known, and holders already expect the cap price. The scheduled path resumes the next year. Two approximations are involved: the statutes' forecasts do not condition on price, so "supply at the cap price" stands in for potential supply; and very short deferrals can be implemented only through a temporary standard, not by keeping last year's standard for a full compliance period. A rule that allows only whole-year deferral has no stable solution when a program sits at its cap, because a full deferral pulls the price below the cap and the next year's forecast then calls for none.

B.2 The fuel market in year t

Fuel pools are gasoline G, diesel D and jet fuel J. In each region, each pool combines a fossil component with biofuel pathways $i = (f,k,j)$, indexed by fuel f, feedstock group k and region j: corn and sugarcane ethanol in gasoline; biodiesel (BD) and renewable diesel (RD) in diesel; and sustainable aviation fuel (SAF) made by hydroprocessing of the same lipids (HEFA-SAF) or from ethanol (alcohol-to-jet, ATJ, with or without carbon capture). Lipids form two groups: soybean and other vegetable oils, and fats (animal fat, used cooking oil and distillers' corn oil). The year's market is a mixed complementarity problem. Time subscripts are dropped within a year where unambiguous.

Producers are price takers. A pathway produces in region j only if its cost covers its value:

$$0 \le Q_i \;\perp\; c_{f,t}^{op} + \theta_{i,t}P_k + \phi_g(u_g) + \rho_g + \nu_{f,j} + \tau_i + w_i^{A1} + \mathbf{1}\{i \in \mathcal{H}\}\big(\psi_t + \rho_t^{SAF}\big) + \mu_i^{cc} - \big[\varphi_f P_{F(f),j}^{conv} + V_i\big] \;\ge 0, \tag{31}$$

where c^{op} is the variable operating cost per gallon (falling with cumulative output for hydroprocessing), θ_i the feedstock yield (8.2 lb of lipids per gallon of RD, 9.35 per gallon of HEFA-SAF, 7.55 per gallon of BD), P_k the price of feedstock group k, ϕ_g and ρ_g the utilization cost and capacity rent of technology group g, ν the shadow price of a blend wall, τ_i the cost of delivering RD to an LCFS state, w_i^{A1} the marginal cost of departing from the observed feedstock mix (below), $\mathcal{H}$ the HEFA-SAF pathways, ψ_t and ρ_t^{SAF} the marginal cost of shifting hydroprocessing output toward jet and the rent on SAF-capable capacity (below), μ_i^{cc} the rent of California's cap on crop-oil-based fuels (from 2028), and φ_f the value of a gallon relative to the fossil fuel it replaces. The policy value of a gallon is

$$V_i = \mathrm{EV}_i\big(P_{D4},\ P_{D6} + \mu_t^{ADV} \text{ or } P_{D6}\big) + P_j^L\,(\sigma_{F,j,t} - CI_i^L)\,ED_i \cdot 10^{-6} + z_{i,t}^{45Z} + z_{i,t}^{state}, \tag{32}$$

the RINs it generates (EV_i per gallon), its LCFS credits or deficits at the program's carbon intensity for that fuel and feedstock group CI_i^L, the federal clean fuel production credit and state credits.

Renewable diesel delivered to a clean fuel state bears a cost that rises with the volume delivered there,

$$\begin{gathered}\tau_j = c_j + d_j X_j + \mathbf{1}\{j = CA\}\, e\, E_{CA}, \qquad X_j = \textstyle\sum_{k \in RD_j} Q_k + IMP_j^{RD}, \\ 0 \le E_{CA} \ \perp\ E_{CA} - X_{CA} + \bar{s}\, Q_{CA}^D \ge 0,\end{gathered} \tag{33}$$

which stands in for terminal capacity, logistics and contracts that the model does not represent. Without it, RD would move all-or-nothing to whichever program pays most. The slopes d_j are estimated on quarterly deliveries, 2023Q1–2025Q4 for California and Oregon and 2024Q3–2025Q4 for Washington. In California the slope steepens, by $e = \$2.396$/gal per billion gallons, once RD exceeds $\bar{s} = 72.13\%$ of the state's diesel pool Q_{CA}^D, the highest share in 29 quarters of CARB data (2025Q1). I interpret the kink as competition from other outlets for renewable diesel (Canada, Europe, Oregon and Washington) and logistics into the state, not as a technical blend limit: renewable diesel can replace fossil diesel entirely. Without the kink, the model fills California's diesel pool with RD from 2027 and prices California diesel below the rest of the country, which the state has never seen; a run without the kink is a sensitivity. The slopes d_j are the least-anchored parameters in the fuel market. Calibrated to one year's deliveries, d_j is close to that year's LCFS premium spread over that year's RD volume, and it decides whether additional national support raises deliveries to the clean fuel states or lowers their credit prices (cases (iii) and (iv) of Proposition 1). Results are therefore also reported at half and at twice the estimated slopes.

Fossil gasoline, diesel and jet fuel are supplied perfectly elastically at the national wholesale price $PC_{F,t}$, which follows a single rule that links observed 2026 prices, EIA's Short-Term Energy Outlook and the Annual Energy Outlook. The rack price of fossil gasoline and diesel adds a state

price differential and the RFS and LCFS compliance costs per gallon,

$$P^{conv}_{F,j} = PC_{F,t} + \Delta^{fos}_{F,j} + w^{RFS}_t + \mathbf{1}\{j \in \mathcal{L}\}\, P^L_j\, (CI_{F0,j} - \sigma_{F,j,t})\, ED_F \cdot 10^{-6}, \qquad F \in \{G, D\}, \quad (34)$$

where $\mathcal{L} = \{CA, OR, WA, NM\}$, P^L_j is program j's credit price (\$ per t CO_2e), $CI_{F0,j}$ the fossil carbon intensity and $\sigma_{F,j,t}$ the standard (gCO_2e/MJ), ED_F the energy density (MJ/gal) and w^{RFS}_t the RFS compliance cost per gallon (equation (29)). The differential $\Delta^{fos}_{D,j}$ is the observed wholesale premium of fossil diesel in each state over the national benchmark (\$0.070/gal in California, \$0.031 in Oregon and Washington, \$0.099 in New Mexico, zero elsewhere); it is a real logistics and specification cost, so it enters welfare as a resource cost. The gasoline differential is zero. Fossil jet fuel carries neither obligation. In the four LCFS states fossil diesel enters as a complementarity, so renewable diesel can in principle fill the whole diesel pool.

Capital is sunk: producers pay operating cost, not full cost, on existing capacity, and the return to capital is the capacity rent $\rho_g \geq 0$, paired with the capacity limit as a complementarity. Running a plant harder costs more as utilization rises,

$$\phi_g(u_g) = a_g \left(\frac{u_g - u_{0,g,t}}{u_{ref,g,t} - u_{0,g,t}} \right)^+, \quad (35)$$

for hydroprocessing (RD and HEFA-SAF together, exports included), and a two-sided version for biodiesel that lets plants run below their reference rate while covering only variable cost. The curves are fitted to monthly U.S. production, and a fixed operating cost is charged on nameplate capacity whether or not it runs.

Each plant can in principle use either lipid group, and with constant costs the cheapest group would take a whole state's production, switching from year to year between corners. Observed feedstock mixes move slowly. I therefore add a quadratic cost of departing from the observed 2025 mix, in the spirit of positive mathematical programming (Howitt, 1995; Heckelei and Wolff, 2003). For each nest $n = (f, j)$ of a fuel $f \in \{\text{BD}, \text{RD}, \text{HEFA-SAF}\}$ in region j, with volumes Q_k of the two groups, total $Q_n = \sum_k Q_k$ and observed 2025 shares s_k,

$$C_n = \sum_k a^{A1}_k Q_k + \frac{g_n}{2} \sum_k \big(Q_k - s_k Q_n\big)^2, \qquad g_n = \frac{1}{2\, c\, \bar{Q}_n}, \qquad \textstyle\sum_k s_k a^{A1}_k = 0, \quad (36)$$

and $w^{A1}_k = \partial C_n / \partial Q_k$. The intercepts a^{A1}_k reproduce the observed 2025 shares, and $\bar{Q}_n$ is the nest's observed 2025 volume, fixed in every year and counterfactual. The cost penalizes changes in the mix, not in scale: total volume in a state is still set by the delivery cost and the utilization curve. With two groups the fat share responds to the gap between the margins of the two groups as $\partial s / \partial(\text{gap}) = c\, \bar{Q}_n / Q_n$, so at reference volumes every state's share moves by c per \$1/gal of margin gap. One response c is shared by RD and BD and estimated on the history (Appendix D.6). Within

each group the composition (for example soybean against canola oil) is held at its 2025 level, and the group's credited carbon intensity is backed out of the programs' quarterly data. C_n is a resource cost; payments in excess of it are producer rent.

Hydroprocessing plants can shift output between RD and jet only if they have SAF-capable equipment. National HEFA-SAF output is limited by SAF-capable capacity,

$$\textstyle\sum_{i\in\mathcal{H}} Q_i + \text{slack}_t^{SAF} = K_t^{SAF}, \qquad 0 \le \rho_t^{SAF} \perp \text{slack}_t^{SAF} \ge 0, \tag{37}$$

which is 0.46 billion gallons in 2025 (U.S. Energy Information Administration, 2025) and grows with announced projects to 2028 and with the investment rule afterwards. Within that capacity, raising the jet share of a plant's output has a rising cost, $\psi_t = (h/H_t)\sum_{i\in\mathcal{H}} Q_i$, with H_t hydroprocessing nameplate capacity and h = \$0.34/gal; it represents the hydrogen cost of the jet-heavy configuration, since yield losses are already in θ_i. A product-slate limit keeps the jet share at or below that of the jet-maximizing HEFA configuration in Baldwin et al. (2022, Fig. 1), where jet is about 55% and renewable diesel about 25% of total product.

Demand for gasoline, diesel and jet fuel is isoelastic in the retail price,

$$Q_{F,j,t}^{dem} = \xi_{F,t}\, A_{F,j}\, \frac{B_F(s_{j,t}^{EV})}{B_F(s_{j,2025}^{EV})}\left(P_{F,j,t}^{dem} + W_{F,j,t}\right)^{\beta_F}, \tag{38}$$

where P^{dem} is the energy-weighted blend of fossil and biofuel prices in the pool, W a fixed real wedge for taxes, retail margins and state premiums (so compliance costs pass through fully to retail prices), and $B_F(\cdot)$ the share of the fleet using fuel F that still burns liquid fuel, evaluated at the electric shares from the vehicle block. The elasticities are -0.37 for gasoline and -0.07 for diesel (Coglianese et al., 2017; Dahl, 2012), and -0.26 for jet fuel on the rack price. They are long-run utilization responses with the fleet held fixed; changes in fleet composition come through B_F. Applied to the oil-price spike of 2026, a long-run elasticity would cut 2026 gasoline demand by about a tenth, which did not happen. The national multiplier $\xi_{F,t}$ therefore sets 2026–27 demand at the levels of EIA's Short-Term Energy Outlook and returns linearly to one by 2030; it does not vary across counterfactuals.

Ethanol's blend value falls steeply above a kink in its share e_j of the gasoline pool, $P_{E,j} = \varphi P_{G,j}^{conv}\left[1 - \mathbf{1}\{e_j > k_j\}(2^{(e_j-k_j)/w} - 1)\right]$, with $k_j = 0.10 + 0.05a_j$, where a_j is the E15 share of gasoline and $w = 0.0036$ is calibrated to the observed blend share; a floor keeps the share at or above 10%. Biodiesel is limited to a share of each region's diesel (7.5% in California, 12% in the other LCFS states, 20% elsewhere). State biodiesel mandates and incentives outside the LCFS states are included as step complementarities.

Feedstock markets are national. Supply of soybean and other vegetable oil to fuel is isoelastic with elasticity 1.5; North American animal fat has elasticity 3 and grows 0.8% a year to 2035 and

0.5% after; distillers' corn oil adds supply to the fat market with elasticity 0.5. Imported fats and used cooking oil are a separate residual supply with elasticity 5. Since 2026, fuel made from imported feedstock earns no 45Z credit, so imported fat trades at a discount to North American fat equal to the credit it forgoes. Price floors at pre-RFS real levels ($0.35/lb for soybean oil, $0.28/lb for fat) keep feedstock prices from falling below what other uses would pay. Corn and imported sugarcane ethanol have their own supply curves.

Imports of RD, BD and SAF enter as fixed volumes: they count toward pools, blend walls, RIN supply and LCFS balances, but use no U.S. feedstock or capacity. Exports of RD (mostly to Canada) and BD are exogenous and do use U.S. feedstock and capacity.

The 45Z clean fuel production credit pays $z_{i,t}^{45Z} = \bar{z}_t \max\{0, (47.39 - CI_i^{Tax})/47.39\}$ per gallon for fuel made from North American feedstock, with $\bar{z}_t \approx \$1.03$ (2024 dollars) and carbon intensities from 45ZCF-GREET without indirect land use change, as the One Big Beautiful Bill Act (OBBBA) provides from 2026. Under current law the credit ends after 2029, there is no SAF premium, and fuel from imported feedstock and sugarcane ethanol does not qualify. ATJ with carbon capture can claim 45Q instead. State SAF credits (Illinois, Minnesota, Nebraska, Iowa, Hawaii) are step schedules with budget caps, solved as complementarities on each step.

B.3 The year-t complementarity problem

Table 10 lists the conditions of the year-t fuel market and the variable paired with each. Given the state at the start of the year and the credit prices (LCFS prices from holders' rule, or from (16) when the year clears; RIN prices from the storage problem), the problem is a mixed complementarity problem in the listed variables.

Table 10: Conditions and paired variables of the year-t market

Condition	Variable	Equation
Zero profit of pathway i	output $Q_i \geq 0$	(31), (32)
Pool balance, fuel F, region j	retail price $P_{F,j}^{dem}$	(46), (38)
Fossil supply at the rack price	fossil volume $Q_{F0,j} \geq 0$	(34)
Feedstock balance, group k	price $P_k \geq \underline{P}_k$	(47)
Capacity of technology group g	rent $\rho_g \geq 0$	Section 4.2
SAF-capable capacity	rent $\rho_t^{SAF} \geq 0$	(37)
Utilization cost	ϕ_g	(35)
Feedstock-mix cost	w_i^{A1}	(36)
Delivery cost and California kink	τ_j, $E_{CA} \geq 0$	(33)
Blend walls and biodiesel share limits	shadow price $\nu_{f,j} \geq 0$	Appendix B.2
Crop-oil cap (California, from 2028)	rent $\mu^{cc} \geq 0$	Section 4.2
RFS requirements: BBD, advanced, total	$P_{D4} - P_{D6} - \mu^{ADV}$, μ^{ADV}, P_{D6}	(48)
LCFS balance, program j	P_j^L, surplus s_j^+, issuance s_j^-	(16), (49)
State SAF credit steps	step prices	Appendix B.2

B.4 Capacity investment

Hydroprocessing capacity (RD and HEFA-SAF) is endogenous from 2029; earlier capacity is under construction. Four types of investment add or convert capacity: (a) conversion of a petroleum refinery, (b) expansion of an operating biofuel plant, (c) a greenfield plant, and (d) conversion of operating hydroprocessing capacity to produce SAF. In each year, each potential investor of type τ decides whether to commit now or wait. This is competitive, single-agent optimal stopping with logit shocks of scale σ, as in Rust (1987), with entry costs estimated from 46 capacity decisions. Investors take prices as given, which fits a national commodity market whose prices are set by the RFS and the credit markets; strategic interaction among entrants is not modelled.

The value of committing per gallon of annual capacity is

$$V^{c}_{\ell,t} = \sum_{s=L}^{L+19} \delta^{s} \Big[u^{e}_{t+s} \big(\mathbb{E}_t \, m_{t+s} + F_\ell / u_0 \big) - F_\ell \Big] - \kappa_{\ell,t}, \qquad \ell \in \{a, b, c\}, \tag{39}$$

with a construction lag $L = 3$, a 20-year life, $\delta = 1/1.07$ and κ_ℓ the entry cost (capital cost plus the hurdle the data imply, falling with capital learning). $F_\ell = \varphi \cdot capex_\ell$ is the fixed operating cost, charged on nameplate capacity at $\varphi = 4\%$ of capital cost a year (National Renewable Energy Laboratory, 2024; Davis et al., 2018); it is written so that the value is unchanged at the reference utilization $u_0 = 0.70$, where the history has the plants, and falls when expected utilization is lower. The margin m_t is the RD operating margin (fuel value plus RIN, LCFS and production credits, less feedstock, operating and delivery cost and the feedstock-mix cost w^{A1}) averaged over destinations with weights equal to each region's share of RD deliveries, net of the 2026–28 transitional terms and of the gap between the 2026–28 utilization ramp and the long-run utilization curve. Expected utilization u^{e}_{t} is the discounted average of the utilization the model solves for over the plant's life, so additional capacity lowers the value of further capacity. RIN and feedstock prices are expected to stay at current values; the LCFS component follows the path the rule implies and the scheduled standards; statutory credits that are due to expire are expected to continue with probability p, so $\mathbb{E}_t z_{t+k} = z^{law}_{t+k} + p\,(z_t - z^{law}_{t+k})$. The main specification sets $p = 1$ while a credit is in law and 0.18 during a lapse; estimated, they are 1 [0.67, 1.00] and 0 [0, 0.31] (Section 5.2).

Waiting has value because the margin may rise. Write $V^{c}_{\ell,t}(z)$ for the value of committing as a function of the margin index $z = m_t$. The value of waiting solves

$$V^{w}_{\ell,t}(z) = \delta \, \mathbb{E}\Big[\zeta \ln\Big(e^{V^{c}_{\ell,t+1}(z')/\zeta} + e^{V^{w}_{\ell,t+1}(z')/\zeta} \Big) \Big], \qquad z' = z + \Delta_{t+1} + s_z \, \varepsilon, \quad \varepsilon \sim N(0,1), \tag{40}$$

where Δ_{t+1} is the expected drift of the margin from the scheduled LCFS standards, holders' LCFS price path and the statutory credits, and s_z is the standard deviation of the margin index in the history (the rest of the margin is a random walk). Once the drifts end, V^{w} is stationary. A potential

investor commits with probability

$$P_{\ell,t}(z) = \Big[1 + \exp\big(-(V^{c}_{\ell,t}(z) - V^{w}_{\ell,t}(z))/\zeta\big)\Big]^{-1}, \tag{41}$$

and expected commitments in year t are the sum over the pools of potential investors of the commitment probabilities times the expected capacity per commitment.

The gain from converting a gallon of hydroprocessing capacity to SAF is the margin of SAF over RD actually earned on SAF sales, net of the RD output given up,

$$g_t = \max\big(0,\ S_t - \vartheta R_t\big), \qquad V_{saf,t} = u_{saf} \sum_{s} \delta^s\, \mathbb{E}_t g_{t+s} - \kappa_{saf,t}, \tag{42}$$

where S_t is the SAF-over-RD margin weighted by realized HEFA-SAF sales (zero when no SAF is sold), R_t the RD margin, $\vartheta = 1.125$ gallons of RD capacity lost per gallon of SAF capacity (Gerveni et al., 2025) and $u_{saf} = 0.70$. Every operating hydroprocessing plant is a potential converter, the conversion takes two years, and the decision follows the same logit rule. Conversion and new capacity share σ.

New capacity is not chosen to be SAF-capable in the model. Half of the refinery conversions of 2017–23 were built SAF-capable at a time when SAF earned nothing over RD, which no margin index can rationalize; in the data the choice looks like a nearly costless design option. Future refinery conversions are therefore SAF-capable in the historical proportion (0.4535 of conversion capacity), with $h_b = 0.2$ of their nameplate able to produce jet, the capacity-weighted share of Rodeo, Great Falls and Kapolei. SAF-capable capacity then evolves as

$$K^{SAF}_{t} = \min\Big(K^{H}_{t},\ K^{SAF}_{t-1} + A^{saf}_{t-2} + h_b \cdot 0.4535 \cdot A^{a}_{t-3} + \Delta^{proj}_{t}\Big), \tag{43}$$

where A^{saf} and A^{a} are conversion and refinery-conversion commitments, Δ^{proj}_{t} projects already announced (2026–28) and K^{H}_{t} hydroprocessing nameplate capacity.

B.5 Vehicles and charging

The vehicle block extends Cole et al. (2023) to light-, medium- and heavy-duty vehicles, five regions and the channels through which LCFS revenue reaches vehicle buyers. The electric share of new sales of class v in region j is logit in the ratio of total costs of ownership and, for light-duty vehicles, in the charging network:

$$\ln\frac{x_{jvt}}{1 - x_{jvt}} = \alpha_{jv} - A^{ACT}_{jvt} + \beta_{1v} \ln\frac{TCO^{EV}_{jvt}}{TCO^{ICE}_{jvt}} + \mathbf{1}[v = LDV]\Big(\beta_2 \ln\frac{N^{L2}_{j,t-1}}{S^{EV}_{j,t-1}} + \beta_3 \ln N^{DC}_{j,t-1}\Big), \tag{44}$$

Vehicle classes follow FHWA gross-weight classes: light-duty is classes 1–2 (including class 2b

vans, with light-duty parameters), medium-duty classes 3–6, and heavy-duty classes 7–8, with heavy-duty costs dominated by class 8 tractors. Medium-duty costs, subsidies, sales, shares and mileage weight class 3 and classes 4–6 by sales. The charging block and the electricity credits use the same classes. For given prices each agent's choice has a closed form or a scalar root.

In equation (44), S^{EV} is the electric stock and N^{L2}, N^{DC} the Level-2 and DC fast public stations. For light-duty vehicles, β_1 and β_3 are estimated with $\beta_2 = \beta_3/2$ (Section D.6). For trucks, $\beta_{1v} = \eta_v / w_{p,v}$ maps the literature price elasticity η_v into the TCO coefficient, where $w_{p,v}$ is the purchase price's share of TCO: the elasticity of the electric truck's TCO with respect to its purchase price, computed by the model at the 2025 anchor (0.779 for medium-duty and 0.752 for heavy-duty trucks). The implied purchase-price elasticity is therefore η_v. The medium-duty η of -2.5 is the approximate median of light-duty estimates chosen by Cole et al. (2023, p. 318 and fn. 2), applied to medium-duty trucks; the heavy-duty η of -2.65 is the midpoint of the range of -2.5 to -2.8 in Weber (2024, p. 16). A^{ACT} removes the component of truck sales in California, Oregon and Washington attributable to the Advanced Clean Trucks rule, which is no longer enforced. The TCO includes the purchase price net of incentives and resale value, discounted fuel or electricity cost, maintenance and registration fees. Households discount at 15% over 15 years, an assumption; 7% and 25% are sensitivities. Fleets discount at 7% over a five-year holding period, as in Zhao et al. (2026), who take these values from Ledna et al. (2024, Table 1); Basma et al. (2023, p. 7) use the same rate and period for class 8 trucks. For heavy-duty trucks the five years are Ledna et al.'s horizon for classes 7–8. For medium-duty trucks they are an assumption: Ledna et al. (2024) use three years for class 3 and four for classes 4–6. Residual values after five years are, for heavy-duty trucks, 35% for diesel trucks and, for battery-electric trucks, 49% for the battery (the model-year 2030 value of Basma et al., 2023) and 35% for the rest of the truck (Zhao et al., 2026, Supplementary Table 11). For medium-duty trucks they are 55% for diesel trucks, derived from the depreciation parameters of Burnham et al. (2021, Table 3.11), and 40% for battery-electric trucks, an assumption. Federal purchase credits end after September 2025. Rebates funded by utilities' residential LCFS credits lower the purchase price. State purchase incentives (California's HVIP, Washington's sales-tax exemption and WAZIP, Oregon's rebates, New Mexico's tax credit) enter only to the extent of announced funding and are zero after the last funded year. A truck's combined purchase incentives are capped at 90% of the electric truck's price, the HVIP stacking rule; credits the owner earns directly are outside the cap. The intercepts α_{jv} match 2025 sales by region and class with the incentives in force in 2025 and are held fixed in every counterfactual.

Public Level-2 and DC fast stations enter freely at the regional level, as in Zhou and Li (2018) and Springel (2021), until a station's annual margin, including the LCFS credits it earns, equals

the user cost of building now rather than next year,

$$\frac{D^k_{j,t}}{N^k_{j,t}+X^k_{j,t}}\Big[\mu^k_j (N^k_{j,t})^{-\eta_k}+P^L_{j,t}e_{j,t}\Big]=(1-\tau)\Big[(C^k_t-g^k_{j,t})-\frac{(1-\varrho)(C^k_{t+1}-g^k_{j,t+1})}{1.07}\Big],\qquad \eta_k=\frac{1}{\upsilon_k}-1, \tag{45}$$

where D^k is public charging demand, e credits per kWh, C^k the installed cost, g^k grants and tax credits, ϱ depreciation and X^k stations built outside this market. Next year's cost is evaluated under the law known in year t: the federal charger credit (30C) is expected to continue in 2025 and earlier years, as the Inflation Reduction Act then provided, and to end in mid-2026 from 2026 on, after OBBBA. The entry elasticities υ_k are estimated by simulated method of moments (Section D.6). Depot chargers for trucks have no entry decision; their capital cost is reduced by announced grants (30C through mid-2026, California's EnergIIZE and utility make-ready programs, Oregon's program).

The electric stock accumulates sales by vintage. The block returns the electric shares of the fleet, which enter fuel demand (38), and electricity credits $\Gamma^{EV}_{j,t}$, computed per kWh as $(\sigma^*_{j,t}-CI^{el}_{j,t}/EER_v)\,EER_v\cdot 3.6\cdot 10^{-6}$ at the standard in force (after acceleration or deferral), which enter the LCFS balance (16). Electricity credits are about a fifth of California's credit generation, so the bank after 2030 depends on the pace of electrification.

In the setting of Section 3 with one region, suppose electric vehicles generate E credits. If $\lambda=\bar{P}$, the fuel price, biofuel and fuel use are set by $\bar{P}$ alone, and a marginal change in E changes only the shortfall issued at the cap: $\partial X^+/\partial E=-1$ and $\partial Z/\partial E=0$. Below the cap, $\partial\lambda/\partial E<0$ and, by equation (9),

$$\frac{\partial Z}{\partial E}=1+\sigma\,D'\left(c_0-\sigma\right)\frac{\partial\lambda}{\partial E}\;>\;1.$$

Holding the electric fleet's direct effect on fuel demand fixed, each credit from electricity permits at least one more unit of fuel-cycle emissions below the cap, because it also lowers the credit price and the fuel price; at the cap it has no effect on fuel emissions.

B.6 Equilibrium, solution and welfare

Given the state of the economy at the start of year t (capacity K^H_t and K^{SAF}_t, vehicle fleets, LCFS banks and expected exhaustion years, the RIN bank and the standards in force), the fuel market clears when the zero-profit conditions (31) hold for every pathway, fuel demand is (38), and the following balances hold, each paired with its price. For each fuel pool F and region j,

$$Q_{F0,j}+\textstyle\sum_{i\in F,j}\varphi_{f(i)}\,Q_i=Q^{dem}_{F,j}. \tag{46}$$

For each feedstock group k, use equals supply, with the floor price $\underline{P}_k$ (Appendix B.2),

$$0 \le P_k - \underline{P}_k \;\perp\; S_k(P_k) - \textstyle\sum_{i:\,k(i)=k} \theta_i Q_i - \theta X_k^{exp} \;\ge 0. \tag{47}$$

For each technology group g, output is within capacity, $0 \le \rho_g \perp K_{g,t} - \sum_{i \in g} Q_i - X_g^{exp} \ge 0$. The three nested RFS requirements clear with the year's draw on the RIN bank, $\Delta B_t = B_{t-1} - B_t$, which counts toward the total requirement:

$$\begin{aligned} 0 \le P_{D4} - P_{D6} - \mu_t^{ADV} \;&\perp\; \textstyle\sum_{i \in BBD} \kappa_i Q_i - \pi_t^{BBD} \mathrm{Base}_t \ge 0, \\ 0 \le \mu_t^{ADV} \;&\perp\; \textstyle\sum_{i \in ADV} \kappa_i Q_i - \pi_t^{ADV} \mathrm{Base}_t \ge 0, \\ 0 \le P_{D6} \;&\perp\; \textstyle\sum_{i} \kappa_i Q_i + \Delta B_t - \pi_t^{RF} \mathrm{Base}_t \ge 0, \end{aligned} \tag{48}$$

with RIN generation net of exports and $BBD \subset ADV$. Each LCFS program clears by (16), with net credit generation

$$\begin{aligned} N_{j,t} = \; & \textstyle\sum_{i \in j} (\sigma_{F(i),j,t} - CI_i^L)\, ED_i\, Q_i \cdot 10^{-6} \\ & - (1 - \iota_{j,t}) \textstyle\sum_{F \in \{G,D\}} (CI_{F0,j} - \sigma_{F,j,t})\, ED_F\, Q_{F0,j} \cdot 10^{-6} + \Gamma_{j,t}^{EV} + \Gamma_{j,t}^{fix}, \end{aligned} \tag{49}$$

where Γ^{fix} are credits from fixed sources. Appendix B.3 lists the full problem, variable by variable.

Because LCFS holders do not have rational expectations, the object I compute is a temporary equilibrium path: each year's prices clear the year's markets given expectations formed from the past, rather than a rational-expectations path.

Definition 2. Given the 2025 state and the exogenous paths (oil prices, standards as scheduled, statutory credits, exemption scenario), a temporary equilibrium path is a sequence of annual allocations, prices, expected exhaustion years, banks, capacity and fleets for 2026–2045 such that

(i) in every year, the fuel market clears as above, given that year's capacity, fleets, banks and standards;

(ii) from 2027, each LCFS price satisfies holders' rule (17)–(19) given their expected exhaustion years, and these and the banks are updated from the realized net credits and any issuance or surplus, $B_{j,t} = B_{j,t-1} + N_{j,t} + s_{j,t}^- - s_{j,t}^+$;

(iii) the RIN bank path solves the storage problem (27) on the net-supply curves of the path;

(iv) capacity follows commitments (21) evaluated on the path, with the construction lag;

(v) fleets and stations follow (44) and the entry conditions at the path's prices; and

(vi) acceleration and deferral decisions (30) are those the rules imply on the path.

Each year's fuel market is a mixed complementarity problem, solved with PATH (Dirkse and Ferris, 1995); Gurobi then checks the solution against every constraint of the original formulation. The feedstock-mix cost (36) keeps the problem linear with a symmetric positive semidefinite Jacobian, and removes the ties in feedstock choice that made solutions depend on the starting point. Within a year, each LCFS price is a one-dimensional fixed point found with the price pinned, and the four programs are swept in turn. The blocks are coupled by damped iteration over the whole 2026–45 path: prices from the previous round give vehicle sales and credits, learning effects on cost, and capacity; the RIN storage problem is re-solved on the updated supply curves; and every year is re-solved with the new RIN path. The iteration stops when, between rounds, LCFS prices move by less than \$0.5/t, rack prices by less than \$0.005/gal, capacity by less than 0.02 billion gallons, deferral amounts by less than 0.02 years and the RIN storage conditions hold within \$0.05 per RIN. Acceleration and deferral decisions are then re-checked on the converged path; if one changes, the iteration resumes. Prices are unique in every check; some quantities are not (when two uses of a fuel are equally valued), so I report aggregates.

C Data and Parameters

Table 11 lists the parameters that matter most for the results, ordered by my judgment of their influence on the headline numbers (based on the sensitivity runs). For each it gives where the value comes from: estimated on the history, calibrated to a single year, exactly identified by 2026 targets, borrowed from the literature, set by statute, or assumed. Appendix C describes the data sources.

The estimated investment parameters in force in every reported run are the joint estimates with the expansion and greenfield entry costs pooled ($\zeta = 2.79$, $\kappa_a = 20.25$, $\kappa_b = \kappa_c = 10.5$, $\kappa_{saf} = 3.0$); the SAF conversion cost falls with capital learning, $\kappa_{saf,t} = 3.0 - 1.38\,(1 - \ell_t)$, to 2.75 in 2045. The two engineering-cost rows of Appendix G.7 replace them with capital costs (1.9, 3.44, 3.4, 1.38).

Four sets of parameters are calibrated to 2025, the last complete year, by matching observed quantities at observed prices: the level of the hydroprocessing utilization curve, the feedstock-mix intercepts a^{A1}, the corn oil supply anchor and the 2025 RIN bank. A single year identifies a level, not a slope; in a calibrated model with quadratic costs, one year's first-order conditions are an underdetermined system, and different slope choices that fit the base year equally well imply different simulation behavior (Heckelei and Wolff, 2003, pp. 29–30). I therefore checked each one-year parameter against its own history, and report the multi-year value as a sensitivity where the history is informative.

The RD delivery slopes have a structural break around 2022. California's annual slope was \$2.1–3.5 per gallon per billion gallons in 2016–21 and \$0.02–0.18 in 2023–25; Oregon's fell from 67–89 to 2.5–7.1. The annual slopes move with the year's credit price: the elasticity of the log slope with respect to the log of credit value per gallon delivered is close to or above one in the annual

Table 11: Parameters: sources and identification

Parameter	Value	Origin	Data and precision	Sensitivity
Expectations				
Exhaustion horizon K	2 (Rule B)	Estimated, weakly	CA and OR, quarterly 2016–25 (2.33 [1.58, 3.14]); annual data do not identify it under Rule A (Section 5.2)	$K = 1.58$, 3.14, 3.7; Rule A
Response to balance surprises γ_{CA}; pooled g (OR, WA, NM)	0.753; 0.109	Estimated	Same	$\gamma = 0.32$ (model-free)
Investors' extension probability p; lapse value	1; 0.18	Set; estimated as 1 [0.67, 1.00] and 0 [0, 0.31]	46 decisions, 2014–25 (15 in lapses); lapse value from RIN prices	$p = 0.5$; $p = 0$
Fuel market				
California RD kink	share 0.7213, slope 2.396	Observed maximum; fitted	CARB quarterly 2019Q1–2026Q1; slope rests on 2 quarters (s.e. about 1.0)	No kink; slope $\times\frac{1}{2}$, $\times 2$; kink share as a parameter with an interval
2026 transitional terms (CA, OR, WA absorption; imported fat supply)	CA 0.131, OR 7.52, WA c 0.058	Exactly identified	Four 2026 targets	Kept permanently; faded to 2023–25 values
End-2025 RIN bank	3.21 bn (D4 1.40)	Calibrated	EMTS (21 Sept 2026); EPA has not published the bank	2.27–3.42; 6.0
Feedstock share response c (RD and BD)	0.283	Estimated	CA, WA, NC fat shares 2016–25 (about 39 observations)	RD response at the ends of its two-parameter interval
Hydroprocessing and BD utilization curves	HEFA a 1.381; BD a 2.761, u_0 0.655, kink 0.83	Estimated; level calibrated	HEFA slope: EIA monthly 2022–26 with year effects; BD level: 2016–24	HEFA slope 0.10; BD slope 0.285 and 0.017
RD delivery slopes d_j (CA, OR, WA)	0.100; 3.83; 0.68	Estimated	Quarterly 2023Q1–2025Q4 (WA 2024Q3–2025Q4), intercepts fixed	2025 single-year values
Feedstock supply elasticities	soy 1.5, fat 3, imports 5, corn oil 0.5	Estimated / literature	Out-of-sample fit of price paths; residual-supply formula for imports (Baker and Bresnahan, 1988)	Imports 2, 10
Fuel demand elasticities	G −0.37, D −0.07, J −0.26	Literature	Coglianese et al. (2017); Dahl (2012)	Retail −0.20
Oil price path	STEO–AEO rule	Data, EIA	—	AEO high and low price; Monte Carlo
Investment				
Logit scale ζ; entry costs κ_a, κ_{bc}	2.79; 20.25; 10.5	Estimated	1,492 decision-years, 46 decisions	Capacity fixed; $p = 0$
SAF conversion cost κ_{saf}	3.0 [−1.35, 9.70]	Estimated	6 conversion events; bootstrap relative s.e. about 100%	Engineering cost (about $1.38 plus lost RD capacity)
Fixed O&M share φ	0.04	Literature	TEA, 2.7–6.5% (National Renewable Energy Laboratory, 2024; Davis et al., 2018); not identified by history	0.03; 0.05
SAF-capable share of new conversions h_b; conversion loss ϑ	0.2; 1.125	Calibrated; literature	Rodeo, Great Falls, Kapolei; Gerveni et al. (2025)	$h_b = 0.5$; $\vartheta = 1$
Vehicles				
LDV price and network coefficients b_1, b_3	−4.92 (0.35); 0.31 (0.10)	Estimated (SMM)	2021–25 sales shares and station counts	Price elasticity −2.8
Truck price elasticities	MDV −2.5, HDV −2.65	Literature	MDV: light-duty median, Cole et al. (2023, p. 318); HDV: midpoint of −2.5 to −2.8, Weber (2024, p. 16)	−6; −10
Statutory parameters (45Z, 45Q, RIN values, RFS percentages, LCFS schedules, caps, AAM)	Section 2	Statute	Federal Register; state codes	45Z, 45Q extensions are counterfactuals

data. A slope calibrated to a single year is therefore largely that year's LCFS premium spread over that year's RD volume, a residual rather than a fixed transport cost. The main specification instead estimates the slopes on quarterly data after the break (California 0.100, Oregon 3.83, Washington 0.68), and the 2025 single-year values are a sensitivity. On quarterly data the slope still co-moves with credit value per gallon (elasticity 1.29 in California, R^2 0.986), so when California's price rises toward its cap a fixed slope understates the marginal cost of absorbing more RD there; in the forward runs the kink in equation (33) carries this.

2025 is not a representative year for biodiesel. It was the first year of 45Z, which replaced the $1 blenders' credit and excluded imported feedstock, and biodiesel utilization fell to 3.6 standard deviations below its 2016–24 mean. The level of the biodiesel utilization curve is therefore set from 2016–24, and 2025 is treated as a one-off shock year. The 2025 feedstock-mix intercepts for biodiesel also break from their 2016–24 values; their 2021–24 means are a sensitivity. The other one-year anchors (corn oil, the California RD feedstock intercept) are in line with their recent history.

The 2026 LCFS prices of California, Oregon and Washington are pinned (Section 4.1). Four 2026-specific terms are then exactly identified by four 2026 targets: the RD delivery terms of California, Oregon and Washington and the supply level of imported fat, which absorb the one-off shocks of 2026 (the oil price spike, OBBBA's feedstock-origin rule and tariffs, California's July 2025 amendments). They fade out by half in 2027 and fully in 2028. Holders' and investors' expectations include the fade-out as a known change. The other 2026 observations remain out of sample (Section 5.4). The RFS is not pinned: the D4 and D6 prices of 2026 are outcomes, and the RIN storage problem iterates with the fuel market until the 2026 D6 price used in calibration and the price the run produces agree within $0.05. I have not found a study in this literature that treats a shock year with terms that exist only in that year; the closest analogues start from an observed state and let the model take over (Borenstein et al., 2019; Bushnell et al., 2020; Fowlie et al., 2016).

EPA has not published the RIN bank after 2024. EPA's final 2026–27 rule uses an effective 2024 carryover of 3.60 billion RINs, which it assumes carries unchanged into 2026–27 (U.S. Environmental Protection Agency, 2026b, Table 1.8.2-1). The main specification uses 3.21 billion, with 2.27–3.42 and 6.0 as sensitivities; the 2026 D6 price is sensitive to this number. Expected small-refinery exemptions for decisions in 2025 are 0.78 billion RINs, from the September 2025 supplemental proposal. From 2026 the main specification uses the 2023–25 average exempted volume, 10.07 billion gallons; the final rule's 1.11 billion RINs (7.55 billion gallons) is a sensitivity (Appendix G.5) (U.S. Environmental Protection Agency, 2025, 2026c).

A few inputs have no public source and are labelled as assumptions with sensitivities: the share of new chargers in census tracts eligible for 30C (0.60; for reference, about two-thirds of the population lives in eligible tracts, U.S. Department of the Treasury, 2024), the split of 2025 HVIP redemptions by truck class (the 2023–24 request mix; public data give the request year, not the

redemption date), the shares of depot-charger grants going to electric trucks, and the class 3 counts outside California and Washington.

D Estimation

D.1 The price-expectation rule

From 2022 California's credits exceeded deficits every year and the bank quadrupled, yet the price stayed positive and then fell. A bank growing at a positive price is what Hotelling predicts when holders expect the standard to bind later, so the question is quantitative. Read through the cap and a holding cost, the price implies an expected exhaustion year, $t + \ln(\bar{P}/P_t)/\ln(1+r)$, and that year moved by decades over 2016–26 (Section 2.1). None of the three standard descriptions of expectations reproduces this history. A static market is rejected by inspection: the bank is positive and the price is not zero. Perfect foresight fails whether or not it is given hindsight: its quarterly price error over 2016–25 is 84.5 when holders know only the rules adopted by then and 80.9 with the rules as finally adopted, against 34.0 for the rule below. With the rules of each year it implies prices of \$34 in 2024–25, against \$63 and \$59; once the 2024 amendments are known it implies \$177 for 2025 and \$186 for 2026, against about \$63. Before the amendments holders priced more scarcity than the rules of the day implied, and after them much less than the amended rules imply. The third description, a market that prices the chance that the regulator relaxes or tightens the standard, does not fit, and the reason is itself informative (below).

I therefore describe holders' expectations with the rule (18)–(19), and I treat it as a forecasting rule for the credit price in the sense of Krusell and Smith (1998), as Gillingham et al. (2022) do for used-car prices: the rule maps the credit balance and the scheduled standards into the price holders expect, it is judged by whether it gets the direction and size of price responses to the balance right, and the equilibrium path is computed with it. Its parameters (γ_{CA}, the pooled small-program response g, K and the initial expected exhaustion years) are estimated by maximum likelihood on the observed prices and banks of California and Oregon, 2016–25, with normal errors on log prices and on banks. For a parameter vector ω, the historical model gives each program's price $P^L_{j,t}(\omega)$ and bank $B_{j,t}(\omega)$, and

$$\ln L(\omega) = \sum_{j\in\{CA,OR\}} \sum_t \Big[\ln\phi\Big(\frac{\ln P^{obs}_{j,t} - \ln P^L_{j,t}(\omega)}{s_P}\Big) + \ln\phi\Big(\frac{B^{obs}_{j,t} - B_{j,t}(\omega)}{s_{B,j}}\Big) - \ln s_P - \ln s_{B,j}\Big], \quad (50)$$

with ϕ the standard normal density and the error scales estimated with ω. The credit supply comes from the historical fuel market solved at each candidate price.

Three annual shift terms (2018, 2023 and 2024) enter the history only; they are not used in forward years. They fall in rulemaking years, but no dated announcement moved prices by the

amounts they imply (below), and they are not the rational response to the changes announced in those years: the 2018 term moves the expected exhaustion year by −23.5 years, while the announced change moves the perfect-foresight exhaustion year by +0.18. I describe them as the years in which the gap between the price and the statutory path opened. The forward runs, which give the paper's results, do not use them; the annual replay of the full model in Appendix E includes them in the history years (Table 16). The sample ends in 2025.

For the horizon K, on quarterly data, 2016–25 (40 observations for California), Rule B gives $\hat{K} = 2.33$ with a 95% interval of [1.58, 3.14]. The interval rests on extrapolation: no program in the sample came close to exhausting its bank. Without the three annual shift terms the estimate is $\hat{K} = 3.68$, [2.93, 4.76], which excludes the $K = 2$ used in the forward runs; that value lies within the interval of the estimate with the shift terms. The holding cost r is set, not estimated. Across r from 0.25% to 10%, $\hat{K}$ stays between 2.26 and 2.84, but $\hat{\gamma}$ moves roughly in proportion to $1/r$ (0.76 at 5%, 1.30 at 2.8%, 0.50 at 8%), and the likelihood rises without an interior maximum as r falls, so the data do not separate r from γ. What they pin down is close to the product $\gamma \ln(1+r)$, the response of the log price to a balance surprise, which is what the counterfactuals use. I fix r at 5% and report γ conditional on it. Evidence from other bankable-credit markets points to a short horizon. Prices are forward-looking and react to news within a day or two, but large rises come as jumps once scarcity falls within a limited horizon: in the 2013 RFS episode, D6 RIN prices jumped about one to two years before the RIN bank was projected to run out (Irwin and Good, 2013; Lade et al., 2018b); Acid Rain Program prices moved about five years ahead of the Clean Air Interstate Rule (Schmalensee and Stavins, 2013); rolling-horizon models of the EU ETS fit with horizons of about 10–13 years (Quemin and Trotignon, 2021; Sitarz et al., 2024); and RECLAIM, with no banking, is the $K = 0$ limit (South Coast Air Quality Management District Governing Board, 2001).

The main specification is Rule B (expected net generation as the three-year average, Nerlove, 1958) with $K = 2$, applied to all four programs. Rule A is the alternative; it fits the history about as well (AIC 9.9 against 10.6) but predicts $179 for California in 2026. Oregon, Washington and New Mexico share one response per unit of obligation. Oregon's bank never fell in the sample; the rule turns small surprises in a small credit pool into large revisions, and its backtest is worse than a random walk. I treat Oregon's price path as the least reliable of the four. Per percentage point of the obligation, the responses estimated separately for each program are similar: 0.134 in California, 0.112 in Oregon and 0.107 in Washington.

The holding cost is set at 5%; the data identify the product $\gamma \ln(1 + r)$, which is what the counterfactuals use. OPIS quotes for adjacent December deliveries imply a nominal carry of 5.7% a year ([5.3, 6.1], 2022–26). Net of expected inflation over the same days (the five-year breakeven rate, 2.3%) the real carry is 3.4% ([2.9, 3.8]); with the Cleveland Fed's one-year expectation it is 3.0%. Both are below the 5% used here. The spread is a one-year horizon, and the breakeven rate is

a five-year rate that includes liquidity and risk premia. Because the rule enters the counterfactuals through $\gamma \ln(1+r)$, a lower r with the same product leaves the results under the rule unchanged; it changes the perfect-foresight paths and RIN storage, and puts the price-implied exhaustion year later. From the forward quotes it is 2057–2076 at a real carry of 3.4% (from CARB's daily transaction prices, 2073 in 2025) and 2047–2060 at 5%, against a first year at the cap of 2032 in the model. Re-estimating the rule at $r = 3.4\%$ gives $\gamma_{CA} = 1.08$ and $K = 2.29$. With $r = 3.4\%$ in the rule (product unchanged), in perfect foresight and in RIN storage, the share of the 45Z extension reaching fuel buyers changes by 0.002 and the cost per tonne of current law by less than 0.1% (Appendix G.7).

Table 12 compares the descriptions of expectations by the error of rolling forecasts of California's price from each quarter's observed state, at horizons of one to twelve quarters, against a no-change forecast. As usual for asset prices, none beats no-change at one quarter in a way the data can detect (Meese and Rogoff, 1983; Alquist et al., 2013; Irwin et al., 2020). Given the net credits that actually occurred, however, the rule forecasts prices far better than no-change (errors of 30, 40 and 39 at four, eight and twelve quarters, against 50, 87 and 117). It maps the credit balance into prices well; its weakness is forecasting the balance, and its forecasts of the level are biased upward, as in its forecast of 2026 from the end of 2024 ($90–105 against $63). The counterfactuals need a mapping from the balance to prices, which a no-change forecast cannot supply, and the upward bias is one reason every result is also reported under perfect foresight and with the response at its model-free value.

Table 12: Rolling forecasts of California's credit price from the observed state

	No change	Rule B (main)	Rule A	Regulatory risk	Rolling horizon
1 quarter	14.8	13.9	14.5	67.7	52.4
4 quarters	49.9	44.7	46.3	69.1	52.4
8 quarters	87.4	88.9	86.6	77.9	63.8
12 quarters	117	132	128	110	103
Rule B given realized net credits		30 (4 q), 40 (8 q), 39 (12 q)			

Notes: Root mean squared error, $/t, 2016–25, forecasts from each quarter's observed price and bank. Rule B does not differ significantly from no change in California (Diebold–Mariano); in Oregon and Washington it is worse at one quarter ($p = 0.03$); the rolling horizon beats no change at eight quarters ($p = 0.02$). The static model and the perfect-foresight benchmarks, which are solved as paths rather than rolled forward, are compared with the history in the text above. Path errors of the full replay are in Section 5.4.

D.2 The rule as a constant-gain filter

This appendix shows that the rule has the form of a steady-state Kalman filter. The equivalence is formal: it says which updating problem the rule would solve, not that holders solve it, and nothing in the main text depends on it.

Let T_t^* be the year in which the bank would run out. It is not known and receives persistent

shocks: $T_t^* = T_{t-1}^* + \varepsilon_t$, $\varepsilon_t \sim N(0, q)$. Holders read the year's net surplus as a noisy measure of how far the scarcity date implied by this year's market is from their previous expectation, $y_t \equiv T_{t-1} + \lambda\big(s_t - E_{t-1}[s_t]\big) = T_t^* + v_t$, $v_t \sim N(0, \varsigma)$, with λ years per Mt. This is a local-level model. The Kalman filter updates $T_t = T_{t-1} + G_t(y_t - T_{t-1}) = T_{t-1} + G_t\lambda\big(s_t - E_{t-1}[s_t]\big)$, and its gain converges to

$$G = \frac{\Sigma}{\Sigma + \varsigma}, \qquad \Sigma = \tfrac{1}{2}\Big(q + \sqrt{q^2 + 4q\varsigma}\Big),$$

where Σ is the steady-state prior variance. In the steady state the update is the estimated rule with $\gamma = G\lambda$; only the product is identified. The gain rises with the signal-to-noise ratio q/ς, and the data show no sign of a gain that declines over time or rises with the bank.

D.3 What moves prices

To see what the rule captures and what it misses, I collected 59 dated LCFS events and 71 RIN events for 2016–26, each checked against its primary source, and 43 CARB quarterly data releases, and classified them as scheduled changes in the standards, news about the regulator's ability to tighten, or data on the credit balance. News about the regulator moves prices most. California's price fell 16% in the week of the Office of Administrative Law's rejection of the amendments (18 February 2025) and rose 14% in the week of their approval (27 June 2025); RIN prices rose 4.6% on average over ten trading days after such news (95% interval [1.9, 7.3]). Scheduled changes move prices when they carry new information (absolute daily LCFS returns 4.7 times those of ordinary days, $p = 0.02$) and not when they are procedural steps already expected; the largest is the Washington Senate's passage of HB 1409, which raised Washington's price by 26% (0.23 log points) within a week. Quarterly data releases move prices little: the response of California's price to the surprise in a release is -0.013 per Mt (s.e. 0.018).

Proposition 2(c) holds when announced changes do not move the expected exhaustion year. Allowing an announcement to move it by a share θ of its effect on the perfect-foresight exhaustion year, the quarterly estimate for California is $\hat{\theta} = 0.15$, 95% interval $[-0.30, 0.59]$. California's proposals did not move prices (model-free θ near zero), while Washington's statute did (near one, from a single trade in a thin market). I keep $\theta = 0$ and read Proposition 2(c) as one end of a range.

The model gives a check on the size of these responses (Table 13). If holders priced the whole announced change in the statutory path ($\theta = 1$), the rule would overstate the price moves on the two Office of Administrative Law rulings six- to thirteenfold, and perfect foresight would overstate them by a similar order of magnitude. With $\theta = 0.15$, the value estimated from California's other scheduled announcements, the rule reproduces the rejection and overstates the approval (0.26 against 0.14). For Washington's statute the reverse holds: full pass-through matches the observed move and $\theta = 0.15$ does not. On these three events, holders priced about a sixth of a regulator's action and nearly all of a legislature's. In the full model, a 45Z extension beyond 2029 announced

in 2026 does not move the state credit prices in 2027 under any description of expectations and moves the D4 price only from 2030, which is consistent with the absence of state price responses on federal dates.

Table 13: Observed and model-implied price responses to three events (change in log price)

	Observed	Rule, $\theta = 1$	Rule, $\theta = 0.15$	Perfect foresight
OAL rejects the 2024 amendments (CA)	−0.16	−1.03	−0.16	−1.35
OAL approves the amendments (CA)	+0.14	+1.76	+0.26	+1.45
Senate passes HB 1409 (WA)	+0.23	+0.27	+0.04	+0.63

Notes: θ is the share of the announced change in the perfect-foresight exhaustion year that enters the expected exhaustion year; $\theta = 0.15$ is estimated from California's other scheduled announcements, not from these events. Perfect foresight is an offline approximation.

D.4 A market that prices regulatory risk

A natural alternative is a rational market that prices regulatory risk: holders expect the standard to be tightened when the bank is large and relaxed or deferred when it is short, and the credit price is the probability-weighted value of the Hotelling paths to the cap under each regime. Disciplined by the record (the shape of the hazard from the 2011–26 regulatory record, the chance that a proposal becomes law from the two 2025 rulings, the holding cost from market interest rates, and only the level of the hazard from prices), such a market cannot reach the observed price level. Its quarterly price error is 74.6 in California against 34.0 for the rule, it predicts 222 for California in 2026 against 63 observed, and the estimated level of the hazard is zero. The reason is informative. The tightenings and relaxations in the record roughly offset, so a hazard disciplined by the record cannot lower prices this far; only a holding cost well above market rates, or the expectation of a permanent relaxation larger than any in the record, could (Salant, 2016). Table 14 compares the price-implied years to exhaustion with the fundamentals projection by program; Washington's market is thin and new, and its rows are illustrative.

D.5 The investment rule

I observe 46 investment decisions in 2014–2025: 16 refinery conversion or co-processing announcements (8 of them SAF-capable) among 1,268 refinery-years at 124 refineries; 5 expansions and 6 SAF retrofits at existing renewable diesel plants, both among the same 106 plant-years at 21 plants; and 19 greenfield announcements over 12 years, each year with a set number of potential entrants. All four types enter one likelihood and share ζ. Before 2023 the SAF margin is zero in every row, so κ_{saf} is identified mainly by the 56 plant-years and 5 retrofits of 2023–25. For each investor-year the margin index m of equation (20) is computed from the historical model at observed LCFS prices, with LCFS expectations on the path the estimated rule implies and statutory credits valued as

Table 14: Price-implied years to exhaustion, $\ln(\bar{P}/P)/\ln(1+r)$, and the fundamentals projection

Programme	Year	P	$r = 0.4\%$	$r = 2.8\%$	$r = 5\%$	Fundamentals ($r = 2.8\%$)
CA	2016	135.9	163.9	23.7	13.4	3.3
CA	2017	113.2	209.7	30.3	17.2	2.3
CA	2018	192.8	76.4	11.0	6.3	2.1
CA	2019	234.0	27.9	4.0	2.3	3.2
CA	2020	241.0	20.5	3.0	1.7	4.9
CA	2021	217.6	46.0	6.7	3.8	7.2
CA	2022	135.8	164.1	23.7	13.4	21.0
CA	2023	77.8	303.8	43.9	24.9	$\geq$38.8
CA	2024	63.3	355.5	51.4	29.1	24.5
CA	2025	58.9	373.4	54.0	30.5	15.3
OR	2016	67.1	333.2	48.2	27.3	2.0
OR	2017	60.8	357.7	51.7	29.3	3.1
OR	2018	95.4	244.9	35.4	20.0	2.4
OR	2019	181.0	84.5	12.2	6.9	3.4
OR	2020	159.0	116.8	16.9	9.6	5.6
OR	2021	144.7	140.4	20.3	11.5	3.1
OR	2022	127.8	171.7	24.8	14.0	1.9
OR	2023	133.7	160.4	23.2	13.1	5.4
OR	2024	82.6	280.8	40.6	23.0	9.1
OR	2025	79.6	290.3	42.0	23.7	3.2
WA	2023	95.1	237.7	34.4	19.4	9.4
WA	2024	49.3	402.3	58.2	32.9	20.1
WA	2025	26.1	561.2	81.1	45.9	10.8

Notes: Price-implied columns: $\ln(\bar{P}/P)/\ln(1+r)$ at the year's mean price, in years from the end of the calendar year. Fundamentals: the regulatory-risk model's projection under the schedule in force, starting from the reported end-of-quarter bank (final-vintage data through 2026Q1), measured from each quarter's end (on average 0.375 years earlier) and averaged over the year's quarters. Projected net credits are the mean of the last four quarters' deseasonalized net credits at the coming standard, plus the deficits added by the scheduled tightening at fixed fuel energy, plus the three-year trend in net credits extrapolated for three years, plus a supply response to the price along the Hotelling path; California's acceleration mechanism and the Oregon and Washington deferral apply as written. While an amendment is pending (CA 2018Q2–Q4 and 2024Q1–2025Q2, OR 2020Q2–2022Q3, WA 2025Q2) the date is the mean of the dates under the proposal and under the law. r enters the fundamentals column only through the supply response; at $r = 5\%$ it changes by at most 1.7 years. Without the trend term, California's column is 4.6, 8.3, 10.0 and 7.3 in 2022–25. $\geq$: three of the four quarters reach the 40-year horizon.

they stood at the decision date. Information is dated at the time it became public, including the LCFS standards (by their adoption date) and the announcements of SAF projects. The four types share the logit scale ζ and are estimated jointly by maximum likelihood,

$$\ln L(\zeta, \kappa_a, \kappa_b, \kappa_c, \kappa_{saf}, p) = \sum_r \big[y_r \ln P_r + (n_r - y_r) \ln(1 - P_r) \big], \tag{51}$$

where P_r is the probability that investor-year r commits under the optimal stopping rule (Rust, 1987) and $\delta = 1/1.07$ is fixed.

Table 15 reports the estimates. Equality of the expansion and greenfield entry costs is not rejected (LR 0.83, $p = 0.36$), so they are pooled. The entry costs are well above engineering capital

cost: about 11 times capital cost for refinery conversions and 3 times for new plants. The excess over capital cost reflects risk, option value and permitting, and it is not counted as a social cost. Estimated on 2014–19, the rule predicts 2020–25 better than a constant hazard (log predictive score +13.3) and than a static logit on current margins (+6.9); most of the gain comes from SAF conversions, the rule alone does about as well as the static logit for renewable diesel, and it under-predicts the 2020–21 conversion wave by about half.

The data reject entry costs at engineering capital cost: with κ fixed there, the likelihood rises without bound in ζ, toward investment that ignores margins, and the likelihood-ratio statistic against the estimates exceeds 283 (three restrictions) at every ζ; at $\zeta = 80$ the rule would predict 165 refinery conversions against 16. The estimated entry costs correspond to real hurdle rates of about 51% for conversions, 22% for expansions and new plants and 18% for SAF retrofits (constant margins, 20-year life, three-year construction, two for SAF), which include option value and permitting and policy risk as well as the cost of capital. The SAF conversion cost rests on six events and is imprecise (bootstrap relative standard error about 100%); the engineering cost of a conversion lies inside its confidence interval, and a run with κ_{saf} at engineering cost is a sensitivity.

Table 15: Investment rule, maximum likelihood, 2014–2025

	Estimate	95% profile interval
Logit scale ζ	2.788	[2.01, 4.11]
Entry cost, refinery conversion κ_a (\$/gal-yr)	20.25	[16.25, 27.50]
Entry cost, expansion and greenfield κ_{bc}	10.5	[7.75, 15.00]
Entry cost, SAF conversion κ_{saf}	3.0	[−1.35, 9.70]
Extension probability p	1	at the upper bound
Log likelihood	−179.75	
Decision-years; decisions	1,492; 46	

Notes: Expansion and greenfield entry costs pooled (LR test of equality 0.83, $p = 0.36$). Parametric bootstrap 90% intervals: ζ [2.01, 3.73], κ_a [16.50, 25.50], κ_{bc} [7.75, 13.76], κ_{saf} [−0.90, 8.53] (200 replications). Profile intervals: likelihood within 1.92 of its maximum. Decision-years: 1,268 refinery-years, 106 plant-years for expansions and the same 106 for SAF retrofits, and 12 greenfield years; decisions 16 conversions, 5 expansions, 19 new plants and 6 SAF retrofits. Fixed: conversion loss 1.125; jet share of SAF-capable capacity $h_b = 0.2$. The estimate of p is at its upper bound of 1 (profile interval [0.67, 1.00]). Estimated on 2014–19 alone (ζ 3.09, κ_a 25.0, κ_{bc} 11.25, κ_{saf} 4.15), the rule beats the benchmarks by +13.3 and +6.9 in log predictive score over 2020–25; with the price-expectation rule also refitted on data before 2019, by +9.5 and +3.0.

The main specification sets the probability at 1 while a credit is in law and at 0.18 during lapses, the restoration probability implied by RIN prices. Letting the probability differ in the months when the blenders' credit had lapsed (15 decisions, 609 decision-months) and when it was in law (31, 883) estimates both: 1 in law (profile interval [0.67, 1.00], [0.74, 1.00] adjusted for the boundary) and 0 in lapses ([0, 0.31]). The data reject equal values and do not reject the main specification's pair. Freeing the lapse value moves the other estimates little: σ from 2.79 to 2.62, κ_a from 20.25 to 18.75, κ_{bc} from 10.5 to 9.25. Almost all of the lapse variation comes from 13 decisions in 2017–19 (2014–15 has two; the 2010 and 2012 lapses precede the panel), so the comparison is in effect five

lapse years against seven years in law, and a shock to investment in the lapse years would load on the lapse value. I have not bootstrapped these intervals.

Eight of the 16 refinery conversions of 2017–23 were built able to produce SAF, when SAF earned nothing over RD. The margin index cannot explain this choice, which looks in the data like a nearly costless design option. Rather than estimate it, I set future SAF capability at build to the historical proportion; SAF value enters investment only through conversions. The SAF conversion rule predicts 4.5 conversions in 2023–25 against 5 observed.

With margins from observed prices, the rule reproduces observed capacity (RMSE 0.47–0.72 billion gallons across versions). In a fully coupled backtest, where capacity feeds back on margins, capacity rents spike in 2018–19 and the rule overbuilds; the main backtest therefore uses observed capacity. The logit rule also implies a positive announcement probability at zero expected profit, so idle and SAF-capable capacity grow slowly even when margins are nil. I bound its welfare cost by a run with capacity fixed at the main path (lower bound) and the main run with endogenous capacity (upper bound).

D.6 Other estimated blocks

The share response c in equation (36) is identified from how the fat share of renewable diesel and biodiesel moved in California, Washington and the rest of the country over 2016–25 (about 39 state-fuel-year observations), as relative feedstock margins changed. I estimate it by simulating the historical model for a grid of values and minimizing the squared errors of the fat shares, with a bootstrap over years. Quarterly panel regressions of the fat share on the margin gap are not informative. Estimated separately, the response of biodiesel is precise ($c_{BD} = 0.446$, 95% interval [0.33, 0.76]) and that of renewable diesel is not identified (0.094, [0.012, 1.28]), because California's history favors a small response and Washington's a large one. The main specification uses one response for both fuels, estimated jointly, with the renewable diesel response at the two ends of its separate interval as sensitivities. Two limits: the response is identified from margins, so quantity shocks such as the 2026 drop in imported used cooking oil and tallow under OBBBA's origin rule and tariffs are not captured by it; and within each group the feedstock composition is fixed at 2025.

The light-duty price coefficient β_1, the network coefficients $\beta_2 = \beta_3/2$ and the station entry elasticities $\upsilon_{L2}, \upsilon_{DC}$ are estimated jointly by simulated method of moments on 2021–25, at observed fuel, electricity and credit prices. The moments are the logit sales share, the demeaned growth of the electric stock, and the log Level-2 and DC fast station counts, by region and year. Market stations exclude Tesla, Electrify America and NEVI/CFI stations, which are exogenous. Charger costs in each year use the law known in that year (the 2021 credit, its retroactive restoration in 2022, the Inflation Reduction Act's version from 2023), and next year's station size is this year's.

The estimates are $\beta_1 = -4.918$ (0.353) and $\beta_3 = 0.313$ (0.095), with entry elasticities $\upsilon_{L2} =$

0.281 and $\upsilon_{DC} = 0.443$ in the LCFS states and 0.724 elsewhere. For the aggregate price elasticity of light-duty electric demand, the literature anchor of −2.8 is a sensitivity. The fast-charging network coefficient is close to the local estimate of Springel (2021) (0.42). For trucks the price elasticities are taken from the literature (medium-duty −2.5, the approximate median of light-duty estimates in Cole et al., 2023, p. 318 and fn. 2, applied to medium-duty trucks; heavy-duty −2.65, the midpoint of the range of −2.5 to −2.8 in Weber, 2024, p. 16, an own-price elasticity for class 7–8 diesel trucks transferred to electric trucks) and the intercepts are set to 2025 sales. The truck block is conservative: intercepts are fixed, there is no diffusion or model-availability term, and there is no Advanced Clean Trucks mandate. Elasticities of −6 and −10 are sensitivities.

E Fit to History

The backtest solves the model year by year over 2016–2025 with the policies then in force. In the version reported in the main text, hydroprocessing capacity is set to its observed level; RIN banking follows its observed path; and small-refinery exemptions and RFS volumes are as finally determined.

Table 16: Annual replay of the main specification, 2016–2025: root mean squared error

	Model		Random	DM p	2016	AEO 2016	AEO each
	2016–25	2017–25	walk	vs RW	held	path	year
CA LCFS price ($/t)	38.4 (0.91)	40.4	46.9	0.59	73.0	—	—
OR LCFS price ($/t)	38.5 (0.44)	40.3	36.3	0.58	63.9	—	—
WA LCFS price, 2023–25 ($/t)	34.9 (0.91)	34.9	36.8	—		—	—
D4 / D6 RIN ($)	0.39 / 0.28	0.33 / 0.22	0.46 / 0.46	0.24 / 0.06	0.42 / 0.47	—	—
CA net credits (Mt)	1.46	1.54	4.63	0.24	4.74		
CA renewable diesel (bn gal)	0.181	0.191	0.335	0.04	1.23		
U.S. renewable diesel / biodiesel (bn gal)	0.121 / 0.103	0.122 / 0.100	0.51 / 0.35	0.08 / 0.25	1.36 / 0.49	0.88 / 0.36	0.47 / 0.47

Notes: Main specification with the three shift terms in the history years, observed hydroprocessing capacity. Correlation of model and observed prices in parentheses. The random walk needs the previous year's value, so it and the Diebold–Mariano tests cover 2017–25; Washington's covers 2024–25 and allows no test. Before AEO 2023, renewable diesel is proxied by AEO's "other biomass-derived liquids" and biodiesel is consumption, while the observed series are EMTS generation; the AEO columns are indicative only.

Table 16 reports the replay from a single start in 2016 under the main specification, with the three shift terms in the history years. California's error is 38.4 over 2016–25 and 40.4 over 2017–25, against 46.9 for a random walk, with an upward bias of about $24; the difference is not significant (Diebold–Mariano $p = 0.59$). Of all the comparisons with the random walk, only California's renewable diesel volume is significant at 5% ($p = 0.04$), with the D6 price and U.S. renewable diesel at 10%. Neither replay is the test reported in the main text. The test in Section 5.4 runs the rule as the forward runs use it, without shift terms, from four start years (2017, 2019, 2021 and 2023), with

the starting point inferred from the price in the starting quarter. There California's pooled error is 29.6 against 43.9 for a random walk, with a bias of +4.2 ($p = 0.15$, clustered by year), and Oregon's is 43.0 against 34.3. Adding the 2018 shift term in its own quarter only gives 27.4 against 46.9 for California, and the price still reaches the cap in 2019 ($261 against $235 observed). Carrying the event terms into later years and starting from a different point, two choices the forward runs do not make, gives 50 against 47. With γ_{CA} converted to annual units by matching the long-run rather than the first-quarter response (0.94 instead of 0.75), the backtest from four start years gives 26.6 against 43.9 for California, and Oregon's error, 37.1, falls within the criterion of 1.25 times the random walk (42.8) though it remains above the random walk itself; in the single replay California's error is 30.5 against 46.9, and its price still reaches the cap in 2019. Washington's price is evaluated over 2024–25 rather than 2023–25, because the random walk needs the previous year's price and Washington's program began in 2023.

The model tracks California's credit balance and fuel quantities with errors a third to a quarter of the naive benchmarks', and its fuel quantities better than the AEO. Oregon is the exception: the model's Oregon price errors exceed those of a random walk, and its prices are only weakly correlated with the observed ones (0.44). Oregon's credit pool is small, so its price-expectation rule turns small surprises into large revisions of the expected exhaustion year, and its price path is sensitive to the starting point. I treat Oregon's forward price path as the least reliable of the four. The backtest validates the fuel-market block and the forecasting performance of the price-expectation rule, not a model of how holders form beliefs. The backtest is not a fully independent test. RIN banking is set to its observed path, 2016–18 capacity is observed, the gain in the RIN price fit is partly mechanical, and the specification was chosen after earlier backtests had been seen. It does not test the coupling of RIN storage with investment: with capacity from the investment rule, the coupled backtest does not converge.

Table 17 compares the model's 2026 with the observed values that were not targets. Six of the eight test variables are within tolerance: the three programs' net credits, California's electricity credits, U.S. biodiesel and U.S. SAF. Two are outside: the January–August average RIN price, which returned within tolerance after EPA's exemption decisions of 31 August, and the fat share of California's renewable diesel. The fat share fell in 2026 because imported used cooking oil and tallow left the market after OBBBA's origin rule and new tariffs, while the relative price moved toward fat; a model in which shares respond to margins predicts a rise. The earlier SAF shortfall had two causes, both corrected: a coding error that gave HEFA-SAF the reference feedstock mix of renewable diesel, and a comparison with an EMTS-based observed value that is an upper bound on SAF use. The 2026 D6 price depends on the end-2025 RIN bank, which EPA has not published: the observed price corresponds to a bank of about 6 billion RINs, against 3.21 in the main specification.

Before pinning the 2026 prices, I predicted them from the end-2025 state with the estimated rules. This is the most direct test of how the price-expectation rule responds to a large deficit, and

Table 17: 2026 out of sample: model against observed

	Observed	Tolerance	Model	Within
D6 / D4 RIN ($)	1.49 / 1.51	±0.25	2.29 / 2.29	no (after 31 August: yes)
CA net credits at pinned price (Mt)	−10.7	±4	−9.60	yes
OR / WA net credits (Mt)	+0.44 / +0.10	±1.5 / ±1.0	+0.07 / +0.20	yes
Fat share of CA renewable diesel	0.577	±0.08	0.787	no
U.S. biodiesel production (bn gal)	1.50	±0.21	1.56	yes
U.S. SAF (bn gal)	0.29	±0.15	0.17	yes
CA EV credits, LDV and MDV (Mt)	7.26	±15%	8.18	yes
NM LCFS price ($/t)	none	$0 < P < \bar{P}$	0	boundary
U.S. HEFA output (bn gal)	3.57 (STEO)	report only	3.97	

Notes: Observed: January–August 2026 unless noted. California's fat share is 2025Q3–2026Q1. Biodiesel is production (consumption plus exports), seasonally adjusted, January–August (range 1.43–1.55); the earlier test value of 1.39 (January–June consumption) is superseded. RIN prices are EMTS volume-weighted January–August averages in 2024 dollars (July–August: D6 2.02, D4 2.00); after EPA's decisions on 2025 exemptions on 31 August, Argus assessed D6 at 2.13 and D4 at 2.27. The three programs' 2026 LCFS prices are pinned at their January–August averages and are calibration, not tests.

it is not used in choosing the specification. Table 18 reports it.

Table 18: One-step prediction of 2026 LCFS prices from the end-2025 state

$/t	CA	OR	WA	NM
Observed (January–August)	63.3	145.8	45.6	—
Rule A, K at the lower end of its interval	160 / 179	170 / 48	45.5	0
Rule B, $K = 2$	83	89		

Notes: Rule A: coupled run / offline replay with net credits held fixed. Rule B: offline replay. The two Rule A numbers for Oregon differ widely, another sign of the sensitivity of Oregon's rule to its starting point.

Under Rule A, California's prediction is more than twice the observed price. The mechanism is the bank-years term: with expected net credits of −11.2 Mt against a bank of 39.7 Mt, the bank would last about 3.5 years, which puts a large weight on it and pulls the expected exhaustion date forward. Under Rule B, the three-year average includes the large surpluses of 2023–24, the bank would last about 7.7 years, the weight is small, and the prediction is closer to the observed price. The market itself priced December 2026 futures at up to $72/t in January 2026 (Argus Media, 2026).

In the 2026–30 runs the aggregate RIN bank stays below 10% of the next year's obligation, so the 20% carry-over limit does not bind in aggregate; deficits are repaid the next year. In the final run the RIN bank stays below 5.1% of the next year's obligation, and the deficits of 2027 and 2030 are repaid the next year. California's acceleration mechanism does not trigger: in 2027 its bank ratio is 2.29 (trigger above 3) and credits are 0.83 of deficits (trigger above 1). No program defers in 2026–30. New Mexico's forecast margin is negative in 2029 and 2030 (−0.85 and −0.20 Mt), but its deferral also requires the expected price to reach the cap, and the expected prices are $155 and $211 against a cap of $252.

The coupled forward model, with all blocks endogenous, cannot be tested on history. The backtest validates the fuel market and the credit-holder rules at observed capacity; the investment rule is validated on the build-out given historical margins (estimated on 2014–19 and tested on 2020–25, Section 5.2); the vehicle block is validated separately at observed prices. The 2026 test is one year.

F Additional Results and Proofs for Section 3

F.1 The static model in figures

Figure 11 shows the two markets that matter for national support. Panel (a) is the allocation of national biofuel between the regions, drawn as a box whose width is national output: the line is the extra value of moving one more unit to S, $(c_0 - c_1)\lambda - \tau(q_S)$, and the outlet requires $q_S \geq q - \bar{q}_N$. Panels (b) and (c) are the state credit market of Lemma 1, with credit supply $(\sigma - c_1)q_S + \Gamma$ and deficits $(c_0 - \sigma)(Q_S - q_S)$ drawn separately; net generation b is the horizontal distance between them.

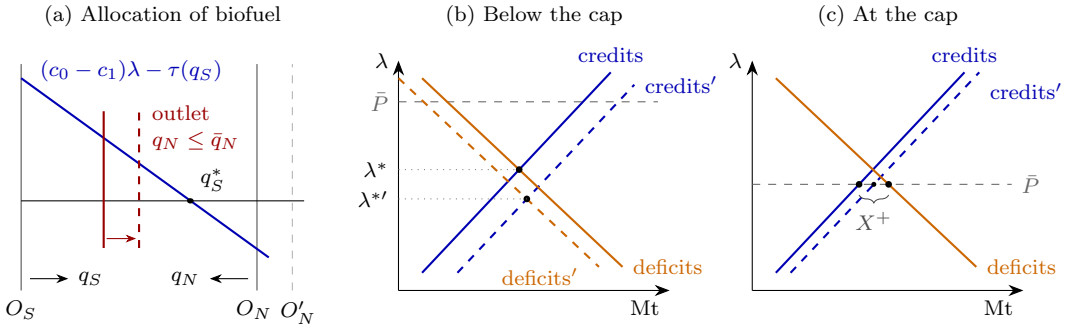


Figure 11: Where the extra biofuel goes, and what it does to the state credit market

Notes: (a) The box has the width of national biofuel output, from O_S to O_N; q_S is measured from the left and q_N from the right. The line is the extra value of a unit sold in S rather than N. Biofuel goes to S up to q_S^*, unless the outlet in N is full (the vertical line), which forces more into S. A subsidy widens the box (O_N to O'_N) and moves the outlet line right; while it stays left of q_S^*, all the extra biofuel goes to N (Proposition 1(iii)). (b) Once the outlet binds, the extra biofuel in S shifts the credit supply curve out and the deficit curve in, and the credit price falls from λ^* to $\lambda^{*\prime}$ (case (iv)). (c) At the cap, the price stays at $\bar{P}$ and the shortfall the state issues, X^+, shrinks (case (v)). Static market without a bank; Section 3.4 adds the bank.

F.2 When the transfer arrives under each description of expectations

Suppose that from year t_0 a permanent subsidy raises the state's net generation by Δb a year at given prices (case (iv) of Proposition 1, the outlet binds below the cap).

Proposition 2 (When the transfer arrives)**.** *At given prices:*

(a) Perfect foresight. *The expected exhaustion year moves out by*

$$\frac{dT}{d\Delta b} = \frac{T - t_0 + 1}{-b_T(\bar{P}) + \ln(1+r)\sum_{t=t_0}^{T} \lambda_t \, \partial b_t / \partial \lambda} > 0, \tag{52}$$

and the credit price falls by the factor $(1+r)^{-dT}$ *in every year from* t_0*: the transfer to the state's fuel buyers starts in the first year.*

(b) Adaptive rule, the subsidy is a surprise. *Under Rule A,* T *moves out by* $\gamma\Delta b$ *once, in* t_0*; from* t_0+1 *the subsidy is in the baseline and causes no further surprise. Under Rule B the revision is spread over three years and totals* $2\gamma\Delta b$*. The price falls from* t_0 *by the factor* $(1+r)^{-\gamma\Delta b}$ *(Rule A).*

(c) Adaptive rule, the subsidy is anticipated *(*Δb *is in* Δ^{known}*). There is no surprise;* T *and the price do not change, and the extra credits accumulate in the bank. The price falls only when the larger bank matters: through the bank-years term if* $K > 0$*, whose weight rises as exhaustion approaches, and in any case in the years around* T*, when the bank that would have run out does not.*

Part (c) is a property of the form of the rule and does not come from the data. Holding the policy fixed, the same accounting can be used to compare any two descriptions of expectations.

Figure 12 illustrates the three cases with a numerical example. Under perfect foresight the subsidy lowers the credit price from the first year; under the adaptive rule, the same subsidy lowers it by less if it is a surprise, and not at all for many years if it was announced, because holders who already expected it see nothing new in the credit balance. Under the adaptive rule, the part of the subsidy that holders do not treat as news first accumulates in the bank and lowers the price only as the bank runs down. The timing of the transfer therefore depends on what holders count as news, and for this reason the paper reports results under alternative descriptions of expectations.

F.3 Proofs

The equilibrium conditions are (3)–(6). Write $a = (c_0 - \sigma)/(c_0 - c_1)$. When the standard binds below the cap, $b = 0$ in (2) gives $q_S = a\,Q_S - \Gamma/(c_0 - c_1)$.

Proof of Lemma 1. With $\nu = 0$, (7) gives $q_S = \tau^{-1}((c_0 - c_1)\lambda)$, and $Q_S = D_S(m + \pi\rho + (c_0 - \sigma)\lambda)$; neither involves s. Differentiating (2) gives (8). Substituting q_S into $Z_S = c_0 Q_S - (c_0 - c_1)q_S$ gives (9).

Proof of Proposition 1. (i) With $q = H$ and $\rho = \nu = 0$, (3) gives $\psi = m + s - C'(H)$, so $d\psi/ds = 1$, and nothing else depends on s.

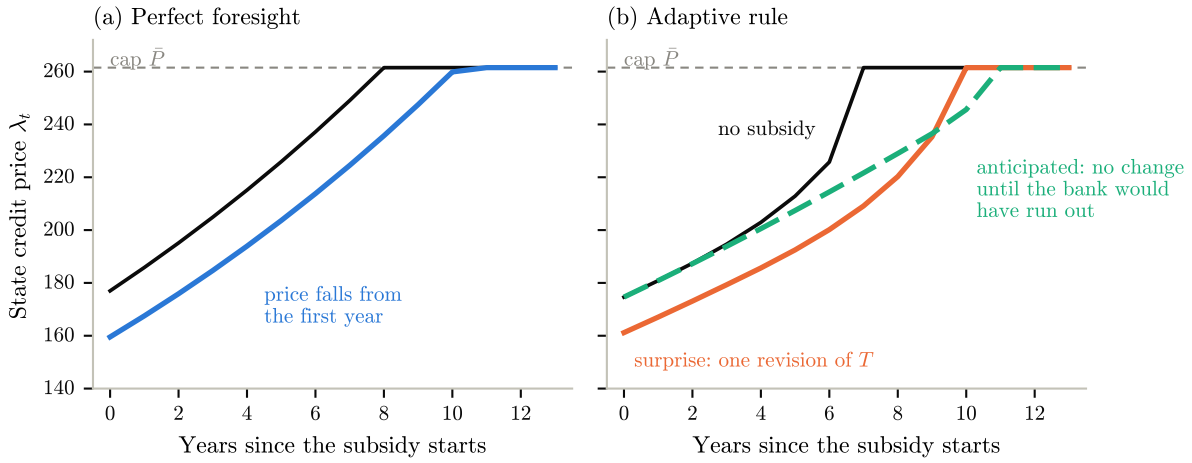


Figure 12: When the transfer arrives: a numerical example

Notes: Credit price paths with and without a permanent subsidy that raises net credit generation by 3 Mt a year from year 0. (a) Perfect foresight, equation (14). (b) Adaptive rule, equation (15) with Rule A, $\gamma = 0.75$ years per Mt and $K = 2$; "surprise" and "anticipated" differ only in whether the subsidy is in Δ^{known}. Prices include the within-year response of generation to the price. Illustrative parameters of California's order of magnitude: $b_t(\lambda) = -14 - t + 0.06\lambda$ Mt, initial bank 40 Mt, $\bar{P} = \$261.5$/t, $r = 5\%$.

(ii) With $q = M$ and $p_N = m + \pi\rho$, (3) gives $C'(M) = m + (1+\pi)\rho + s$, so $d\rho/ds = -1/(1+\pi)$ and fuel prices fall by $\pi/(1+\pi)$ per unit. Fuel buyers gain $\frac{\pi}{1+\pi}\,ds$ per unit on $Q^F + M$ units with $Q^F = M/\pi$, that is $M\,ds$. Effects on the state market, with the outlet slack: $\tau(q_S) = \lambda(c_0 - c_1)$ gives $d\lambda = \tau' dq_S/(c_0 - c_1)$; with $dQ_S = D'_S[\pi\,d\rho + (c_0 - \sigma)d\lambda]$, $dq_S\big[1 - aD'_S(c_0 - \sigma)\tau'/(c_0 - c_1)\big] = aD'_S\pi\,d\rho \geq 0$, and the bracket is positive, so $dq_S \geq 0$, $d\lambda \geq 0$ and $dq_N = -dq_S$. The price in S changes by $\pi\,d\rho/(1+x) < 0$ with $x = -a^2 D'_S\tau' \geq 0$, so Q_S and, by (9), Z_S rise; in N, p_N falls and q_N falls, so emissions rise. These effects are of the order of the demand response to the RIN pass-through and are small when demand is inelastic.

(iii) With $\rho = \psi = \nu = 0$, (3) gives $dq/ds = 1/C''$. By Lemma 1, λ and q_S do not depend on s, so $dq_N = dq$; N's demand is unchanged because $p_N = m$, so emissions fall by $(c_0 - c_1)/C''$. At the cap the same holds: $q_S = \tau^{-1}((c_0 - c_1)\bar{P})$ is fixed, and so is X^+.

(iv) With $q_N = \bar{q}_N$ and $\pi\rho = 0$, (4) gives $(C'' + \tau')\,dq_S = (c_0 - c_1)\,d\lambda + ds$, and $b = 0$ gives $dq_S = aD'_S(c_0 - \sigma)\,d\lambda = -\varpi\,d\lambda$. Solving gives (10); $dZ_S/ds = \sigma(c_0 - \sigma)D'_S\,d\lambda/ds > 0$ by (9), and N does not change.

(v) With $\lambda = \bar{P}$, p_S and Q_S are fixed and (4) gives $(C'' + \tau')\,dq_S = ds$; fossil use in S falls one for one, so $dZ_S = -(c_0 - c_1)\,dq_S$, and differentiating $b + X^+ = 0$ gives $dX^+ = -(c_0 - c_1)\,dq_S$.

Proof of Proposition 2. (a) Let $\Phi(T, \Delta b) = B_{t_0-1} + \sum_{t=t_0}^{T}\big[b_t(\lambda_t(T)) + \Delta b\big]$ with $\lambda_t(T) = \bar{P}(1+r)^{-(T-t)}$, treating T as continuous. Then $\partial\Phi/\partial\Delta b = T - t_0 + 1$ and $\partial\Phi/\partial T = b_T(\bar{P}) - \ln(1+r)\sum_t \lambda_t\,\partial b_t/\partial\lambda < 0$, and the implicit function theorem gives (52). (b) Under Rule A, $\bar{b}_{t-1} = b_{t-1}$,

which includes Δb from t_0+1; under Rule B, $\bar{b}_{t-1}$ includes $\frac{1}{3}\Delta b$, $\frac{2}{3}\Delta b$ and Δb in t_0+1, t_0+2 and t_0+3, so the surprises are Δb, $\frac{2}{3}\Delta b$, $\frac{1}{3}\Delta b$ and zero after, which sum to $2\Delta b$. (c) With Δb in $\Delta_{t_0}^{known}$ (and, under Rule B, the part of Δb not yet in the average in $\Delta_{t_0+1}^{known}$ and $\Delta_{t_0+2}^{known}$), the surprise is zero in every year. The bank is larger by $(t-t_0+1)\Delta b$ in year t, which raises Y_t and lowers the weight $e^{-Y_t/K}$ on the bank-years term. Within-year feedback: the price that clears a year satisfies (13) with T_t from (15), and a lower price lowers b_t and hence the surprise, so the revision is smaller than $\gamma\Delta b$ but of the same sign.

Proof of Corollary 1. Before either bank runs out no credits are issued at the cap, so $B_t = B_{t_0-1} + \sum_{k=t_0}^{t} b_k(\lambda_k)$. Net generation is increasing in the price: by Lemma 1 when the outlet is slack, and with the outlet binding because then $\partial b/\partial\lambda = (c_0-c_1)^2/(C''+\tau') - (c_0-\sigma)^2 D_S' > 0$. For $t < \min(T^e, T^{e'})$, $\lambda_k^e \le \lambda_k^{e'}$ for every $k \le t$, so $B_t^e \le B_t^{e'}$. If $T^e < T^{e'}$ there is nothing to prove. Otherwise $T^{e'}-1 < \min(T^e, T^{e'})$, so $B^e_{T^{e'}-1} \le B^{e'}_{T^{e'}-1}$, and at $T^{e'}$ the bank under e' cannot cover the shortfall at the cap, $B^{e'}_{T^{e'}-1} + b_{T^{e'}}(\bar{P}) < 0$. Then $B^e_{T^{e'}-1} + b_{T^{e'}}(\bar{P}) \le B^{e'}_{T^{e'}-1} + b_{T^{e'}}(\bar{P}) < 0$, so the bank under e cannot cover it either, and $T^e \le T^{e'}$.

F.4 Notes on the numerical example

Panel (b) of Figure 12 has two further features that come from the adaptive rule. Without the subsidy, the price rises from 226 to the cap in one year (year 6 to 7), much faster than the 5% a year at which holders are willing to store credits; holders with perfect foresight would have bought earlier. The bank-years term raises the price only as the bank runs low, and the last step comes when the bank is overdrawn and the market clears at the cap. Prices under the rule therefore do not follow Hotelling's rule near exhaustion. In addition, the surprise path reaches the cap a year before the anticipated path. The surprise lowers the price at once, so less credit is generated and the bank runs down sooner; when the subsidy is anticipated, the price and credit generation do not change and the extra credits stay in the bank.

Part (c) follows from the form of the rule, not from data: an announced change enters expected net credits and so never moves T. It illustrates one end of a range rather than a finding. A more general rule lets an announcement move T by a share θ of its effect on the perfect-foresight exhaustion year, $T_t = T_{t-1} + \gamma(b_t - E_{t-1}[b_t]) + \theta\,\Delta T_t^{PF}$. Part (c) is the case $\theta = 0$, and with $\theta = 1$ an announced subsidy moves the price at once, as in (a). Appendix D.3 estimates θ close to zero but imprecisely; the main specification keeps $\theta = 0$, and Section 7 reports the other end.

The state rules that respond to the bank work through the same channel: a dollar that adds to California's bank makes its acceleration mechanism more likely to tighten the standard, and a dollar that adds to the other states' banks makes their deferral rules less likely to relax it. Perino (2024) shows that a supply rule that responds to the bank, the EU's Market Stability Reserve, makes prices more sensitive to expectations that depart from fundamentals.

G Additional Results

G.1 The current-law path

For medium- and heavy-duty trucks in the clean fuel states, the electric share of new sales peaks in 2032–34 and then falls, as each kilowatt-hour earns fewer credits. Washington's path depends on whether Ecology lifts the statutory 20% limit. The main specification keeps the limit; lifting it to 45%, as described below, adds 30 Mt of abatement over 2030–45.

The D6 price is $2.29–2.52 in 2026–28 and falls to about $1.35 in 2029, when hydroprocessing utilization completes its ramp to the long-run reference rate and about 0.5 billion gallons of capacity move to the flat part of the utilization-cost curve; the ramp is applied as a single step, so I report 2028–29 as an average ($1.94). It then drifts down to about $1.14 in 2045; capacity earns a rent above the annualized conversion cost only in 2026–28, and in every year some hydroprocessing capacity is idle.

SAF output is 0.14–0.21 billion gallons in 2026–32. Almost all of it is made outside the clean fuel states, with Illinois's SAF credit. That credit ends in 2032, and SAF output falls to almost nothing. SAF earns about the same credits per gallon as renewable diesel. But it sells at the price of jet fuel, which carries no clean fuel standard obligation, and it costs $0.4–0.6 a gallon more to make. In California the gap between the diesel and jet prices widens from $0.5 to $3.2 a gallon, as the credit price passes into diesel. A gallon of feedstock therefore earns more as road fuel in every year (Wu et al., 2026).

California's shortfall at the cap grows as the electric share of the light-duty stock rises to 36%. Electricity credits peak at 15 Mt in 2034 and then fall, because each kilowatt-hour earns less as the standard approaches zero. CARB's own projection for the 2024 amendments (California Air Resources Board, 2024) has similar deficits until the mid-2030s and lower ones after, because it assumes the Advanced Clean Cars II rule, whose federal waiver Congress revoked in 2025, and credits from direct air capture, hydrogen and other projects after 2040, which the model does not include.

The model closes each year by issuing the shortfall as credits at the cap, the safety valve a regulator would have to adopt. The statute instead carries an unmet obligation forward with 5% interest and requires it to be retired within five years (17 CCR §95485(c)(5)), which cannot happen when the cap binds every year. Carrying the deficit forward instead of issuing changes the resource cost within 2045 by −$0.1 billion (Appendix G.4). With issuance the state receives the money. Under the statute fuel sellers collect it in fuel prices and owe it as an unpaid obligation, which becomes a windfall if it is never enforced.

Direct air capture does not change California's shortfall under current law, because 45Q pays only for facilities whose construction begins before 2033; with 45Q extended, it would take California off its cap from 2041 (Figure 14). New Mexico reaches its cap in 2033; its deferral then holds it

close to balance, and its standard reaches only about 30% by 2045.

From 2030 the RIN bank sits at its floor. Because the one-year deficit carry-forward applies to the industry as a whole in the model, the market can borrow only every other year, and RIN prices alternate between years; firms' cycles need not coincide, so I report RIN prices from 2030 as two-year averages, with a version in which obligated parties are grouped as a sensitivity.

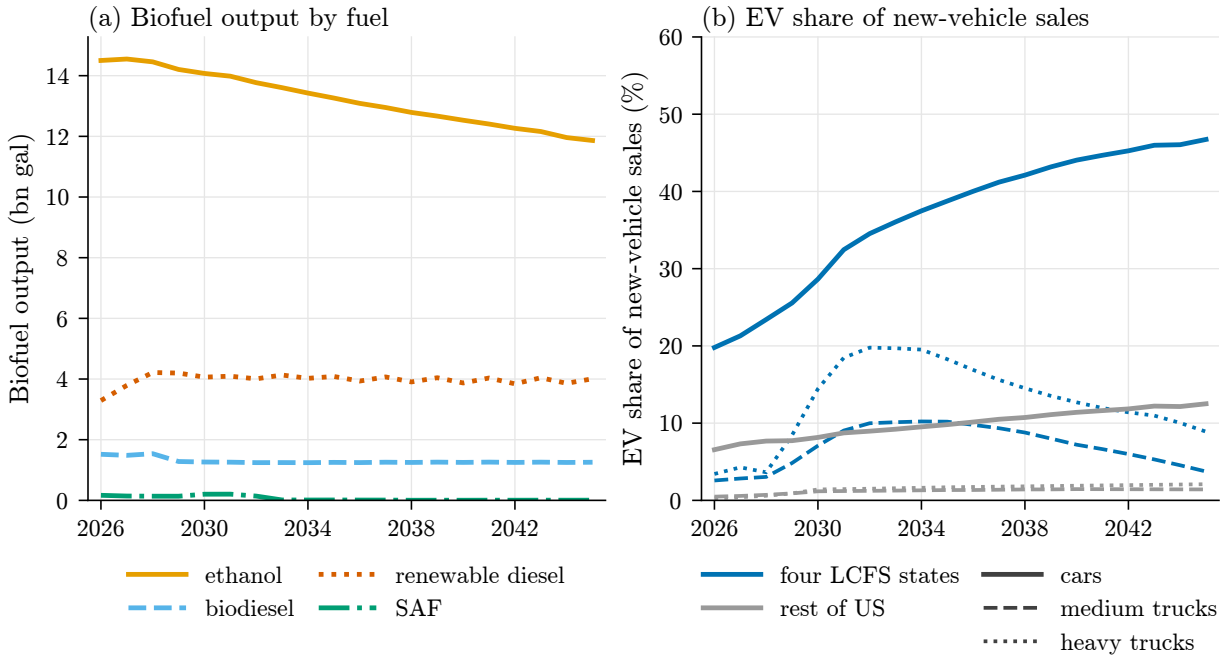


Figure 13: Current law, 2026–2045: biofuel output and electric vehicle sales

Notes: Biofuel output by fuel, and the electric share of new car, medium- and heavy-truck sales in the four clean fuel states and in the rest of the country. Main run.

Issuance at the cap and the statute. The model closes each year by issuing the shortfall as credits at the cap, and this closure is a maintained assumption of the main scenario. The statute instead carries an unmet obligation forward with interest and requires it to be retired within five years, which cannot happen when the cap binds every year. The two treatments give the same prices and quantities under two conditions. First, the cap binds in every year from 2032 to 2045, so obligations carried forward add to a demand for credits that already exceeds supply, and the price stays at the cap either way. Second, fuel sellers treat the cost of a deficit as the cap price, which holds if they expect carried-forward obligations to be settled at the cap or never enforced. If they instead expected the obligations to be enforced in full, with interest, the effective cost of a deficit could differ from the cap and fuel use in California would respond. Under these conditions the choice changes the resource cost within 2045 by −$0.1 billion and only determines whether the money is held by the state or remains an unpaid obligation of fuel sellers. A regulator facing a permanent shortfall could also relax the standard. Appendix Figure 14 shows how much of the gap

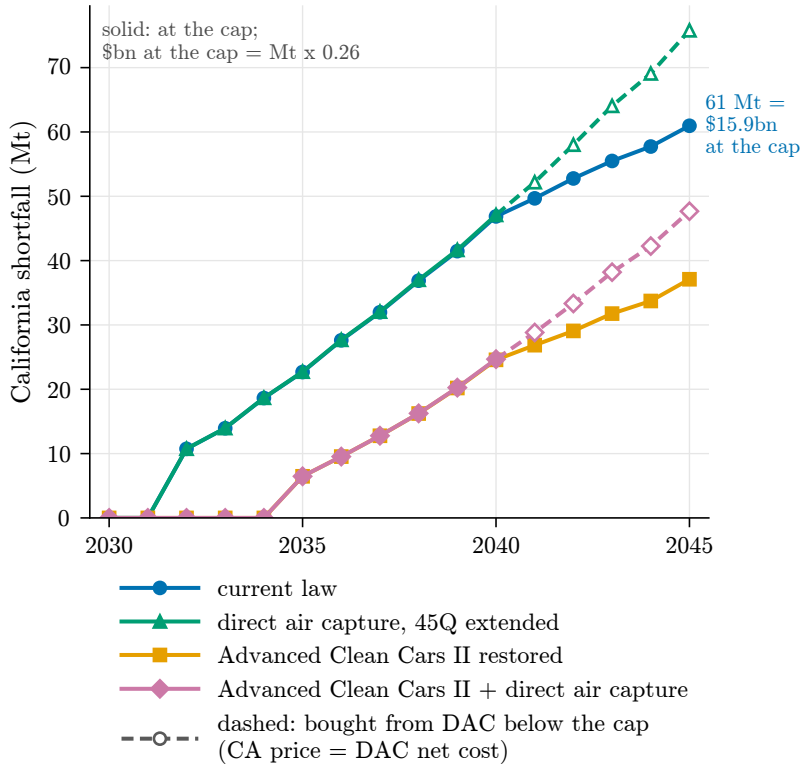


Figure 14: From 2032 California cannot meet its standard at the cap

Notes: Shortfall at the cap, met by credits issued at the cap. Under current law the shortfall in 2045 is 61 Mt, $15.9 billion at the cap. With Advanced Clean Cars II restored, California first reaches the cap in 2035 and the shortfall is 37 Mt in 2045 (248 Mt over 2026–45, against 527 Mt under current law). With direct air capture under an extended 45Q, California is at its cap from 2032 to 2040; from 2041 DAC's cost net of 45Q ($230 a tonne, falling to $102 in 2045) is below the cap and sets the price. Solid lines show the shortfall at the cap (252 Mt over 2032–40 with DAC; 90 Mt over 2035–40 with Advanced Clean Cars II and DAC). Dashed lines show credits bought from DAC below the cap, where California's price equals DAC's net cost (319 Mt and 190 Mt over 2041–45).

restoring Advanced Clean Cars II or direct air capture under an extended 45Q would close.

Washington's path depends on whether Ecology lifts the statutory 20% limit. Under the law as it stood in October 2026 (the main specification), its standard stops tightening in 2030 while electricity and renewable fuel credits keep growing, so its bank grows from 3 Mt in 2032 to 22 Mt in 2045 and its price falls from about $50 to $6. This is the consequence of a standard frozen at 20% and continued growth in credit supply, not a forecast. If Ecology makes the demonstration the statute requires, which its 2027 rulemaking may do, the standard follows the path to 45% and Washington's price reaches its cap in 2033; the stack then abates 30 Mt more over 2030–45 (1,705 against 1,675 Mt) at $135 a tonne on average. I report Washington's results under both, as a range set by that one administrative step.

Direct air capture does not change this under current law. CARB's own market model lets DAC credits set the price once their cost net of the federal 45Q credit falls below the cap (California Air

Resources Board, 2024), but 45Q pays only for facilities whose construction begins before 2033 (26 U.S.C. 45Q(d)(1)), and on CARB's cost path DAC costs more than the cap in every year to 2045 with or without it ($544 a tonne in 2035, $273 in 2045, 2024 dollars).[12] If 45Q were extended to later facilities, DAC would take California off its cap from 2041, with a price of $102 in 2045 and 319 Mt removed by 2045 (Figure 14).

G.2 The cost of current law and who pays

Cost per tonne under other definitions. The main text uses resource cost including forgone fuel-tax revenue, in present value at 2%, over undiscounted tonnes including indirect land-use change. Other definitions give other numbers, and the 2030–45 values are reported with the comparison of instruments below. With tonnes discounted at the same rate (levelized) it is $168. Over 2030–45 it is $132, and $118 with forgone fuel-tax revenue counted as a transfer. Excluding indirect land-use change, current law abates about 1,900 Mt over 2030–45 at $116 a tonne, the federal mandate at $92 and the state standards at $266. On the definition of Wu et al. (2026), which adds government spending to the extra cost of buying the no-policy quantities at policy prices, the cost is $340 a tonne. Counting the change in congestion, accident and local pollution costs as well, it is $111 (range $91–123). If the whole wedge between the delivered price and the absorption cost of renewable diesel into California is a resource cost rather than a rent (Appendix H), it is $160; if the cap-and-trade charge on fuel is counted as allowance revenue, it is $149. In the run without fuel policy and the run without the RFS the market chooses the ethanol share; if the 10% blend floor is kept in those runs and the loss it imposes on blenders is counted, current law costs $143 a tonne (Appendix H).

Counting the lower congestion, accident and pollution costs of using less fuel lowers the threshold to about $126, 0.45 times EPA's central value, and to 0.71 times EPA's value at 3%.

At a 3% discount rate current law gains $31 billion net of climate damages, the share of the first-best gain is 10% and the share of a 45Z extension reaching fuel buyers 0.726.

The duties on imported fat treat the whole gap between the import and domestic price of fat as tariff, which the duties in force in 2026 exceed. Counting only the most-favored-nation tariff and the Section 301 duty, without the temporary Section 122 surcharge of February–July 2026, the resource cost is $261 billion ($135 a tonne) and the federal cost $2.0 billion higher. The model holds that gap at its 2026 value and does not follow later changes in tariffs.

Capital costs are a range: endogenous capacity (main run) and capacity fixed at observed levels.

I allocate changes in federal spending and revenue by each quintile's share of federal taxes in 2022: 0.3, 3.9, 8.6, 16.8 and 70.2% from the lowest quintile to the highest (Congressional Budget Office, 2026, supplemental Table 12). The credits issued at the caps are state revenue that I do not

[12] CARB's cost path is taken to be in 2021 dollars; its source does not state the year.

allocate to households; under the statute fuel sellers would hold the same amount: for the state standards, governments go from +$80.6 to −$21.7 billion and fuel sellers gain $102.3 billion as an unpaid obligation, while the resource cost is $125.7 billion either way. Without the credits issued at the cap, the taxpayer channel of the state standards costs the highest income quintile 0.02% of income a year and the lowest almost nothing.

The fuel-price figures leave out the credit revenue that returns to households in the four states. In the model, electricity credits there are worth $59.7 billion over 2026–45. Of this, $33.0 billion comes from residential charging and passes through utilities, which spend it on vehicle rebates ($12.2 billion for light-duty and $12.6 billion for trucks), charger grants and administration; $26.7 billion goes to owners of non-residential stations.

G.3 The blenders' credit and the RIN price

Proposition 1(ii) implies that a credit on top of a binding mandate is capitalized one for one into the RIN price. I test this on the $1 blenders' credit for biomass-based diesel, which lapsed and was restored retroactively several times. The regression is monthly, January 2014 to July 2021, of the D4 price on the credit in force per RIN (the credit divided by 1.5 RINs per gallon) and the gap between the biodiesel and diesel prices per RIN, with Newey–West standard errors, in the spirit of Irwin et al. (2020). Full capitalization gives a coefficient of −1 on the credit. Daily event windows around each lapse, restoration and extension from 2010 to 2026 complete the evidence.

Over 2014–21, when the $1 blenders' credit lapsed and was restored, a dollar of credit moved the D4 price by −0.41 (s.e. 0.13, 86 months in which the mandate bound), well short of the −1 that full capitalization implies. Two readings fit this number. Every lapse was later reversed retroactively, so traders may have treated a lapse as only partly a loss of the credit; if they expected restoration with probability q, full capitalization gives $-(1-q)$, and the estimate implies $q = 0.59$. Or capitalization is partial. Two other measures of q point to the second reading. The decomposition of the D4 price used for the model gives a median of 0.18 over the lapse months, and investors' own decisions imply 0 ([0, 0.31]; Section 5.2). At q between 0 and 0.18 the regression implies that only 40–50% of the credit was capitalized. A third measure points the other way. In EPA's weekly transaction prices, current-vintage D4 RINs barely move in the weeks around each lapse and restoration, which under full capitalization implies $q = 0.98$ ([0.79, 1.18], nine events). The design is weak: the lapses on 1 January were scheduled, and the RIN vintage changes in the same weeks. Taken together, RIN traders appear to have treated restoration as nearly certain over a few weeks, investors did not, and the monthly regression lies between. Other evidence does not separate the two readings: the coefficient is −0.10 in 2014–16 and −0.59 in 2017–21, the D4–D6 spread responds significantly only in 2014–19, and neither the December 2019 restoration nor the 2022 and 2025 extensions moved it detectably in the following ten trading days (Appendix G.3).

G.4 The 45Z extension

Table 19: Incidence of a 45Z extension under alternative specifications, 2030–2045 (shares of the federal cost; present value at 2%)

	Main	Perfect foresight	$\gamma = 0.32$	$p = 0.5$	With EPA offset
RIN transfer to fuel buyers in the model	1.081	1.092	1.107	1.118	−0.378
Fuel buyers, other (biofuel prices and quantities)	−0.338	−0.353	−0.360	−0.375	0.268
Transfer through the state credit prices	−0.016	−0.005	−0.006	−0.015	0.034
Share to fuel buyers in the model	*0.727*	*0.735*	*0.742*	*0.728*	−0.076
RIN transfer to non-road diesel buyers (outside the model)	0.115	0.116	0.118	0.119	−0.041
Feedstock suppliers	0.185	0.186	0.184	0.184	0.463
Biofuel producers (net of capital costs)	0.050	0.045	0.041	0.032	0.246
State revenue from credits issued at the caps	0.014	0.020	0.020	0.014	−0.013
Other (duties and other government lines, holders, exempt refiners, vehicle buyers)	−0.025	−0.031	−0.030	−0.025	0.004
Net resource cost	−0.066	−0.071	−0.075	−0.050	0.418
Federal cost ($ billion)	53.5	53.4	53.5	53.5	57.6
Emissions change (Mt)	+64.3	+65.2	+65.7	+64.0	−102.2
of which outside the clean fuel states	+59.7	+60.5	+60.0	+60.0	−90.4
Resource saving per extra tonne / cost per tonne abated ($)	55	58	61	42	235
Binding case of Proposition 1	ii (v)	ii (v)	ii (v)	ii (v)	ii (v)

Notes: Extension of 45Z from 2030 to 2045 at its current-law rate, against current law (each column against its own reference: perfect foresight against current law under perfect foresight; $p = 0.5$ against current law with $p = 0.5$). "With EPA offset": the extension together with an RFS requirement raised by 6.4%, the increase that returns the D6 price to its current-law path, against current law. Shares sum to one. Case (v), California at its cap, applies in the state market from 2032. Under the alternative RIN deficit pattern the share to fuel buyers changes by 0.1% and the RIN transfer by 3%. Memo: 45Z booked on fuel from imported fat, borne by foreign fat suppliers through a lower import price, 0.077 of the federal cost (main column).

The extension with an EPA offset. Two runs separate what the credit does from what EPA does. If EPA raised the requirement by enough to return the D6 price to its current-law path, 6.4%, the extension and the higher requirement together would abate 102 Mt at $235 a tonne. Fuel buyers would lose (−8% of the federal cost), and feedstock suppliers (46%) and biofuel producers (25%) would receive most of the federal dollar (last column of Table 3). A higher requirement on its own, raised by 3.8%, abates the same 100 Mt at $163 a tonne. The abatement of the pair therefore comes from the requirement, while the credit changes who pays for it and raises its cost: with the extension, cheaper fuel raises fuel use, so the 6.4% increase abates only 102 Mt. The same 6.4% increase without the extension abates 171 Mt at $163 a tonne and raises welfare by $14.0 billion once climate damages are counted. Measured against that run, the extension does what it does

under current law: 70% of its cost reaches fuel buyers, emissions rise by 68 Mt and welfare falls by $13.0 billion. Any abatement attributed to a 45Z extension would come from an RFS increase made in response to it, and EPA could raise the requirement without the extension at lower cost. With the requirement raised by 2.65%, about where the extra emissions disappear (−2.6 Mt), fuel buyers receive 39% of the federal cost and feedstock suppliers 31%. Counting climate damages, the extension and the higher requirement together change welfare by −$12.2 billion with no offset, −$7.1 billion at 2.65% and +$1.1 billion with the full offset. That gain comes from the higher requirement. Figure 6 shows the whole range. As the offset rises from zero to 6.4%, the share reaching fuel buyers, the change in emissions and net welfare all move almost in a straight line. Emissions stop rising at an increase of about 2.6%, and the pair's net welfare turns positive only near the full offset. The dotted line marks the size of the revisions EPA has made to its own proposals: from 2014 to 2020 its final percentage standards were on average 2.4% above what it had proposed (1.9% on total volumes, 2014–21; Appendix Table 8). At that size of revision the extra emissions roughly disappear, and the pair still lowers welfare.

Under perfect foresight the share reaching fuel buyers is 0.735 and emissions rise by 65 Mt; with a lower gain in the price-expectation rule ($\gamma = 0.32$, against 0.75) they are 0.742 and 66 Mt; and with a probability of extension of $p = 0.5$ they are 0.728 and 64 Mt (Table 3).

The result does not depend on the biomass-based diesel requirement binding on its own. Under current law it binds separately only in 2030, 2032 and 2034, with D4 above D6 by at most $0.14, and in every other year the two prices are equal, so the fall in the RIN price reaches gasoline as well as diesel. Nor does it depend on which years the market borrows RINs: with the other deficit pattern the solver can settle on, the share reaching fuel buyers changes by 0.1% and the RIN transfer by 3%.

Oregon, Washington and New Mexico defer their standards when credits run short, and California tightens its standard when the bank is large. These responses could offset part of the extension. They do not: with the responses switched off, the extension raises emissions by 63.8 Mt, against 64.3 Mt with them, and the share reaching fuel buyers is 0.72 against 0.73. Once the federal mandate binds, the extension barely moves the state banks, so the rules have nothing to respond to (Appendix Figure 15).

Table 20 splits the full offset into the higher requirement and the extension. The requirement raised by 6.4% without the extension abates 171 Mt at $163 a tonne. Measured against that run, the extension sends 70% of its cost to fuel buyers and raises emissions by 68 Mt, close to its effect under current law (73% and 64 Mt). The split almost adds up. The extension alone and the higher requirement alone together abate 106 Mt, against 102 Mt for the pair. With the higher requirement the extension costs the Treasury $4.1 billion more, because more biofuel earns the credit.

The states have written part of their response to the bank into law. A dollar of national support that adds to Oregon's, Washington's or New Mexico's bank makes their forecast deferral less likely,

Table 20: Decomposition of the full EPA offset, 2030–2045 (present value at 2%)

	Federal cost ($ billion)	Share to fuel buyers	Emissions (Mt)	Net welfare incl. climate ($ billion)
RFS +6.4% alone, against current law	0	—	−170.6	+14.0
45Z extension alone, against current law	53.5	0.727	+64.3	−12.2
45Z extension given RFS +6.4%	57.6	0.699	+68.4	−13.0
Both together, against current law	57.6	−0.076	−102.2	+1.1

Notes: "Given RFS +6.4%": the pair against the run with the requirement raised by 6.4% and no extension. Emissions: positive is an increase.

so their standards stay tighter and part of the dollar becomes abatement in those states rather than a lower credit price. California's acceleration mechanism does not trigger in any run. Figure 15 compares the 45Z extension with the codified responses on and off.

They make no measurable difference: with the responses switched off the extension raises emissions by 63.8 Mt against 64.3 Mt with them, and the share reaching fuel buyers is 0.72 against 0.73; the result is the same under each of the three RIN deficit patterns. Once the mandate binds, the extension barely moves the state banks, so the deferral rules have nothing to respond to. In the rule's version the codified response is the only state response; California's discretionary tightenings of 2018 and 2024 both followed large banks, so this offset is a lower bound on the states' response. The same figure reports two state reforms that move credit value from renewable diesel toward SAF (SAF credits multiplied by 1.5, or renewable diesel credits by 0.5), the treatment of the shortfall at California's cap as an accumulated deficit carried forward at 5% rather than as issuance (within 2045 this leaves resource cost unchanged, −$0.1 billion, but leaves 710 Mt of obligations to be met after 2045, about $125 billion at the cap in present value at 2%), and Washington's 20% limit against its statutory path to 45%: with the path to 45% the extension's share to fuel buyers is 0.737 and emissions rise by 64.5 Mt, against 0.727 and 64.3 Mt.

G.5 Exemptions and the RIN price

At the exemptions in EPA's final rule (7.55 billion gallons, about 1.11 billion RINs), the model's 2026 D6 price is $2.58, against a D6 price of $2.13 after EPA's decision on 2025 exemptions on 31 August 2026 ($2.02 in July–August, $1.49 over January–August) (Section 5.4). The main specification uses the 2023–25 average exempted volume, 10.07 billion gallons, which gives $2.29. The exemption volume at which the model reproduces the market price, holding the 2025 bank at its observed level, is 1.71 billion RINs (11.6 billion gallons) at the Argus D6 quote of 31 August 2026, against EPA's projection of 1.11 billion for 2026–27; for 2025 EPA projected 0.78–0.99 billion exempted RINs and on 31 August 2026 granted 1.76 billion RINs on 29 of 34 petitions. This market-implied volume is a calibration, not an estimate: it absorbs any other 2026 misfit of the RIN block, so I report it next to the 2026 D4 fit.

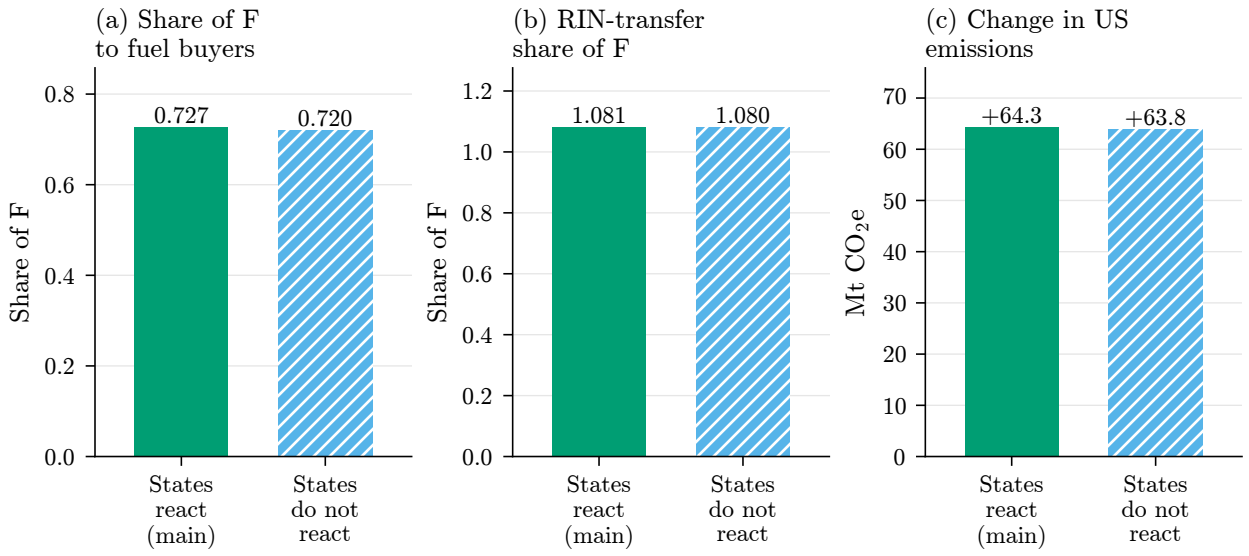


Figure 15: The 45Z extension with the states' codified responses on and off, 2030–2045

Notes: Codified responses: California's automatic acceleration mechanism and the deferral rules of Oregon, Washington and New Mexico.

G.6 A national low carbon fuel standard

This appendix asks what a single national standard would do in place of the current stack. Congress announces it in 2028 and it takes effect in 2030. It replaces the four state programs and repeals the RFS. The state programs run as before in 2028–29, and their banks, 16.75 Mt in all, merge into the national market at the start of 2030. The national standard starts from the 2027 pool carbon intensity under current law and declines at California's scheduled rate, to 87% below that level by 2045. It has one price cap, California's $261.52 a tonne. Credits issued at the cap are federal revenue. 45Z ends after 2029, as under current law; 45Q, the state SAF credits and the vehicle policies are unchanged. Electric vehicles earn credits in every state. I report two versions. In the first (E1a), jet fuel carries no obligation and SAF earns credits against the diesel standard, as in California's program. In the second (E1b), fossil jet fuel is obligated against a national jet standard.

Both versions bind at the cap. The merged bank is used up in 2030, and from 2031 the national market issues credits at the cap in every year. In E1a the shortfall grows from 0.16 Gt in 2031 to 1.16 Gt in 2045, and federal revenue from issuance from $43 billion to $304 billion a year. In E1b the shortfall grows from 0.20 to 1.38 Gt, and revenue from $52 billion to $362 billion a year. A national standard on California's path therefore works as a carbon price of $262 a tonne on transportation fuel, extended to jet fuel in E1b. Its effect comes from the cap price, not from meeting the standard.

Table 21 compares the two versions with current law. Against no fuel policy, over 2030–45, current law abates 1,675 Mt at $132 a tonne. A national standard abates about three times as

much at about 70% of the cost per tonne: 5,177 Mt at $92 in E1a, and 5,691 Mt at $89 in E1b. Measured against current law, moving to E1a abates another 3,502 Mt at $73 a tonne, and moving to E1b another 4,016 Mt at $71. Both gain far more in climate value than they cost. Net welfare including climate damages is +$784 billion for E1a and +$881 billion for E1b, against +$189 billion for current law. Governments gain as well, because the federal government collects the issuance revenue: against current law the fiscal gain is $1,963 billion in E1a and $2,366 billion in E1b, in present value.

The comparison makes the point of Section 6.4 at scale. The current stack pays for fuel inside a binding federal mandate and leaves California's cap to do the pricing in one state. One national market at the same cap prices carbon in every state, and the cost per tonne falls.

Table 21: A national low carbon fuel standard, 2030–2045 ($ billion, present value at 2%)

	Abatement (Mt)	Resource cost ($-W$)	Cost per tonne ($)	Net welfare incl. climate damages
Against no fuel policy				
Current law	1,675	221.2	132	+189.1
National standard, jet not obligated (E1a)	5,177	477.1	92	+784.3
National standard, jet obligated (E1b)	5,691	506.1	89	+880.5
Against current law				
E1a	3,502	254.9	73	+596.2
E1b	4,016	283.9	71	+692.4

Notes: Rows against current law are computed from each pair of runs directly; because consumer surplus is measured at the average demand of each pair, they differ from the difference of the rows against no fuel policy by about $1 billion. The national standard is announced in 2028, takes effect in 2030, replaces the four state programs and repeals the RFS. Abatement is undiscounted and includes indirect land-use change. Resource cost includes forgone fuel-tax revenue. Climate damages use EPA's social cost of carbon at a 2% discount rate (U.S. Environmental Protection Agency, 2023). Over 2026–45 against no fuel policy, E1a abates 5,427 Mt at $95 and E1b 5,941 Mt at $91. Levelized cost per tonne against current law: $97 for E1a and $94 for E1b. Shares of a federal dollar are not reported, because federal outlays barely change (45Z ends after 2029 in both runs). The run without fuel policy has no ethanol blend floor.

G.7 Sensitivity

This subsection reports the headline quantities (California's cap year, the cost per tonne of current law and the 45Z decomposition) under alternative assumptions. The delivery slopes are varied from half to twice their estimated values, a range that contains the 95% confidence interval of California's estimate (0.58 to 1.42 times the estimate) and the estimate from 2023–25 data alone (0.53 times).

The runs at $\gamma = 0.32$ and $p = 0.5$ are in Table 5, and Advanced Clean Cars II and direct air capture with 45Q extended in Figure 14. With a real holding cost of 3.4% in place of 5% everywhere r enters (the rule's balance responses rescaled so that $\gamma \ln(1+r)$ is unchanged), the 45Z extension sends 0.729 of its cost to fuel buyers against 0.727, emissions rise by 63.9 Mt against 64.3, the cost

per tonne of current law changes by less than 0.1%, and California first reaches its cap in 2032 in both. With the delivery slopes at half and at twice their estimates, the extension sends 0.741 and 0.739 to fuel buyers, and the state standards cost $242 and $253 per national tonne against $254 (Section 6.2). With a stronger balance response ($\gamma = 0.94$), current law costs the same per tonne as in the main run, the extension sends 0.740 of its cost to fuel buyers and raises emissions by 65 Mt, and California first reaches its cap in 2032.

The price-expectation rule projects net credits from the last three years. When a standard is amended, past net credits should be restated to the amended standard; otherwise holders learn the same tightening twice, once in the starting price and again as a surprise. The backtest restates them. The forward runs start in 2026 without the restatement of the 2024 net credits to the 2025 amendments. Before running the correction I set five criteria for changing the main specification: the cost per tonne of current law moves by more than $5, its abatement by more than 1%, the share of a 45Z extension reaching fuel buyers by more than 0.01, the extension's extra emissions by more than 2 Mt, or California's bank-out or cap year changes. None is met.

Table 22: Forward runs with and without the restatement of past net credits

	Main specification	With restatement
Cost per tonne, current law, 2026–45 ($, accounting used for this test; see note)	184.41	184.30
Abatement, current law, 2026–45 (Mt, earlier baseline; see note)	1,792.9	1,792.6
Share of a 45Z extension reaching fuel buyers	0.727	0.729
Extra emissions from the extension (Mt)	64.3	64.1
California: bank runs out / first year at the cap	2031 / 2032	2031 / 2032
Oregon / New Mexico: first year of deferral	2034 / 2033	2034 / 2033
California credit price, 2027 / 2028 ($/t)	77.3 / 113.8	82.8 / 117.0
California shortfall at the cap, 2045 (Mt)	61.0	61.0

Notes: The restatement raises California's price in 2027–28, when the expected net credits are lower, and leaves later years almost unchanged ($156.4 in 2029 in both runs). The cost per tonne row uses the welfare accounting of the time of this test; under the current accounting the main specification costs $134 a tonne. The abatement row is against the earlier run without fuel policy, which kept the ethanol blend floor.

H Welfare Accounting

This appendix states how welfare is computed. Every result compares a scenario S with a reference R, year by year, and reports present values at 2% to 2025 over the stated window. The building blocks are standard partial-equilibrium surplus measures (Just et al., 2004): consumer surplus under each demand curve, producer surplus above each supply curve, government revenue, and climate damages. Transfers between groups cancel in the total.

Economic welfare is

$$W = \sum_r \Delta CS_r^{fuel} + \Delta CS^{veh} + \Delta PS^{feed} + \Delta \Pi^{bio} + \Delta \Pi^{sta} + \Delta T + \Delta G^{fed} + \sum_r \Delta G_r^{state}, \quad (53)$$

where T collects the credit-market transfers to holders and other generators. The resource cost of a policy is $-W$. Net welfare adds avoided climate damages, $W - \sum_t SCC_t\,\Delta E_t$, with EPA's social cost of carbon at a 2% discount rate (U.S. Environmental Protection Agency, 2023) and lifecycle emissions E_t including indirect land-use change. The cost per tonne is the present value of $-W$ over undiscounted tonnes abated. An accounting identity checks the total: the credit ledgers (RIN and each LCFS program) close in every state and year, and the run stops if they do not.

Road and jet fuel demand is isoelastic in the retail price, $Q = A(P + w)^{\beta}$, where w is a fixed retail wedge that includes excise taxes. The change in consumer surplus is the exact integral

$$\Delta CS^{fuel} = -\frac{\bar{A}}{\beta+1}\Big[(P_S + w)^{\beta+1} - (P_R + w)^{\beta+1}\Big], \qquad \bar{A} = \tfrac{1}{2}\Big[\frac{Q_S}{(P_S+w)^{\beta}} + \frac{Q_R}{(P_R+w)^{\beta}}\Big], \quad (54)$$

by region and fuel. P is the pool price per gasoline- or diesel-equivalent gallon, so buyers bear the lower energy content of ethanol. The price carries every pass-through: RIN and LCFS compliance costs and the cap-and-trade charge. Averaging the demand level A over the two runs lets the curve shift with the electric share of the fleet; the shift itself is counted with vehicle buyers.

The electric share of new sales is logit. The change in buyers' surplus is the Small and Rosen (1981) log-sum, converted to dollars with the reference run's marginal utility of money. The internal-combustion alternative is held at the reference run's cost of ownership, so fuel-price changes are counted once, with fuel buyers. Purchase credits, LCFS-funded rebates and the charging network enter through the log-sum.

Each feedstock has an isoelastic supply, $Q = \alpha P^{\beta}$, and producer surplus $\alpha P^{\beta+1}/(\beta+1)$, for soybean oil, North American animal fat, corn oil and corn. The surplus of foreign suppliers of imported fat and cane ethanol is reported but not counted in W.

Producers' surplus is revenue less feedstock, operating, utilization and capital costs. Three rules decide what is a resource cost and what is a rent.

- *Rising costs.* Where a cost rises with volume, the area under the marginal-cost curve is the resource cost and the rest of the payment is a rent to producers. This applies to the utilization cost of hydroprocessing capacity and to the cost of delivering renewable diesel into a clean fuel state, $\tau(q)$, on both segments of California's curve: the resource cost is $\int_0^q \tau(x)\,dx$ and the rent is $\tau(q)\,q - \int_0^q \tau(x)\,dx$. Because the curve is calibrated from credit values (Section 5.3), the split between cost and rent is an assumption. Counting the whole payment $\tau(q)\,q$ as a resource cost gives an upper bound, reported with the main results.

- *Capital.* New capacity is charged its capital cost, annualized over a 20-year life and counted only within 2025–45. The part of the estimated entry cost above the engineering capital cost is not a resource cost. Idle capacity carries its fixed operating cost.

- *Freight.* Freight into the clean fuel states is a resource cost.

Changes in credit banks, valued at the year's price, are transfers between deficit payers and holders. End-2045 banks are valued at the 2045 price; the counterparty lies after the horizon, and results are also reported without them. The federal line collects excise taxes, duties on imported fat and carbon revenue, less outlays on 45Z, 45Q and the other credits. State lines collect excise taxes and, when a program is short at its cap, the credits issued at the cap, less state outlays.

Six choices are conventions rather than measurements; the text reports each alternative.

1. *Fuel taxes.* The change in excise revenue is a cost of using less fuel. The cost per tonne is also reported with that revenue treated as a transfer, and with the change in congestion, accident and local pollution costs added (Section 4.5).

2. *Credits issued at the cap* are state revenue. Under the statute they are an unpaid obligation of fuel sellers; prices, quantities and the resource cost are the same either way.

3. *Cap-and-trade* is a fixed charge at 2025 allowance prices, and allowance revenue is not counted. Covered abatement is counted in full (Section 8).

4. *The ethanol blend floor.* The reference runs without fuel policy and without the RFS let the market choose the ethanol share. In those runs a cane-ethanol price floor at 1% of its 2025 price keeps the solver away from a zero price; cane ethanol is foreign and not counted in W. Keeping a 10% floor in both reference runs and counting blenders' loss from it is a sensitivity: current law then costs \$143 a tonne over 2026–45, against \$134, and the blenders' loss is \$20.6 billion over 2030–45 in the run without the RFS.

5. *Price changes are not rounded.* Every price change between two runs enters surplus, however small.

6. *Foreign surplus* is excluded; the terms-of-trade gain on imported fat is counted as a saving of resources.

Social welfare is the sum of the surpluses of fuel users (by state), vehicle buyers, biofuel producers, feedstock suppliers, the federal government, the state governments and owners of capital, less climate damages. Feedstock supplier surplus is the area above the upward-sloping supply curves of North American soybean oil, fat, corn oil and corn; the surplus of foreign suppliers of imported feedstock is reported separately. Transfers between groups (credit payments, tax credits, rents) cancel in the total, which is checked by an accounting identity. Climate damages use EPA's 2023 social cost of carbon at a 2% discount rate, with 1.5% and 2.5%, the interagency value and a domestic value as sensitivities; electricity used by electric vehicles is valued at the long-run marginal emission rates of NREL's Cambium 2024 Mid-case (Gagnon et al., 2025). Costs are resource costs: capital at its capital cost, not the investment hurdle (the part of κ_ℓ above capital cost is a sensitivity), fixed operating cost on idle capacity, the feedstock-mix cost (36) and the state diesel differentials.

The shadow price of the utilization curve is not a social cost. Two capital ledgers are reported side by side: all SAF conversion capital, and only the SAF-capable capacity that is used; the difference comes from the logit rule's positive announcement probability at zero expected profit. When a program is at its cap, issuance at the cap is state revenue; the statutory deficit is reported as an unpaid liability, not added to that revenue. The first-best benchmark is a carbon price at the social cost of carbon on all fuels and vehicles in the model, so each policy can be scored by the share of the first-best welfare gain it achieves; a carbon price that delivers the same cumulative abatement is the cost-effectiveness benchmark. Both are first best within the model's scope: the model is partial equilibrium, and world feedstock markets, indirect land use and foreign emissions are not endogenous. I do not compute marginal values of public funds, because the LCFS's cost does not pass through a government budget; for 45Z I report the outcome per dollar of federal cost.

Other external costs of fuel use. A mile of car travel costs 5.0 cents in congestion and 3.0 cents in accidents (2005 dollars, Parry et al., 2006), and 0.54 cents in local air pollution for a new gasoline car (Holland et al., 2016, 2014 dollars), with the older-fleet value of 2.0 cents as the upper bound; I convert at 21 miles per gallon. A mile of truck travel adds congestion, accident, noise and pavement costs of 95 cents per gallon (2000 dollars, Parry, 2008). Gallons displaced by electric vehicles cause no change in miles; in the upper bound they add the local damages of charging on the 2014 grid (Holland et al., 2016). All values are converted to 2024 dollars with the CPI-U. The other external costs of fuel use are discounted to 2026 rather than 2025, which overstates them by about 2%.